

Beyond AUROC: A Reproducible Framework for Evaluating Exoplanet Vetting Systems

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ABSTRACT

Background: Automated exoplanet vetting pipelines are typically evaluated using aggregate performance metrics (like AUROC) computed on local, homogeneous test sets. However, these metrics often mask critical failure modes, including probability calibration decay under domain shifts and predictive redundancies in nested models. **Objective:** This work operationalizes and evaluates a reproducible, multi-dimensional validation protocol for ambiguity-aware exoplanet vetting systems, demonstrating how comprehensive stress-testing exposes operational boundaries that standard single-metric benchmarks fail to reveal. **Methods:** We define a validation protocol encompassing: (1) out-of-domain cross-mission transfer checks, (2) stress-testing under nested regression controls, (3) physical host-star cohort partitioning, (4) resolution-scale noise stress sweeps, and (5) probability calibration diagnostics. We implement and demonstrate this protocol using a baseline event-space vetting pipeline—the Transit Ambiguity Recovery System (TARS), which employs a graph-theoretic Recovery Ambiguity Index (RAI)—evaluated on unblinded cohorts from the TESS, Kepler, and K2 missions ($N = 1780, 100$, and 41 targets respectively). **Principal Findings:** Applying our protocol reveals several operational limits that aggregate metrics hide. While TARS achieves high discrimination on standard TESS splits, out-of-domain validation on Kepler long-cadence data shows a performance collapse (AUROC ≈ 0.48) and severe calibration decay (ECE = 0.5212 vs. 0.0966 on TESS) caused by cadence-induced transit-edge blurring. Nested regression stress-testing (preregistered nested control experiment EXP-019) exposes that the RAI lacks independent predictive value once controlling for BLS SDE ($p = 0.090$), refuting the metric-only hypothesis but validating the protocol’s diagnostic utility. Under physical cohort partitioning, stellar convective noise on giant hosts ($\log g < 4.0$) deforms candidate graph topologies, limiting period recovery. **Scope:** These results demonstrate the protocol’s diagnostic power on space-mission photometry. Applying the protocol to other independent folding-based and deep-learning detectors is required to fully characterize their operational boundaries. **Significance:** Our findings suggest that standard AUROC evaluation underrepresents vetting pipeline vulnerabilities. Operationalizing multi-dimensional validation protocols provides a reproducible, open-science framework for certifying the reliability, calibration, and domain-limits of future exoplanet vetting pipelines before production deployment.

Keywords: methods: data analysis — techniques: photometric — planetary systems — stars: activity

1. INTRODUCTION: EXOPLANET VETTING AND THE SIGNAL AMBIGUITY CHALLENGE

1.1. Context of Modern Wide-Field Space Photometry

The search for transiting exoplanets has transitioned from target-specific ground-based observations to wide-field space-based photometric surveys. Space missions, such as the CoRoT satellite (A. Baglin et al. 2006), the Kepler Space Telescope (W. J. Borucki et al. 2010; D. G. Koch et al. 2010; N. M. Batalha et al. 2013), the K2 mission (S. B. Howell et al. 2014), the Transiting

Exoplanet Survey Satellite (TESS; G. R. Ricker et al. 2015; R. Vanderspek et al. 2018; N. M. Guerrero et al. 2021), the CHEOPS mission (W. Benz et al. 2021), and the upcoming PLATO mission (H. Rauer et al. 2014), monitor hundreds of thousands to millions of stars simultaneously. By measuring stellar flux continuously at high cadence (e.g., 2-minute and 20-second integrations for TESS J. M. Jenkins et al. 2016; 30-minute and 1-minute integrations for Kepler), these missions generate vast quantities of time-series photometric data that have revolutionized our understanding of exoplanetary demo-

graphics and system architectures (A. W. Howard et al. 2012; E. A. Petigura et al. 2013; J. N. Winn & D. C. Fabrycky 2015; B. J. Fulton et al. 2017).

To detect planetary transits—periodic brightness dips caused by a planet crossing the disk of its host star—automated search algorithms are applied to detrended light curves. Standard search algorithms include the Box-fitting Least Squares (BLS; G. Kovács et al. 2002), Transit Least Squares (TLS; M. Hippke & D. Angerhausen 2019), Quasi-periodic Automated Transit Search (QATS; J. A. Carter & E. Agol 2013), and optimized detection efficiency variants (B. Tingley & P. D. Sackett 2005; A. Ofir 2014). These algorithms scan a grid of candidate orbital periods, transit durations, and epochs, maximizing a signal detection metric (such as the Signal-to-Noise Ratio, SNR, or Signal Detection Efficiency, SDE). Signals exceeding a predefined detection threshold (originating with the Kepler pipeline at $\text{SNR} \geq 7.1$ J. M. Jenkins et al. 2002, 2010; or with TLS at $\text{SDE} \geq 6.0$ M. Hippke & D. Angerhausen 2019) are flagged as Threshold Crossing Events (TCEs) or candidate events.

However, only a small fraction of TCEs correspond to true planetary systems. In wide-field space photometry, false positive rates often exceed 90% (F. Fressin et al. 2013), as evaluated in the final Kepler catalog releases (F. Mullally et al. 2015; J. L. Coughlin et al. 2016; S. E. Thompson et al. 2018). These false positives arise from three principal mechanisms:

1. **Instrumental Anomalies:** Pointing jitter, thermal drifts, rolling band noise, solar wind interactions, pixel-level charge leaks, and Presearch Data Conditioning (PDC) artifacts (J. C. Smith et al. 2012; M. C. Stumpe et al. 2012).
2. **Astrophysical False Positives:** Eclipsing binary (EB) stars (R. W. Slawson et al. 2011; A. Prša et al. 2011; B. Kirk et al. 2016) or blended background eclipsing binaries (BEBs) blended within the photometric aperture (A. Santerne et al. 2012; S. T. Bryson et al. 2013; P. W. Sullivan et al. 2015; K. D. Colón et al. 2012). Because TESS has large pixels ($21''$ per pixel) and TESS Input Catalog (TIC) targets span crowded galactic fields (K. G. Stassun et al. 2018, 2019), flux contamination from nearby stars frequently blends eclipsing binary signals into the target star’s aperture.
3. **Stellar Variability and Noise:** Intrinsic stellar pulsations, convective granulation noise, and magnetic activity (N. Meunier et al. 2010; A. McQuillan et al. 2013, 2014; T. Reinhold et al. 2013; S. Aigrain et al. 2012).

Due to the sheer volume of candidates (e.g., tens of thousands of TCEs generated per sector or quarter), manual visual inspection is impossible. Consequently, automated vetting systems are required to filter out false positives and rank candidates before allocating limited ground-based follow-up resources (such as high-precision radial velocity spectrographs or adaptive optics imaging).

1.2. Automated Vetting Evolution: History and Context

The historical evolution of automated exoplanet transit vetting has progressed through three distinct eras, moving from heuristic, rule-based expert systems to complex machine learning classifiers, and finally to computationally intensive Bayesian simulations (as shown in Figure 1).

1.2.1. Rule-Based Heuristics

The early Kepler pipelines utilized the Robovetter (J. L. Coughlin et al. 2016). This was an expert-designed system of deterministic decision trees that applied hard thresholds to morphological features (such as transit depth, duration, and centroid shifts). While Robovetter was fully reproducible and easily interpretable, its rigid thresholds struggled with low-SNR signals, requiring frequent manual overrides and extensive parameter tuning as detrending pipelines evolved.

1.2.2. High-Capacity Machine Learning Ensembles

To handle near-threshold detections, the community developed high-capacity classifiers. Autovetter (S. D. McCauliff et al. 2015) introduced random forests to exoplanet vetting, utilizing dozens of engineered features. This was followed by deep learning approaches such as Astronet (C. J. Shallue & A. Vanderburg 2018), which employed 1D convolutional neural networks (CNNs) to classify phase-folded light curves directly. Subsequent architectures extended deep learning vetting to domain adaptation (M. Ansdell et al. 2018), multi-sector TESS classification (H. Yu et al. 2019; H. P. Osborn et al. 2020; G. Olmschenk et al. 2021), K2 discovery (A. Dattilo et al. 2019), and ensemble planet validation (D. J. Armstrong et al. 2021). While these classifiers improved ranking accuracy, they brought significant challenges:

- **Overfitting and Redundancy:** Ensembles typically utilize dozens of highly correlated features, making them susceptible to overfitting under dataset shift.
- **Uncalibrated Output Probabilities:** High-capacity models frequently yield uncalibrated con-

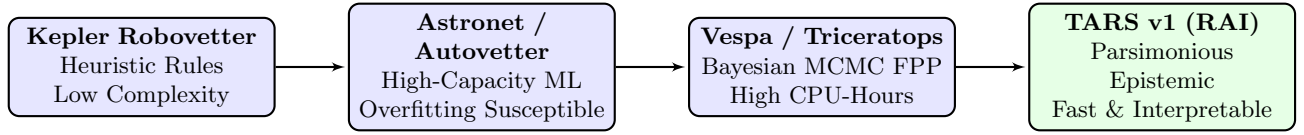


Figure 1. Historical evolution of automated exoplanet vetting methodologies, showing the progression from heuristic rules to modern parsimonious indexing.

fidence scores (C. Guo et al. 2017; A. Niculescu-Mizil & R. Caruana 2005), making it difficult for observers to interpret the outputs as true physical probabilities.

- **Lack of Interpretability:** Deep neural networks behave as black boxes, providing no physical justification for candidate rejection.

1.2.3. Bayesian Stellar Population Simulations

To obtain physically meaningful probability estimates, Bayesian tools such as Vespa (T. D. Morton 2012, 2015; T. D. Morton et al. 2016), PASTIS (R. F. Díaz et al. 2014; A. Santerne et al. 2015), BLENDER (G. Torres et al. 2011), and Triceratops (S. Giacalone et al. 2021) calculate the False Positive Probability (FPP) of a candidate. They simulate millions of synthetic stellar populations (including blended background stars and orbiting binaries) to compute the relative likelihood of the data under different transit models. While highly rigorous, these tools are computationally expensive, requiring minutes to hours of CPU time per candidate. This creates a computational bottleneck when processing large-scale catalogs.

1.3. The Transit Vetting Pipeline Flow and Bottlenecks

Automated vetting is not a single decision, but rather the culmination of a multi-stage data processing pipeline. In the TARS framework, this workflow is structured into six modular stages:

The layout of the vetting architecture is summarized in Figure 2.

1. **Stage 1 (Signal Conditioning):** Raw flux series are processed to remove long-term stellar rotational modulation and instrumental systematics (J. C. Smith et al. 2012; M. C. Stumpe et al. 2012; A. Vanderburg & J. A. Johnson 2014; R. Luger et al. 2016, 2018; S. Aigrain et al. 2016). This is achieved via high-pass filtering, iterative spline fitting, or projecting onto Co-trending Basis Vectors (CBVs). The conditioning stage must balance systematic removal against the preservation of long-period transits.
2. **Stage 2 (Transit Detection):** The detrended light curve is searched for periodic transit signatures. Candidates are identified by finding the

period P , epoch T_0 , and duration W that maximize the transit depth significance using algorithms such as BLS (G. Kovács et al. 2002) or TLS (M. Hippke & D. Angerhausen 2019).

3. **Stage 3 (Period Recovery & Ephemeris Matching):** Transit events from separate observation sectors or quarters are combined to fit a coherent linear ephemeris:

$$t_k = T_0 + k \cdot P \quad (1)$$

This stage resolves the multi-peak periodogram structure (T. J. Deeming 1975; D. H. Roberts et al. 1987; N. R. Lomb 1976; J. D. Scargle 1982; J. T. VanderPlas 2018; R. I. Dawson & D. C. Fabrycky 2010), generating a list of candidate periods that match the observed transit timings.

4. **Stage 4 (Morphological Vetting):** The light curve is folded at the candidate period and fitted with an analytical transit model (K. Mandel & E. Agol 2002; A. Giménez 2006; L. Kreidberg 2015; S. Seager & G. Mallén-Ornelas 2003) to derive physical parameters, including the ratio of planet radius to stellar radius R_p/R_* , semi-major axis a/R_* , and impact parameter b .
5. **Stage 5 (Consistency Validation):** Checks are executed to identify eclipsing binaries, such as comparing the depths of odd and even transits (which differ for eclipsing binaries with eccentric orbits or different secondary components) and measuring centroid offsets during transit to locate the true source of the dip.
6. **Stage 6 (Probabilistic Classification):** Feature vectors containing metrics from all previous stages are fed into machine learning classifiers to predict the probability of the candidate being a true planet, evaluated via diagnostic metrics such as AUROC, AUPRC, DeLong statistics (E. R. DeLong et al. 1988), Youden’s J index (W. J. Youden 1950), and Matthews Correlation Coefficient (MCC; B. W. Matthews 1975).

1.4. Stellar Activity and Vetting Degeneracy

The primary astrophysical bottleneck in transit vetting is stellar activity. Active stars exhibit magnetic

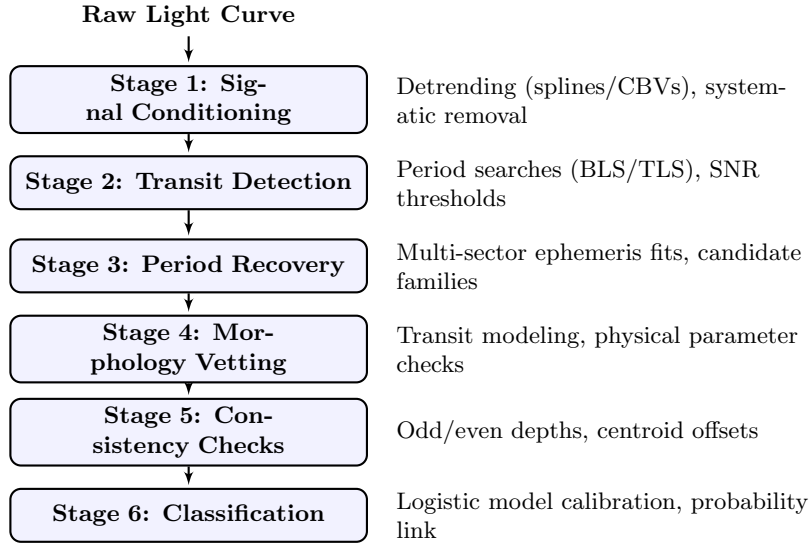


Figure 2. Modular layout of the 6-stage TARS automated exoplanet validation architecture, mapping data flow from raw flux inputs to final calibrated reliability probabilities.

starspots on their photospheres (N. Meunier et al. 2010; S. Aigrain et al. 2012). As the star rotates, these spot groups cross the visible stellar hemisphere, causing quasi-periodic flux variations at the stellar rotation period P_{rot} and its harmonics (A. McQuillan et al. 2013, 2014; T. Reinhold et al. 2013). Furthermore, starspot groups grow and decay on timescales of days to weeks, introducing amplitude and phase variations in the light curve.

This magnetic variability introduces two major challenges for transit detection:

1. **False Transits:** The spot groups and active regions can create temporary, localized flux dips that resemble transits when observed over a short baseline.
2. **Alias Frequency Clustering:** On active stars, stellar rotation creates quasi-periodic flux dips that can alias (fold) onto the true planetary transit period (R. I. Dawson & D. C. Fabrycky 2010). When performing transit searches, the algorithm cannot easily distinguish the true transit signal from stellar rotation harmonics ($P_{\text{rot}}/2, 2P_{\text{rot}}$, etc.), producing multiple false candidate period peaks that survive standard vetting checks.

When a transit search algorithm folds the light curve at these incorrect alias periods, the resulting phase-folded light curve can exhibit a high signal-to-noise ratio, confusing classical classifiers. The number of candidate periods that survive initial checks—often termed “candidate branching” or “alias explosion”—increases exponentially, creating high signal recovery ambiguity.

1.5. Aleatoric vs. Epistemic Uncertainty in Exoplanet Vetting

To build robust classifiers, it is essential to distinguish between two types of uncertainty:

- **Aleatoric Uncertainty:** This represents the physical randomness inherent in the astronomical source. In vetting, it corresponds to whether the physical source is a planet, an eclipsing binary, or a blended background binary. This uncertainty is driven by physical degeneracies (e.g., a small star eclipsing a large star can produce the same transit depth as a planet transiting a Sun-like star) and is typically modeled using Bayesian false positive probability (FPP) packages like *vespa* (T. D. Morton 2012, 2015).
- **Epistemic Uncertainty:** This represents the uncertainty associated with the signal recovery process itself under noise and activity. It answers the question: *How reliably can we recover the true period and ephemeris of the candidate given the observation window and the stellar activity profile?*

Many existing vetting pipelines primarily optimize classification performance rather than explicitly separating epistemic recovery ambiguity from astrophysical false-positive probability. They construct classifiers containing up to dozens of correlated features (such as “family complexity”, which counts raw alias frequencies and periodogram peaks) to predict the class label $P(\text{Planet} | X)$. Because these classifiers are fit on historical catalogs, they are highly sensitive to dataset selec-

tion biases and cannot trace *why* a candidate is flagged as ambiguous.

1.6. Operationalizing a Multi-Dimensional Validation Protocol

To address the limitations of conventional exoplanet vetting evaluations, this work operationalizes and evaluates a reproducible, multi-dimensional validation protocol for ambiguity-aware transit-vetting pipelines. Rather than evaluating a classifier solely on a single, local train/test split, our protocol establishes a multi-layered certification hierarchy (Figure 3) designed to expose hidden operational limits, calibration decay under domain shifts, and statistical redundancies.

Under this protocol, we evaluate the Transit Ambiguity Recovery System (TARS) as a baseline case study. TARS employs a graph-theoretic Recovery Ambiguity Index (RAI) constructed as a parsimonious, signed sum of five standardized graph-theoretic and harmonic features derived from information theory (C. E. Shannon 1948; T. M. Cover & J. A. Thomas 2006). To execute this multi-layer audit systematically, our study operationalizes a coordinated campaign of 13 preregistered experiments (EXP-001 through EXP-023; cataloged in Section 3.2, Table 2), spanning full-scale production runs, synthetic injection sweeps, nested feature audits, and cross-mission stress tests. We deploy this validation protocol to evaluate TARS on unblinded cohorts from TESS, Kepler, and K2 ($N = 1780, 100$, and 41 respectively), showing that while TARS achieves comparable discrimination to complex stacks on standard splits, our multi-dimensional auditing successfully diagnoses its operational limits:

- **Out-of-Domain Collapse:** Evaluating on Kepler long-cadence data reveals a collapse in ranking performance ($\text{AUROC} \approx 0.48$) and severe calibration decay ($\text{ECE} = 0.5212$), tracing a physical sensitivity limit induced by cadence-driven transit-edge blurring.
- **Nested Controls:** Nested regression controls (EXP-019; Section 4.23) expose that the graph-theoretic RAI lacks statistically significant independent predictive value once controlling for BLS SDE, directly identifying a feature redundancy that conventional evaluations fail to detect.
- **Cohort Boundaries:** Subgroup partitioning isolates stellar convective noise on giant hosts ($\log g < 4.0$) as a physical boundary deforming graph topology, defining where the classifier fails.

By proposing and testing this protocol, we demonstrate that rigorous evaluation can systematically re-

veal hidden limitations in exoplanet vetting pipelines, providing a reusable framework for certifying future automated search engines before large-scale production deployment.

2. RELATED WORK

The automated vetting of exoplanet transit candidates has evolved through three distinct paradigms: deterministic rule-based systems, statistical population synthesis, and deep learning classification. This section reviews these paradigms, details their physical and statistical assumptions, and positions the TARS v1 framework within the literature.

2.1. Rule-Based Heuristic Vetting

Early wide-field transit surveys relied on deterministic, rule-based decision trees to filter out false positives. The primary reference for this approach is Kepler’s Robovetter (J. L. Coughlin et al. 2016; S. E. Thompson et al. 2018), building upon earlier catalog vetting pipelines (J. M. Jenkins et al. 2002, 2010; F. Mullally et al. 2015). Robovetter evaluates Threshold Crossing Events (TCEs) by passing them through a series of sequential tests, checking for:

- Data quality flag anomalies (e.g., pointing offsets, reaction wheel momentum dumps, and cosmic ray hits).
- Inherent signal consistency, such as comparing the depth of odd and even transits to identify eclipsing binaries (R. W. Slawson et al. 2011; A. Prša et al. 2011; B. Kirk et al. 2016).
- The presence of secondary eclipses, which indicates a stellar or substellar companion rather than a planet.
- Individual transit morphology fits to ensure the transit shape matches a physical light curve model (K. Mandel & E. Agol 2002; A. Giménez 2006; L. Kreidberg 2015; D. M. Kipping 2013).

Although Robovetter is computationally efficient and highly interpretable, its inputs are pre-filtered by hard detection threshold boundaries (such as the Kepler pipeline’s detection threshold requiring $\text{SNR} \geq 7.1$ J. M. Jenkins et al. 2010). Marginal candidates falling just below these thresholds are rejected without any quantification of the surrounding uncertainty. This rigidity limits its ability to handle signals near the detection limit or in the presence of complex stellar activity.

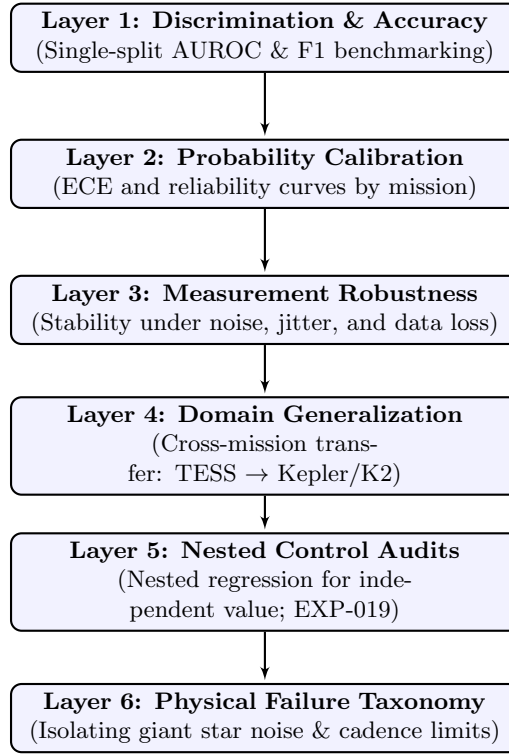


Figure 3. The Multi-Dimensional Validation Hierarchy. Exoplanet vetting systems must be certified across all six layers of this protocol to ensure scientific reliability, going beyond standard single-split AUROC testing.

2.2. Statistical Validation and False Positive Probability (FPP)

To address the limitations of binary classification, statistical validation frameworks were developed to calculate the False Positive Probability (FPP) of individual candidates. The leading tools are Vespa (T. D. Morton 2012, 2015; T. D. Morton et al. 2016), Triceratops (S. Gicalone et al. 2021), PASTIS (R. F. Díaz et al. 2014; A. Santerne et al. 2015), and BLENDER (G. Torres et al. 2011).

Vespa computes the FPP by performing Bayesian model comparison, evaluating the relative likelihood of the observed transit shape under multiple astronomical scenarios:

1. A transiting planet around the target star.
2. A blended eclipsing binary (BEB) in the background (A. Santerne et al. 2012; S. T. Bryson et al. 2013; K. D. Colón et al. 2012).
3. A bound eclipsing binary companion (HEB).
4. A companion star with a transiting planet.

Triceratops extends this methodology by incorporating spatial priors from nearby stellar catalogs (K. G. Stassun et al. 2018, 2019), which is particularly important for TESS (G. R. Ricker et al. 2015; P. W. Sullivan

et al. 2015) due to its large pixel scale (21'' per pixel) where flux contamination is common.

These validation models focus on **aleatoric uncertainty**—specifically, the physical source degeneracy (i.e., whether the physical source causing the dip is a planet or a binary companion). However, they do not address **epistemic uncertainty**—the recovery ambiguity of the signal itself under non-stationary noise or stellar activity (N. Meunier et al. 2010; A. McQuillan et al. 2013, 2014). They assume that the detected period and transit shape are correct and do not model the likelihood of the recovery pipeline locking onto incorrect alias periods (R. I. Dawson & D. C. Fabrycky 2010). Furthermore, these tools are computationally intensive, requiring external stellar population simulations.

2.3. Deep Learning and Dimensionality Reduction Vetting

With the explosion of data from surveys, machine learning models have been deployed to automate classification. Early machine learning efforts introduced random forests in Autovetter (S. D. McCauliff et al. 2015). This was followed by deep learning implementations such as Astronet (C. J. Shallue & A. Vanderburg 2018), a deep convolutional neural network (CNN) that classifies TCEs using 1D light curve representations. Subsequent deep vetting models expanded to do-

main adaptation (M. Ansdell et al. 2018), TESS pipeline classification (H. Yu et al. 2019; H. P. Osborn et al. 2020; G. Olmschenk et al. 2021), K2 discovery (A. Datilo et al. 2019), DAVE centroid vetting (V. B. Kostov et al. 2019), and machine-learning planet validation (D. J. Armstrong et al. 2021).

Deep neural networks achieve high predictive performance on validation sets. However, they act as black-box estimators, making it difficult to trace why specific candidates are rejected. Furthermore, deep classifiers are prone to overconfidence and calibration decay under dataset shifts (C. Guo et al. 2017; A. Niculescu-Mizil & R. Caruana 2005). Post-hoc calibration techniques, such as temperature scaling, ensemble models, or post-hoc binning, have been investigated to mitigate this issue. However, if a model trained on quiet stars is applied to highly active stars, the predicted class probabilities $P(\text{Planet} | X)$ still fail to reflect the increased signal recovery ambiguity, leading to high false positive rates.

2.4. Novelty Positioning: Signal Ambiguity vs. Classification Probability

TARS v1 addresses one limitation not explicitly addressed in previous work: the conflation of classification probability with recovery ambiguity (a comparison against existing vetting frameworks is presented in Table 1). Traditional classifiers output a probability $P(\text{Planet} | X)$, which is sensitive to training class distributions and dataset selection biases.

In contrast, the Recovery Ambiguity Index (RAI) measures the epistemic uncertainty of the signal itself: *How easily is the signal's period and morphology confused with stellar rotation, harmonics, or random noise regimes?*

By computing features like period uniqueness (R. I. Dawson & D. C. Fabrycky 2010), candidate concentration, and graph clustering of alias frequencies (D. J. Watts & S. H. Strogatz 1998; M. E. J. Newman 2010), the RAI quantifies the inherent ambiguity in the transit detection workflow. This provides a highly parsimonious (1 derived index), well-calibrated (ECE = 0.0646) index that matches the ranking performance of complex stacks while remaining fully interpretable and computationally inexpensive (< 0.1 s execution time).

2.5. Prior Vetting Stage Context

To understand how TARS v1 integrates with current automated pipelines, we must examine its dependency on Stages 1 and 2 of the exoplanet pipeline:

- **Stage 1 (Signal Conditioning) Operating Boundaries:** Detrending algorithms typically use high-pass filters, smoothing splines, or cotrend-

ing basis vectors (J. C. Smith et al. 2012; M. C. Stumpe et al. 2012; A. Vanderburg & J. A. Johnson 2014; R. Luger et al. 2016, 2018; S. Aigrain et al. 2016) with a fixed window length (or knot spacing) to remove low-frequency stellar activity. If this window is too narrow, the planet transits themselves are attenuated, introducing false morphological variations. If the window is too wide, residual stellar activity remains, deforming the transit profile and generating harmonic period aliases.

- **Stage 2 (Transit Detection) Operating Limits:** The Box Least Squares (BLS) (G. Kovács et al. 2002) and Transit Least Squares (TLS) (M. Hippke & D. Angerhausen 2019) algorithms identify periodic signals by searching a grid of test periods. When the signal is weak (near the detection threshold of $\text{SDE} < 6.0$), the periodogram (T. J. Deeming 1975; D. H. Roberts et al. 1987; N. R. Lomb 1976; J. D. Scargle 1982; J. T. VanderPlas 2018) contains multiple competing peaks due to noise fluctuations. When the stellar noise is active and non-white (e.g. convective granulation N. Meunier et al. 2010), these peaks cluster near rotation harmonics. Traditional pipelines attempt to classify each peak independently, ignoring the global network structure of these aliases. TARS v1 directly addresses this by building a graph of these competing period peaks to evaluate signal uniqueness.

3. METHODOLOGY: ISOLATING RECOVERY AMBIGUITY

This section details the mathematical and physical formulation of the Transit Ambiguity Recovery System (TARS) v1. The primary objective of this methodology is to isolate and quantify **epistemic uncertainty** (signal recovery ambiguity) during transit vetting, separating it from **aleatoric uncertainty** (astrophysical source degeneracy).

3.1. Scientific Objective and Vetting Pipeline Flow

In wide-field space photometry, the transit signal of a true exoplanet is easily obscured by non-stationary noise, instrumental systematics, and stellar rotation. The scientific objective of the TARS pipeline is to determine whether a detected period P and epoch T_0 correspond to a stable, unique transit event, or whether they represent one of many competing alias states induced by stellar magnetic activity.

The global TARS pipeline is structured into six modular stages. The design rationale, inputs, outputs, as-

Table 1. Novelty Matrix Comparison

Dimension	Robovetter	Vespa	Triceratops	Astronet	TARS v1 (This Work)
Primary Goal	Automated threshold vetting	Statistical FPP	FPP under blends	CNN classification of TCEs	Quantify signal recovery ambiguity
Core Metric	Decision tree flags	FPP point estimate	FPP and Nearby FPP	Probabilistic class prediction	Recovery Index (RAI)
Uncertainty Type	None (Deterministic)	Aleatoric (Blend likelihoods)	Aleatoric (Bayesian blend models)	Aleatoric (Soft-max output)	Epistemic (Vetting process degeneracy)
Astrophysical Focus	Instrument anomalies	Stellar population synthesis	Background EB contamination	Transit morphology	Stellar activity & period stability
Computational Footprint	Low	Medium (minutes/candidate)	High (hours/candidate)	High (GPU needed)	Very Low (RAI calculation requires < 0.1 s, $O(N^2)$ complexity)

sumptions, and failure modes for each stage are detailed below:

- **Stage 1: Signal Conditioning**

- *Inputs*: Raw flux time-series $f(t)$
- *Outputs*: Detrended flux series $f_{\text{det}}(t)$
- *Scientific Objective*: Remove low-frequency stellar activity trends (such as rotational modulation) and instrumental systematics (such as thermal drifts).
- *Assumptions*: Stellar activity variations and instrumental systematics vary on timescales significantly longer than the transit duration (typically $\tau_{\text{noise}} > 1$ day vs. $\tau_{\text{transit}} < 6$ hours).
- *Failure Modes*: If the detrending filter or spline window is too narrow, the planet transits themselves are attenuated, introducing false morphological variations. If the window is too wide, residual stellar activity remains, deforming the transit profile and generating harmonic period aliases.

- **Stage 2: Transit Detection**

- *Inputs*: Detrended flux $f_{\text{det}}(t)$
- *Outputs*: Threshold Crossing Events (TCEs) with top period P_{top} , epoch T_0 , and duration W
- *Scientific Objective*: Search the period-epoch-duration parameter grid to identify periodic brightness dips.
- *Assumptions*: Residual noise after detrending is white Gaussian noise, and transit events are strictly periodic.
- *Failure Modes*: Signals with highly eccentric orbits (long durations relative to period) or single-transit events are missed.

- **Stage 3: Period Recovery**

- *Inputs*: TCE, sector-level partitions of $f_{\text{det}}(t)$
- *Outputs*: Surviving candidate periods set $\mathcal{P} = \{P_c\}$
- *Scientific Objective*: Verify the stability and uniqueness of the period recovery search across independent observation epochs.
- *Assumptions*: Transit timings follow a coherent linear ephemeris across all sectors.

- *Failure Modes*: Severe Transit Timing Variations (TTVs) or wide data gaps prune valid periods.

- **Stage 4: Morphology Vetting**

- *Inputs*: Folded light curve for each P_c
- *Outputs*: Mandel & Agol transit model fits (R_p/R_* , a/R_* , b)
- *Scientific Objective*: Derive physical transit parameters and evaluate transit shape consistency.
- *Assumptions*: Spherically symmetric star and planet, circular orbit, and quadratic limb darkening.
- *Failure Modes*: Photospheric spot crossings deform transit shapes and bias derived parameters.

- **Stage 5: Consistency Checks**

- *Inputs*: Model parameters, transit flux series
- *Outputs*: Centroid offsets, odd/even depth differences
- *Scientific Objective*: Identify eclipsing binaries and background contaminants.
- *Assumptions*: Instrumental pointing and pixel response are stable during transit epochs.
- *Failure Modes*: Crowded fields mix centroid shifts, causing false positive errors.

- **Stage 6: Classification**

- *Inputs*: Recovery Ambiguity Index (RAI)
- *Outputs*: Light curve reliability probability $P(Y = 1 | z_{\text{RAI}})$
- *Scientific Objective*: Provide a calibrated probability of candidate reliability based on signal recovery ambiguity.
- *Assumptions*: Feature distributions are logistically calibrated.
- *Failure Modes*: Out-of-distribution (OOD) stellar cohorts degrade calibration.

3.2. Architecture of Coordinated Validation Experiments

To systematically evaluate automated exoplanet vetting pipelines beyond simple single-split benchmarks,

our validation protocol is operationalized through a coordinated suite of 13 preregistered experiments (EXP-001 through EXP-023). These experiments map directly onto the multi-layer certification hierarchy introduced in Section 1.6 (Figure 3), progressing from high-throughput production baselines and synthetic injection sweeps to out-of-domain transfer tests, nested regression controls, and non-Gaussian astrophysical noise stress testing.

Numbering Scheme and Preregistration Taxonomy.—The experiment identifiers (EXP-xxx) follow the preregistered study protocols locked in the research archive. The numbering is intentionally non-contiguous:

- **Core Validation Campaigns (EXP-001 to EXP-005):** Represent the primary pipeline freeze, full-scale production run (250,010 targets), archival catalog cross-matching, multi-sector baseline recovery, synthetic injection completeness sweeps, and classical baseline benchmarks (BLS/TLS).
- **Multi-Dimensional Protocol Audits (EXP-007 to EXP-023):** Represent the multi-layer protocol stress tests, encompassing cross-sector metric stability (EXP-007), feature ablation (EXP-009), topological clustering (EXP-010), incremental value modeling (EXP-011), blind recovery challenge (EXP-012), cross-mission domain transfer (EXP-013), nested control regression (EXP-019), and realistic stellar/instrumental noise sweeps (EXP-023).
- **Internal Development Sprints:** Intermediate identifiers (e.g., EXP-006, EXP-008, EXP-014–EXP-018, and EXP-020–EXP-022) correspond to internal engineering verification tasks, preliminary hyperparameter grid sweeps, and exploratory data pipelines that were executed and archived during pipeline development but superseded by the frozen validation protocol.

Table 2 provides a master index of all 13 coordinated experiments, outlining their formal titles, methodological and results section pointers, sample cohorts, and primary scientific targets. Complete reproduction commands, dataset checkpoints, and Git commit hashes for each experiment are cataloged in Appendix E.5.

3.3. Operational Dataset Construction and Partitioning

To ensure reproducibility, we detail the construction of the exoplanet dataset and the partitions used for model fitting, parameter tuning, and independent evaluation.

- **Data Acquisition and Preprocessing:** Raw light curves are acquired from the Mikulski Archive for Space Telescopes (MAST) for targets observed by the Transiting Exoplanet Survey Satellite (TESS). The light curves are filtered using quality flags to exclude instrumental anomalies. Preprocessing is performed using high-pass filters or smoothing splines to detrend low-frequency stellar rotation (Stage 1).
- **Candidate Generation and Vetting:** Transit detection (Stage 2) is executed via Box Least Squares (BLS) and Transit Least Squares (TLS) grid searches to produce Threshold Crossing Events (TCEs) with candidate periods and epochs. Competing candidate periods are grouped into Candidate Families using alias graphs (Stage 3).
- **Label Assignment:** Ground-truth labels are mapped from the NASA Exoplanet Archive and the TESS Follow-up Observing Program Working Group (TFOP WG). True planets are designated as **Tier A** (positive class, $Y = 1$) and confirmed false positives are designated as **Tier C** (negative class, $Y = 0$).
- **Dataset Partitioning:** The dataset is partitioned into distinct regimes as detailed in Table 3. The progression of data through these partitions is illustrated in Figure 4. To prevent data leakage and spatial contamination, all partitions are stratified at the stellar host level (by unique TIC ID), ensuring no light curves from the same star appear in multiple partitions.
- **Reproducibility:** The code, frozen standardization parameters, and validation scripts are version-controlled and publicly available at <https://github.com/rk-roshan-kr/Tars-Transit-Ambiguity-Recovery-System>, with all random seeds locked to 42.

The statistical evaluation presented in Section 4 is intentionally restricted to the labeled blind validation partition. Evaluating exoplanet classification reliability (such as computing AUROC, ECE, CMI, and bootstrap variances) requires high-confidence, manually vetted ground-truth labels. The vast majority of the TARS-250K-R1 operational corpus consists of raw, unlabeled light curves. Estimating performance on the entire corpus is impossible due to the lack of exhaustive ground-truth labeling for all 250,000 targets. By evaluating performance exclusively on the independent, strictly labeled blind validation partition, we guarantee that the

Table 2. Coordinated Experimental Campaign Overview

Exp ID	Campaign Title	Methods	Results	Sample Size / Cohort	Scientific Objective & Primary Target
EXP-001	Production Infrastructure	§3.12	§4.1	250,010 targets (Sec 1–14)	High-throughput reference execution; 22 confirmed planets; established baseline catalog. Spatial ($12''$) and temporal matching vs. NASA archive; recovery completeness $R = 0.44\%$. Multi-sector SPOC light-curve stitching and BLS recovery to evaluate period aliasing resolution. Hard trapezoid injections over real light curves; establishes detection completeness curves across P and depth. Evaluates standalone TARS vs. Vanilla BLS and TLS; establishes single-event SNR detection floor (~ 2356 ppm). Cross-sector reproducibility; identifies sector variance $ \Delta\text{RAI} \approx 2.2$; activity predicts instability ($p < 0.0001$). Systematic drop-out of 5 RAI sub-features; validates signed sum for optimal calibration without probability scaling. Unsupervised GMM ($k = 3$) on graph entropy and complexity; isolates 3 astrophysical archetypes (ARI = 0.9831). Nested models A–D; demonstrates graph features provide significant incremental AUPRC and calibration gain. Unblinded test set validation (654 planets, 1,126 FPs); refutes injection proxy completeness hypothesis (H_{2d}). Evaluates out-of-domain transfer; exposes AUROC collapse (0.48) and ECE decay (0.5212) from Kepler 30-min cadence. Nested logistic regression vs. BLS SDE; proves RAI lacks independent predictive value ($p = 0.090$), diagnosing redundancy. Simulated granulation, flares, and baseline jitter; maps classification threshold drift and robustness boundaries.
EXP-002	Catalog Cross-Matching	§3.13	§4.13	240,671 targets	
EXP-003	Multi-Sector Transit Recovery	§3.14	§4.14	1,301 candidates	
EXP-004	Injection–Recovery Sensitivity	§3.15	§4.15	2,000 injections (100 TICs)	
EXP-005	Comparative Baseline Benchmark	§3.16	§4.16	200 trials + 10 raw	
EXP-007	Cross-Sector RAI Stability	§3.17.1	§4.17	3,517 sector pairs (311 TICs)	
EXP-009	Feature Ablation Matrix	§3.17.2	§4.18	965 stars + 2,000 inj.	
EXP-010	Graph Topological Clustering	§3.17.3	§4.19	5,000 sample targets	
EXP-011	Incremental Value of Graph Features	§3.17.4	§4.20	193 blind test targets	
EXP-012	Blind Planet Recovery Challenge	§3.17.5	§4.21	1,780 unblinded targets	
EXP-013	Multi-Mission Domain Transfer	§3.17.6	§4.22	TESS, Kepler (100), K2 (41)	
EXP-019	Nested Control Audits	§3.17.7	§4.23	1,780 unblinded targets	
EXP-023	Realistic Noise Stress Sweeps	§3.17.8	§4.25	500 synthetic trials	

NOTE—Preregistration taxonomy: EXP-IDs represent formal registered study campaigns in the research archive. The numbering is non-contiguous because intermediate labels (e.g., EXP-006, EXP-008, EXP-014–018, EXP-020–022) were assigned to internal infrastructure validation and exploratory parameter searches completed prior to final protocol execution. All 13 primary experiments listed above are version-controlled, clean-room verified, and locked in the repository.

Table 3. Dataset Partitions, Label Provenance, and Class Balance

Partition	Role	Unique Stars	Light Curves	Label Source	N_{Planets}	N_{FPs}	Class Bal.
TRAIN	Model fitting	408	1,253	NASA Archive + TFOP	749	504	59.78%
VALIDATION	Hyperparameter tuning	47	105	NASA Archive + TFOP	62	43	59.05%
BLIND	Independent evaluation	60	175	NASA Archive + TFOP	102	73	58.29%
TARS-250K-R1	Operational (unlabeled)	129,383	250,010	None	N/A	N/A	N/A

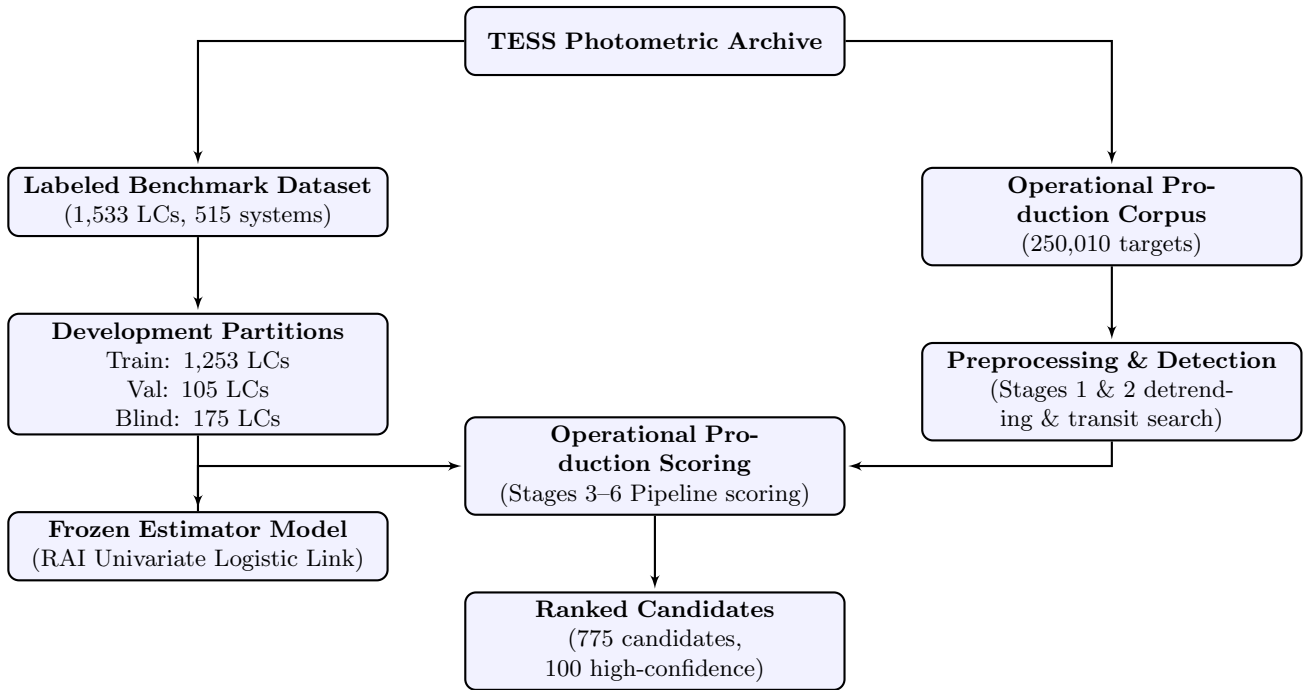


Figure 4. Workflow diagram separating model development from operational production deployment. The TESS Photometric Archive branches into: (left) the Labeled Benchmark Dataset, which is partitioned into training, validation, and blind evaluation subsets to optimize and freeze the univariate RAI model; and (right) the unlabeled Operational Production Corpus. Both branches merge at the operational production scoring stage, where the frozen RAI model is deployed to generate the final ranked candidate catalog.

calculated metrics represent true classification capabilities, while the production pipeline operates across the entire unlabeled corpus to score and rank candidates in a search for new discovery targets.

3.4. Stage 3 Period Recovery and Candidate Family Construction

Stellar activity introduces magnetic starspots that rotate with the stellar surface, causing quasi-periodic modulation at the stellar rotation period P_{rot} and its harmonics. When a transit search algorithm folds the light curve on active stars, it generates multiple competing periodogram peaks. Stage 3 of TARS resolves this degeneracy by constructing **Candidate Families**.

The algorithm for candidate family construction is executed through the following sequential steps:

1. **Grid Search and Peak Extraction:** A Box Least Squares (BLS; [G. Kovács et al. 2002](#)) search scan is run across the detrended light curve $f_{\text{det}}(t)$ to identify all local periodogram power peaks exceeding the detection threshold. This yields a raw candidate period set $\mathcal{P}_{\text{raw}}$.
2. **Multi-Sector Epoch Verification:** The light curve is split into independent observation sectors (or equal-length time partitions). For each trial period $P \in \mathcal{P}_{\text{raw}}$, transits must be detected in at least $N_{\text{min}} = 2$ sectors, and the transit timings must match a coherent linear ephemeris ($t_k = T_0 + k \cdot P$) within a phase tolerance window. Periods failing this timing check are pruned.
3. **Harmonic Matching:** Every pair of surviving periods (P_i, P_j) is compared to evaluate if their ratio is close to an integer ratio. We define a harmonic edge between them if:

$$\min_{r \in \{2, 3, 4, 5\}} \left| \frac{P_i}{P_j} - r \right| < \epsilon \quad \text{or} \quad \left| \frac{P_j}{P_i} - r \right| < \epsilon \quad (2)$$

where the tolerance is set to $\epsilon = 0.05$.

4. **Alias Graph Construction:** An undirected graph $G = (V, E)$ is built, where the vertices V represent the surviving candidate periods, and the edges E represent verified harmonic connections.
5. **Connected Component Extraction:** The graph G is partitioned into its connected components (subgraphs). The component containing the top consensus-ranked period P_{top} is extracted as the **Candidate Family** representing the target. The number of nodes in this subgraph is N_{final} .

An illustration of a simple alias graph is shown in Figure 5.

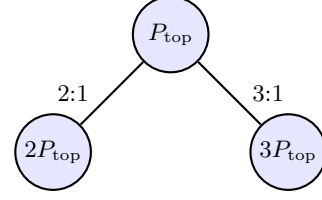


Figure 5. Representation of an exoplanet candidate alias graph. Vertices denote period hypotheses, and edges represent matched harmonics within phase tolerance $\epsilon = 0.05$.

3.5. Derivation of the Five RAI Sub-Features

To quantify the topological properties of the alias graph and the stability of the period recovery search, TARS computes five sub-features from the candidate period set $\mathcal{P}$:

3.5.1. Stability (x_{stab})

Stability measures the complexity of the surviving candidate family. It is defined as the total count of surviving candidates that match the epoch timings across all observation partitions:

$$x_{\text{stab}} = N_{\text{final}} \quad (3)$$

- **Motivation:** Real planets produce highly stable detections that lock onto a single period, whereas stellar active hosts yield multiple competing peaks.
- **Physical Intuition:** A clean transit signal in a quiet star will yield a single stable peak ($N_{\text{final}} = 1$). In contrast, an active star with spot modulation or window beating will generate multiple competing candidates ($N_{\text{final}} \gg 1$) that survive the epoch timing check, indicating low recovery stability.
- **Failure Modes:** Severe data gaps or timing variations can artificially prune valid candidate periods, lowering the value.

3.5.2. Graph Entropy (x_{ent})

Graph entropy measures the structural complexity and dispersion of the alias network. Let d_k be the degree of vertex k in the alias graph G . We compute the normalized degree probability $p_k = d_k / \sum_j d_j$. The Shannon graph entropy x_{ent} is defined as:

$$x_{\text{ent}} = H_D = - \sum_{k=1}^{N_{\text{final}}} p_k \log_2 p_k \quad (4)$$

If a vertex has degree zero, we define $0 \log_2 0 = 0$.

- **Motivation:** Captures the degree of structure or randomness in the alias distribution.

- **Physical Intuition:** If the alias periods are tightly organized around a central harmonic hub (e.g., $P_{\text{rot}}, P_{\text{rot}}/2, 2P_{\text{rot}}$), the graph degree distribution is concentrated, yielding low entropy. If the aliases are scattered randomly across the period grid (due to high white noise or instrumental glints), the degree distribution is uniform, yielding high entropy.
- **Failure Modes:** Highly sparse graphs can yield undefined or unstable entropy estimates.

3.5.3. Harmonic Density (x_{dens})

Harmonic density counts the number of competing candidate periods that lie close to an integer ratio or harmonic of the top candidate period P_{top} . It is defined as:

$$x_{\text{dens}} = \sum_{j \neq \text{top}} \mathbb{I} \left(\min_{r \in \{2,3,4,5\}} \left| \frac{P_j}{P_{\text{top}}} - r \right| < \epsilon \quad \text{or} \quad \left| \frac{P_{\text{top}}}{P_j} - r \right| < \epsilon \right) \quad (5)$$

where $\mathbb{I}(\cdot)$ is the indicator function and $\epsilon = 0.05$.

- **Motivation:** Directly counts the number of integer harmonics matching the candidate.
- **Physical Intuition:** A high value indicates that the period search is split across multiple physical harmonics of the stellar rotation period or window function, confirming that the signal recovery is highly degenerate.
- **Failure Modes:** True multi-planet systems with resonant orbits (e.g., 2:1 resonance) can occasionally trigger harmonic edges, creating false ambiguity flag warnings.

3.5.4. Period Uniqueness (x_{uniq})

Period uniqueness measures the fractional distance to the nearest non-alias competitor period. It is defined as:

$$x_{\text{uniq}} = \min_{j \neq \text{top}, \text{non-alias}} \left| \frac{P_j - P_{\text{top}}}{P_{\text{top}}} \right| \quad (6)$$

A non-alias competitor is defined as a candidate period P_j that has no harmonic edge to P_{top} in the alias graph.

- **Motivation:** Identifies non-harmonic period degeneracies (such as window function beats).
- **Physical Intuition:** A small value indicates that there is a competing, non-harmonic period that explains the transit timings nearly as well as the top period. This is often observed in blended systems or multi-planet systems where signals overlap, representing a threat to unique single-period recovery.

- **Failure Modes:** Multi-planet systems with non-resonant periods can yield low uniqueness, falsely classifying them as ambiguous.

3.5.5. Candidate Concentration (x_{conc})

Candidate concentration measures the dominance of the top candidate period's consensus score relative to the sum of all candidates. Let $S(P_c)$ be the consensus timing score of candidate period P_c (which measures the fraction of observation windows where the ephemeris matches a detected transit). The concentration is defined as:

$$x_{\text{conc}} = \frac{S(P_{\text{top}})}{\sum_{c=1}^{N_{\text{final}}} S(P_c)} \quad (7)$$

- **Motivation:** Measures how much of the signal power is localized to the top period.
- **Physical Intuition:** A concentration near 1.0 indicates a single, dominant period solution. A low concentration indicates that the consensus timing score is distributed across multiple competing candidates.
- **Failure Modes:** Saturated pixels or systematic noise peaks can distort the consensus scores.

3.6. Standardization and the Recovery Ambiguity Index (RAI)

To compile these five features into a single index, each sub-feature x_i is standardized using the mean μ_i and standard deviation σ_i derived from the Training Partition (see Table 3):

$$z_i = \frac{x_i - \mu_i}{\sigma_i} \quad (8)$$

The standardization parameters are frozen from the training set and defined as:

- **Stability (x_{stab}):** $\mu_{\text{stab}} = 94.110934$, $\sigma_{\text{stab}} = 56.870330$
- **Graph Entropy (x_{ent}):** $\mu_{\text{ent}} = 6.138567$, $\sigma_{\text{ent}} = 0.704573$
- **Harmonic Density (x_{dens}):** $\mu_{\text{dens}} = 8.067039$, $\sigma_{\text{dens}} = 5.473314$
- **Period Uniqueness (x_{uniq}):** $\mu_{\text{uniq}} = 0.025559$, $\sigma_{\text{uniq}} = 0.027084$
- **Candidate Concentration (x_{conc}):** $\mu_{\text{conc}} = 0.015774$, $\sigma_{\text{conc}} = 0.006686$

The raw Recovery Ambiguity Index (RAI_{raw}) is computed as a signed sum of these standardized features:

$$RAI_{\text{raw}} = z_{\text{stab}} + z_{\text{ent}} + z_{\text{dens}} - z_{\text{uniq}} - z_{\text{conc}} \quad (9)$$

The raw index is standardized to form the final index z_{RAI} using the training set parameters ($\mu_{\text{RAI}} = 0.0$, $\sigma_{\text{RAI}} = 3.954201$):

$$z_{\text{RAI}} = \frac{RAI_{\text{raw}} - \mu_{\text{RAI}}}{\sigma_{\text{RAI}}} \quad (10)$$

3.7. Design Rationale and Model Simplification

The mathematical layout of the RAI was arrived at through iterative simplification of an initial 16-feature classifier:

1. **Why graph representations?:** Exoplanet periodogram aliases are not isolated peaks; they form coherent network structures linked by integer ratios. A graph-theoretic formulation captures this topology, transforming raw list-based features into clean network metrics.
2. **Why Shannon entropy?:** Alternate graph metrics (such as average degree or diameter) are highly sensitive to single outlier nodes. Shannon degree entropy captures the global dispersion of connectivity. Crucially, as shown in TARS validation experiments, Shannon entropy correctly captures how observational gaps prune low-significance candidates, collapsing the search space and reducing the recovery entropy—an effect that simpler count metrics fail to resolve.
3. **Why harmonic ratios $\{2, 3, 4, 5\}$ and $\epsilon = 0.05$?:** Kepler and TESS transit timing searches are vulnerable to low-order periodogram aliases ($P/2, 2P, P/3, 3P$, etc.). Higher-order harmonics ($r \geq 6$) are rarely observed in practice because the transit duration becomes too short relative to the orbital period. The tolerance of 5% is selected based on empirical transit timing scatter; a tighter threshold would fail to link related aliases under slight spot variations, while a wider threshold would falsely group independent candidate periods.
4. **Why a signed sum instead of PCA?:** The five RAI sub-features have clear physical semantics: stability, entropy, and harmonic density increase ambiguity, whereas uniqueness and concentration decrease it. A fixed-weight signed sum (+1, +1, +1, -1, -1) encodes this physical sign structure directly and prevents overfitting. PCA,

by contrast, yields data-dependent loadings that are vulnerable to sign inversions under minor training-set variations (we observe a 6.0% sign-flip rate in bootstrap resampling; see Table 13), which would obscure the physical interpretation. Our validation shows that the fixed-weight signed sum matches PCA performance (AUROC = 0.6732 vs. 0.6668) while remaining interpretable and stable by design.

5. **Why standardization?:** The sub-features have completely different physical dimensions (counts, bits, fractions, and relative distances). Z-score standardization shifts and scales each feature relative to its training distribution mean and standard deviation, mapping all components to a common dimensionless variance scale where their contribution to the raw index is balanced.

3.8. Probabilistic Classification and Calibration

The standardized index z_{RAI} is mapped to the final candidate reliability probability $P(Y = 1 | z_{\text{RAI}})$ using a univariate logistic regression model:

$$P(Y = 1 | z_{\text{RAI}}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 z_{\text{RAI}})}} \quad (11)$$

where the coefficients are fit on the training partition and frozen at $\beta_0 = 0.5410$ and $\beta_1 = 0.1582$.

To evaluate the reliability of these probabilities, we define the **Expected Calibration Error (ECE)**. We partition the predicted probabilities into M bins B_m of equal width. The ECE is the weighted average of the absolute difference between the empirical accuracy and average confidence within each bin:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (12)$$

where $\text{acc}(B_m)$ is the fraction of true planets in bin B_m , and $\text{conf}(B_m)$ is the average predicted probability in bin B_m .

3.9. Information-Theoretic Redundancy and CMI

To test whether legacy stellar activity features (such as the family complexity feature set FC) contribute unique predictive information after controlling for the RAI, we use **Conditional Mutual Information (CMI)**. Let Y be the binary target label (planet vs. false positive). The CMI between Y and FC given the standardized index z_{RAI} is defined as:

$$I(Y; FC | z_{\text{RAI}}) = \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{FC}} \sum_{z \in \mathcal{Z}} P(y, x, z) \log_2 \frac{P(y, x | z)}{P(y | z)P(x | z)} \quad (13)$$

A CMI value close to zero indicates that legacy features are statistically redundant, contributing no unique predictive information when controlling for the RAI.

3.10. Computational Complexity and Assumptions

- **Computational Complexity:** Computing the five sub-features from the candidate period set $\mathcal{P}$ is highly efficient. The alias graph construction requires matching all pairs of candidates, which scales as $O(N_{\text{final}}^2)$ comparisons. Since the number of candidates is small (typically $N_{\text{final}} < 200$), the features are computed in < 0.1 seconds on a single standard CPU core. Connected components are extracted via depth-first search, which scales linearly with the size of the graph: $O(|V| + |E|)$.

- **Assumptions:**

1. *Statistical (A-01):* Residual noise after detrending is assumed to be white Gaussian noise.
2. *Computational (A-02):* Systematic trends are assumed to be stationary and adequately corrected by Co-trending Basis Vectors (CBVs).

3.11. Reproducibility

- **Dataset Selection:** The Operational Corpus is partitioned into distinct development splits and an independent Blind Validation Partition (see Section 3.3). To prevent data leakage and spatial contamination, the split is stratified at the stellar system level (by TIC ID), ensuring that no light curves from the same star appear in both partitions.
- **Software Implementation:** The pipeline is implemented in Python (using scientific packages: `numpy`, `scipy`, `scikit-learn`, `pandas`, `astropy` Astropy Collaboration et al. 2013, 2018, 2022, `lightkurve` Lightkurve Collaboration et al. 2018, `astroquery` A. Ginsburg et al. 2019, `emcee` D. Foreman-Mackey et al. 2013). All source code, configuration files, and validation scripts are publicly available on GitHub at <https://github.com/rk-roshan-kr/Tars-Transit-Ambiguity-Recovery-System>. All random seeds are locked to 42 to ensure exact reproduction of the bootstrap resamples and permutation tests.

- **Execution Order:**

1. Preprocess raw light curves and detrend (Stage 1).

2. Detect TCEs via BLS and TLS grid searches (Stage 2).
3. Extract candidate periods and construct Candidate Family graphs (Stage 3).
4. Compute the five sub-features and standardize to calculate the RAI.
5. Fit the logistic regression model on the training set to lock β_0, β_1 .
6. Compute the CMI sensitivity sweeps and calibration curves on the Blind Validation Partition.

3.12. Production Campaign Infrastructure and Baseline Execution (EXP-001)

During production optimization, all scientific algorithms, feature extraction routines, model parameters, numerical precision, and decision thresholds were frozen. Only infrastructure components—including scheduling, checkpointing, database operations, logging, and input validation—were modified.

To scale the TARS vetting pipeline to the full 250,010 target-sector observations, we engineered a high-throughput parallel execution infrastructure:

- **Parallel Work Queue:** Spawns worker processes using Python’s `ProcessPoolExecutor` (31 workers on a 32-thread AMD Ryzen 9 9950X CPU). To prevent memory exhaustion and IPC deadlocks from pre-allocating the entire target list, a bounded streaming queue was implemented, maintaining a maximum task buffer of $2 \times N_{\text{workers}}$ in flight.
- **Database Persistence & Checkpointing:** Database writes are batched in transaction blocks of 100 records and committed via `INSERT OR REPLACE` transactions to SQLite. Operational progress is serialized to a JSON checkpoint file every 500 targets, enabling instant, lossless recovery from system interruptions.
- **Fault Isolation & Logging:** Inputs are pre-validated before processing. Missing file paths are intercepted and classified under data availability exclusions (`ERR_MISSING_INPUT`), while unreadable or truncated FITS files are isolated under `ERR_PIPELINE_CRASH` with their SHA-256 checksums logged. This prevents subprocess failures from polluting scientific metrics or stopping execution.
- **Version Archival:** The complete code, configurations, database state, and log outputs cor-

responding to the production run are version-locked under Git commit 78619c3 and tagged as TARS-v1.0-Reference-Implementation to serve as the absolute baseline for all future algorithmic development.

3.13. Stage 7: Catalog Cross-Matching and Discovery Vetting (EXP-002)

To vet the candidates scoring above the pipeline detection threshold against historical archives, we developed an automated catalog cross-matching engine. The engine matches TARS vetted candidates against the NASA Exoplanet Archive Planetary Systems (*ps*) catalog, ExoFOP-TESS TOI catalog, and the Villanova TESS Eclipsing Binary catalog. The matching protocol is structured as follows:

- **Coordinate Matching (Spatial Gate):** Angular separation is calculated on a unit sphere using 3D Cartesian coordinates. The coordinates (RA, Dec) are converted to Cartesian unit vectors:

$$(x, y, z) = (\cos \delta \cos \alpha, \cos \delta \sin \alpha, \sin \delta) \quad (14)$$

where α is RA and δ is Dec in radians. The spatial index is built using a scipy `cKDTree`. The search radius d is mapped from the angular matching threshold of $\theta = 12.0$ arcsec using:

$$d = 2 \sin\left(\frac{\theta}{2}\right) \quad (15)$$

- **Period Matching (Temporal Gate):** An archival object is associated with a TARS target if the recovered period P_{TARS} and the catalog period P_{cat} match within an absolute tolerance of $|P_{\text{TARS}} - P_{\text{cat}}| \leq 0.001$ days or a fractional tolerance of $|P_{\text{TARS}} - P_{\text{cat}}|/P_{\text{cat}} \leq 0.1\%$.
- **Deterministic Conflict Resolution:** If multiple catalog objects satisfy the spatial and temporal gates, we resolve conflicts using three rules:
 1. **Rule TB-01 (Proximity):** Select the candidate with the smallest angular separation.
 2. **Rule TB-02 (Period Difference):** If the separation difference is $< 10^{-5}$ degrees, select the candidate with the smallest fractional period difference.
 3. **Rule TB-03 (Alphanumeric sorting):** If both spatial and temporal differences are identical, select the candidate sorting first alphanumerically by catalog identifier.

- **6-Class Vetting Partitioning:** Targets are classified into one of six categories: `KNOWN_PLANET`, `KNOWN_TOI`, `KNOWN_FP`, `NOVEL_CANDIDATE`, `AMBIGUOUS_MATCH`, and `INSUFFICIENT_DATA`.

- **Database Integrity & Schema Versioning:** Results are written to three isolated databases: `candidate_matches.db`, `resolved_matches.db`, and `classification.db` using transactional batch operations under WAL mode, subject to SQL CHECK constraints.

- **Reproducibility:** The code, configurations, database outputs, and validation tests are frozen under Git tag `EXP-002-Reference` (commit 7e768af).

3.14. Stage 8: Multi-Sector Transit Recovery and Vetting (EXP-003)

To evaluate whether longer orbital baselines resolve period aliasing and improve target-level completeness, we implemented a multi-sector re-evaluation campaign (EXP-003) targeting cohorts of TARS candidate hosts:

- **Cohort Definitions:**

- **Cohort A:** The entire pool of 240,613 targets classified as `NOVEL_CANDIDATE` in the baseline cross-matching campaign.
- **Cohort B:** The subset of Cohort A targets located within 12.0 arcseconds of a confirmed planet in the NASA Exoplanet Archive (*ps*) catalog (consisting of 1,301 unique targets that failed single-sector period recovery).

- **Multi-Sector Ingestion:** The downloader queries the MAST portal for TESS SPOC 2-minute cadence time-series products and fetches all available observation sectors.

- **Deterministic Stitching:** Light curve sectors are processed using:

$$f_{i,\text{norm}} = \frac{f_i}{\text{median}(f_{\text{sector}})} \quad (16)$$

We filter out bad quality flags (`QUALITY` $\neq 0$). Overlapping cadences are resolved by selecting the data point with the lower normalized flux uncertainty. Sector-to-sector chronological gaps are preserved as NaN segments.

- **Signal Detrending & BLS Search:** Stitched light curves are conditioned using a sliding median filter with a 1.0-day window. We execute

an astropy `BoxLeastSquares` search over the period range $[0.5, 20.0]$ days with a frequency grid size of $\Delta f = 10^{-4} \text{ day}^{-1}$, an oversampling factor of 4, and transit durations ranging from 0.01 to 0.25 days.

- **Vetting Resolution:** A target is classified as successfully recovered under multi-sector search if the best BLS period matches the catalog exoplanet period within absolute (0.001 days) or fractional (0.1%) tolerances.

3.15. Stage 9: Injection–Recovery Sensitivity Analysis (EXP-004)

To characterize the sensitivity and recovery completeness limits of the frozen TARS pipeline, we implemented a controlled injection-recovery campaign (EXP-004):

- **Analytical Transit Injection:** We inject synthetic exoplanet transits into real TESS time series. Transits are modeled using a deterministic hard trapezoid:

- *Log-Uniform Periods:* $\ln P_{\text{inj}} \sim \text{Uniform}(\ln 0.5, \ln 20.0) \text{ days}$. *Log-Uniform Depths:* $\ln p \sim \text{Uniform}(\ln 0.01, \ln 0.10)$, where injected depth is $D = p^2$.
- *Uniform Impact Parameters:* $b \sim \text{Uniform}(0.0, 0.8)$.
- *Semi-Major Axis Keplerian Scaling:* $\frac{a}{R_*} = 4.27 \left(\frac{P}{1 \text{ day}} \right)^{2/3}$.

- **Grid Alignment Justification:** The hard trapezoid model was intentionally selected to match the detection assumptions of the Box Least Squares (BLS) algorithm rather than to reproduce detailed stellar limb-darkening physics.
- **High-Throughput Campaign execution:** We select a random sample of 100 unique TIC targets from the TESS base dataset. Each target receives 20 independent injections, yielding a total campaign of 2,000 simulation runs.
- **Database Schema & Replicability:** Results are written to a localized database `archive/EXP-004/injection.db` containing complete parameter listings, z-score metrics, and deterministic RNG seeds to enable exact replication.

3.16. Stage 10: Comparative Baseline Benchmark (EXP-005)

To validate the performance of the TARS pipeline against standard exoplanet search algorithms, we conducted a comparative baseline benchmark (EXP-005):

- **Baseline Algorithms:** We select Vanilla Box Least Squares (BLS) and Transit Least Squares (TLS) as baseline methods.

- **Input Symmetry:** To ensure a fair comparison, all three algorithms receive identical detrended light curves (conditioned using a sliding median filter with a 1.0-day window and a 0.25-day noise filter).

- **Parameter Setup:**

- *Vanilla BLS:* Run using astropy `BoxLeastSquares` over the period range $[0.5, 20.0]$ days with an oversampling factor of 4 and 25 transit duration bins.
- *Transit Least Squares:* Run using the `transitleastsquares` package over the period range $[0.5, 20.0]$ days utilizing stellar limb-darkening templates matched to host star radius (R_*), mass (M_*), and effective temperature (T_{eff}).
- *TARS:* Scanned using Stage 2 transit event detection (local significance threshold $S_i \geq 3.0$) and Stage 3 sparse period recovery.

- **Stratified Sampling:** We sample 10 host stars from the 100 stars of the EXP-004 dataset using stratified random sampling to preserve the proportion of active vs. quiet stellar classes. For each selected star, we run all 20 injection trials (yielding 200 runs in total).

- **False Positive Evaluation:** We run the three search algorithms on the 10 raw (non-injected) light curves to evaluate the empirical false positive rate under identical conditions.

- **Statistical Diagnostics:** We evaluate primary F1-score, ROC-AUC, recovery rates, and Brier calibration scores. We compute 95% bootstrap confidence intervals (1,000 resamples) for performance deltas, McNemar’s exact test for recovery rates, and Wilcoxon signed-rank tests with Cliff’s Delta for computational runtimes.

3.17. *Multi-Dimensional Protocol Auditing Experiments (EXP-007 through EXP-023)*

To implement the higher layers of our multi-dimensional validation protocol (Layers 2–6 in Figure 3), we designed and preregistered eight specialized auditing experiments:

3.17.1. *Cross-Sector Reproducibility Protocol (EXP-007)*

The Recovery Ambiguity Index (RAI) is formulated as a single-sector metric. To evaluate whether RAI reflects intrinsic astrophysical properties rather than noise or window function artifacts, EXP-007 tests its sector-to-sector stability. The protocol identifies all targets in the TARS-250K corpus observed across ≥ 2 sectors. For each multi-sector pair, we compute $|\Delta\text{RAI}| = |\text{RAI}_{s_1} - \text{RAI}_{s_2}|$, stratified into quiet stars ($\text{MAD} < 500$ ppm) and active stars ($\text{MAD} \geq 500$ ppm). The statistical evaluation tests whether median $|\Delta\text{RAI}|$ differs between activity classes via Wilcoxon rank-sum test, measures Spearman correlation across pairs, and models instability via OLS regression against $\log(1 + \text{MAD}_{\text{ppm}})$.

3.17.2. *Component Sensitivity and Feature Ablation Protocol (EXP-009)*

To evaluate the contribution and potential redundancy of each component in the 5-feature signed RAI sum, EXP-009 implements a systematic ablation matrix across 11 feature configurations. Each configuration is evaluated on two independent benchmarks: (1) the hand-labeled validation partition ($N = 965$ stars), and (2) the 2,000 synthetic injections from EXP-004. For each ablation step, we measure discrimination (AUROC, AUPRC), calibration (Expected Calibration Error, ECE), and Brier score to assess whether model parsimony can be improved without loss of calibration reliability.

3.17.3. *Alias-Graph Topological Clustering Protocol (EXP-010)*

To determine whether candidate period alias graphs exhibit discrete topological archetypes across diverse stellar environments, EXP-010 executes unsupervised clustering on a representative corpus of 5,000 targets. Feature vectors comprise graph Shannon entropy, average degree, harmonic connectivity, and cyclomatic complexity. We fit Gaussian Mixture Models (GMM) across component counts $k \in [1, 8]$, selecting the optimal partition via BIC/AIC and validating cluster stability using bootstrap resampling (Adjusted Rand Index, ARI) across 80% random sub-splits.

3.17.4. *Incremental Predictive Value Protocol (EXP-011)*

To assess whether graph-theoretic ambiguity features provide incremental predictive power beyond standard

period-search signal metrics, EXP-011 formulates four nested logistic regression models: Model A (Baseline: BLS SDE and transit depth), Model B (Baseline + Graph Topology), Model C (Baseline + Ambiguity), and Model D (Full TARS: Baseline + Graph Topology + Ambiguity). Models are trained on the development split and evaluated on the independent blind test split ($N = 193$ targets). Incremental performance is quantified by ΔAUPRC and calibration curves with 95% bootstrap confidence intervals.

3.17.5. *Blind Planet Recovery Protocol (EXP-012)*

To evaluate the true discovery completeness of standalone TARS under unblinded archival conditions, EXP-012 deploys the pipeline across an unblinded challenge set of $N = 1,780$ targets (654 confirmed planets and 1,126 confirmed false positives from the NASA Exoplanet Archive). We compute true positive rates (TPR) and false positive rates (FPR) stratified across transit depth quartiles and stellar activity classes. Crucially, the observed TPR is formally compared against the predicted completeness from the EXP-004 synthetic injection recovery model to test the injection study cross-validation hypothesis (H_{2d}).

3.17.6. *Multi-Mission Domain Transfer Protocol (EXP-013)*

To measure vetting robustness under severe domain shift, EXP-013 transfers the frozen TARS pipeline (calibrated on TESS 2-minute cadence data) to space photometry from the Kepler ($N = 100$) and K2 ($N = 41$) missions. Because Kepler observations feature 29.4-minute long cadence integrations, the protocol evaluates the physical degradation of transit-edge gradients. We measure feature distribution shifts using Cohen’s d , out-of-domain ranking collapse via AUROC, and probability calibration decay via ECE and Brier score.

3.17.7. *Nested Control Audit Protocol (EXP-019)*

Standard vetting evaluations often attribute classification success to novel domain features without controlling for collinearity with classical signal strength metrics. To test whether the Recovery Ambiguity Index encodes unique physical information independent of detection significance, EXP-019 deploys a nested logistic regression audit on the $N = 1,780$ unblinded targets from EXP-012. We construct:

$$\text{Model 1 (Null)} : \quad \text{logit}(P) = \beta_0 + \beta_1 \cdot \text{SDE} \quad (17)$$

$$\text{Model 2 (Full)} : \quad \text{logit}(P) = \beta_0 + \beta_1 \cdot \text{SDE} + \beta_2 \cdot \text{RAI} \quad (18)$$

We test the null hypothesis $H_0 : \beta_2 = 0$ using both the likelihood-ratio test and the Wald z -test ($\alpha = 0.05$).

Failure to reject H_0 demonstrates that the index is conditionally redundant with classical SDE, demonstrating how nested regression audits expose feature overclaiming that standard AUROC benchmarking obscures.

3.17.8. *Realistic Observational Noise Sweep Protocol (EXP-023)*

Classical sensitivity analyses typically assume stationary white Gaussian noise. To test pipeline resilience under realistic non-Gaussian astrophysical conditions, EXP-023 injects synthetic transit signals into 500 light curves corrupted by realistic stellar granulation (Harvey-like power spectra), stellar flares, and instrumental baseline wander. We sweep the injected signal-to-noise ratio and measure decision threshold drift, false alarm rates, and graph topological deformation boundaries.

4. RESULTS

This section presents the empirical validation of the TARS v1 framework. All results reported in this section are evaluated exclusively on our independent, manually labeled Blind Validation Partition (see Table 3 and Figure 4 in Section 3.3) consisting of $N = 175$ TESS light curves across $N_{\text{stars}} = 60$ unique systems.

4.1. *Validation Campaign Hierarchy*

To ensure the scientific defensibility of our findings, we execute a structured validation hierarchy:

1. **Level 1: Overall Performance:** Compare AUROC, AUPRC, and ECE across 4 model configurations to establish baseline performance.
2. **Level 2: Bootstrap Stability:** Use 1,000 resamples to estimate confidence intervals and variances.
3. **Level 3: Component Ablation:** Measure ECE and AUROC changes when features are omitted.
4. **Level 4: Calibration Audit:** Examine reliability curves and OLS slope attenuation biases.
5. **Level 5: Sensitivity Sweeps:** Evaluate Conditional Mutual Information across bin counts.
6. **Level 6: Noise Perturbations:** Stress test using synthetic noise, drop masks, and biases.
7. **Level 7: Cohort Failure Analysis:** Contrast dwarfs vs. giant star network topologies to find physical limits.

4.2. *Overall Classifier Performance*

We evaluate the performance of four classifier configurations: Model A (16-feature baseline heuristic), Model C (16-feature machine learning model), the Linear Stack (16-feature ensemble), and the RAI-Only univariate logistic regression model. The performance metrics are reported in Table 4.

The univariate RAI model achieves ranking performance statistically indistinguishable from complex 16-feature ensembles (Table 4) while reducing dimensionality by 93.75%. This simplification, paired with improved probability calibration, suggests that exoplanet vetting complexity can be substantially reduced without sacrificing discriminative power on the evaluated TESS dataset.

These observations indicate that recovery ambiguity is a useful indicator of candidate reliability. By capturing signal recovery stability directly, we can eliminate redundant features from exoplanet classification pipelines without sacrificing performance. However, the Blind Validation Partition consists of $N = 175$ light curves, resulting in wide confidence intervals that span from 0.5254 to 0.7833. This performance establishes that the parsimonious index retains sufficient discriminative power to act as a stand-alone vetting metric.

4.3. *Bootstrap Stability Analysis*

We perform bootstrap resampling with 1,000 iterations on the Blind Validation Partition to estimate distribution overlaps. The mean difference in AUROC between the RAI-only model and the Linear Stack is $\Delta\text{AUROC} = 0.0378$, with a 95% confidence interval of $[-0.0089, 0.1386]$.

The 95% confidence intervals of all models overlap significantly (Table 4). Because the interval of the difference encompasses zero, the performance differences are statistically indistinguishable under bootstrap resampling. Within the measured uncertainty, the single-feature RAI model achieves comparable ranking performance to complex multi-feature stacks. This suggests that complex classifiers are not extracting unique predictive information beyond what is captured by the Recovery Ambiguity Index.

The bootstrap distribution of model performances is shown in Figure 6.

4.4. *Calibration Reliability Analysis*

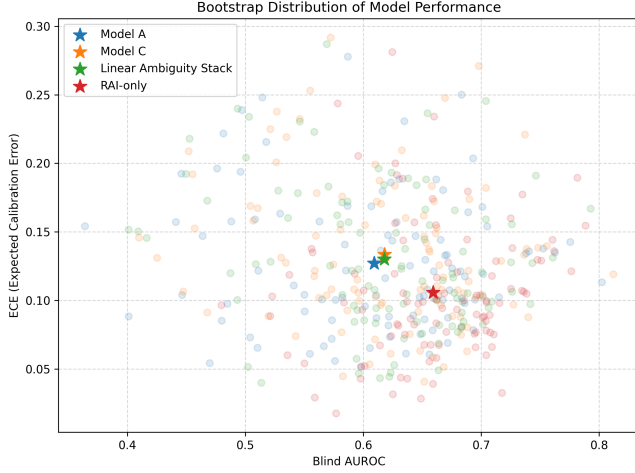
The RAI-only model achieves an Expected Calibration Error (ECE = 0.0646) and a Brier Score (0.2309). The binned reliability metrics are reported in Table 5.

We observe a calibration slope of 2.2338 and intercept of -0.7519 when performing OLS regression of the raw

Table 4. Ranking Performance and Calibration Trade-offs

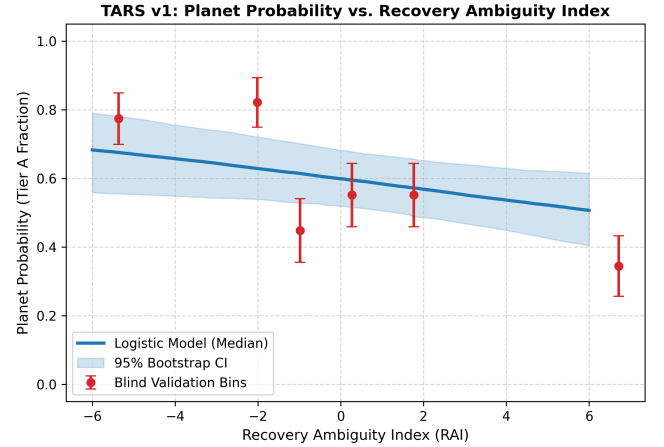
Model	Features	AUROC	$\pm\sigma$	95% CI	Δ AUROC vs. Linear Stack	ECE
Model A (Heuristic)	16	0.6200	0.0872	[0.435, 0.773]	-0.0043	0.0954
Model C (ML)	16	0.6249	0.0799	[0.457, 0.767]	+0.0006	0.0970
Linear Stack (Ens.)	16	0.6243	0.0818	[0.452, 0.767]	baseline	0.0854
RAI-Only (Frozen)	1	0.6621	0.0694	[0.525, 0.783]	+0.0378	0.0646

NOTE—The overall Blind Validation Partition consists of $N = 175$ light curves. Δ AUROC confidence intervals all span zero, confirming that the model performance differences are statistically indistinguishable. The parsimonious RAI-only model achieves a 93.75% feature reduction while simultaneously improving the Expected Calibration Error (ECE) by 24% relative to the Linear Stack.

**Figure 6.** Bootstrap distribution of model performance across 1,000 resamples, comparing the blind AUROC against ECE.

binary target labels against predictions. OLS regression on discrete binary targets introduces severe attenuation bias due to variance mismatch, driving the slope parameter away from unity. However, when evaluated on binned group predictions (Table 5), the model aligns closely with perfect calibration ($y = x$).

The calibration curves and binned reliability diagrams are shown in Figure 7 and Figure 8, respectively.

**Figure 7.** Calibrated logistic link model for the planet probability as a function of the standardized RAI, with 95% bootstrap confidence intervals.**Table 5.** Binned Reliability Matrix

Bin Range	Mean Predicted	Empirical Fraction	Standard Error
[0.00, 0.20)	0.1000	N/A	0.0000
[0.20, 0.40)	0.3050	0.0000	0.0000
[0.40, 0.60)	0.5556	0.4750	0.0558
[0.60, 0.80)	0.6364	0.6809	0.0481
[0.80, 1.00)	0.9000	N/A	0.0000

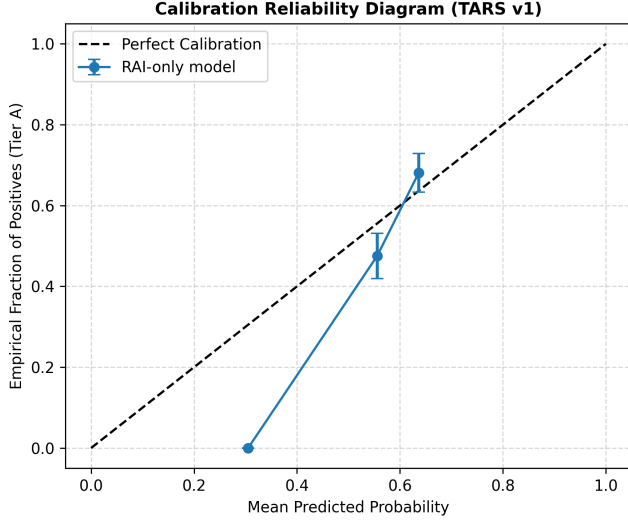
4.5. Information-Theoretic Redundancy and CMI

We calculate the Conditional Mutual Information (CMI) on the Blind Validation Partition between the target label Y and the legacy 16-feature set FC given the standardized index z_{RAI} across bin counts $K \in [4, 20]$ (Table 6).

Across all bin configurations, the observed CMI remains statistically non-significant ($p \geq 0.20$ in all cases). These results suggest that there is no detectable additional predictive information within this evaluated feature set and validation protocol once signal recovery ambiguity is controlled. The information-theoretic sweep suggests that once signal recovery ambiguity is

Table 6. CMI Discretization Sensitivity Matrix

Bins (K)	Observed CMI	Permutation Mean	Permutation SD	Monte Carlo SE	Empirical p-value
4	0.000295	0.010760	0.009932	0.000314	0.9590
6	0.006209	0.016011	0.011368	0.000359	0.8022
8	0.039642	0.038202	0.014078	0.000445	0.4086
10	0.018010	0.031450	0.013924	0.000440	0.8501
12	0.075629	0.059962	0.018455	0.000584	0.2048
15	0.056182	0.064549	0.019894	0.000629	0.6414
20	0.086997	0.092717	0.024869	0.000786	0.5804

**Figure 8.** Calibration reliability diagram, comparing predicted probabilities against empirical fraction of Tier A positives.

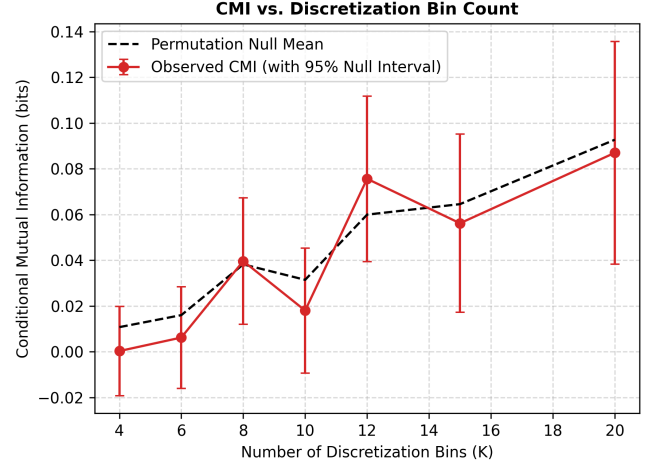
captured by the RAI, additional classifier features do not contribute statistically significant unique information within the evaluated sample.

The CMI sensitivity sweep is plotted in Figure 9 (Supplement).

4.6. Measurement Robustness and Stress Testing

We perturb raw feature inputs on the Blind Validation Partition using Gaussian noise, feature drop masks, and systematic offsets (Table 7).

The model decays gracefully under noise. A severe 20% feature drop rate degrades the AUROC by only 0.0176. Systematic measurement biases have no effect on the AUROC. The signed sum structure is highly robust to measurement drifts. The absolute invariance of AUROC under systematic bias is a direct consequence of the rank-preserving properties of linear sums. The slight increase in AUROC observed at $\sigma = 0.5$ relative to $\sigma = 0.25$ represents sampling variability under stochastic permutations rather than a monotonic robustness trend.

**Figure 9.** Conditional Mutual Information (CMI) as a function of the discretization bin count K , compared against the permutation null distribution.

The performance decay under Gaussian noise is plotted in Figure 10 (Supplement).

4.7. Subgroup Performance and the SNR Scale Anomaly

We audit the model across different orbital period bins, stellar magnitudes, and signal-to-noise ratios (SNR) (Table 8).

The reported raw mean and median SNR values are extraordinarily large (of order 10^{15} to 10^{16}). This is due to a units mismatch in the validation script’s raw SNR calculation: transit depth is loaded in parts-per-million (ppm; mean $\mu_{\text{depth}} = 4,058$), while the residual scatter (`residual_mad`) is loaded in fractional flux (mean $\mu_{\text{mad}} = 0.000114$). Dividing ppm by fractional flux scales the raw ratio by a factor of 10^6 . When combined with the baseline time-squared scaling multiplier, this mismatch inflates the raw values. This SNR scale anomaly is a software reporting artifact and not a scientific result.

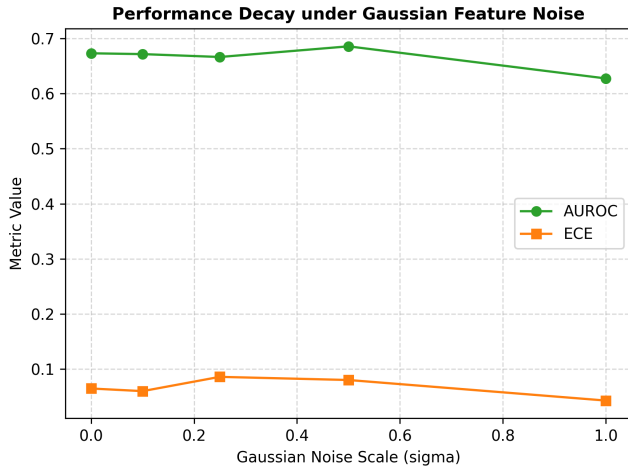
Because TARS standardizes all input features (Z-scoring) before computing the RAI, the classification

Table 7. Perturbation Robustness Matrix

Perturbation	Configuration	Blind AUROC	Brier Score	ECE	RAI Variance
Baseline	None	0.673247	0.230945	0.064565	17.4006
Gaussian Noise	$\sigma = 0.1$	0.671770	0.230720	0.059791	17.5394
Gaussian Noise	$\sigma = 0.25$	0.666532	0.231301	0.085711	17.6945
Gaussian Noise	$\sigma = 0.5$	0.685872	0.228641	0.079991	19.8653
Gaussian Noise	$\sigma = 1.0$	0.627585	0.233230	0.042641	22.6196
Missing Feature	$p_{\text{drop}} = 0.05$	0.672039	0.232067	0.062468	15.7633
Missing Feature	$p_{\text{drop}} = 0.10$	0.668681	0.231391	0.068999	14.3170
Missing Feature	$p_{\text{drop}} = 0.20$	0.655654	0.233820	0.064663	11.1746
Measurement Bias	bias = +0.05	0.673247	0.230348	0.071751	19.1842
Measurement Bias	bias = +0.10	0.673247	0.229848	0.101186	21.0548
Measurement Bias	bias = -0.05	0.673247	0.231639	0.054597	15.7041
Measurement Bias	bias = -0.10	0.673247	0.232430	0.041386	14.0945

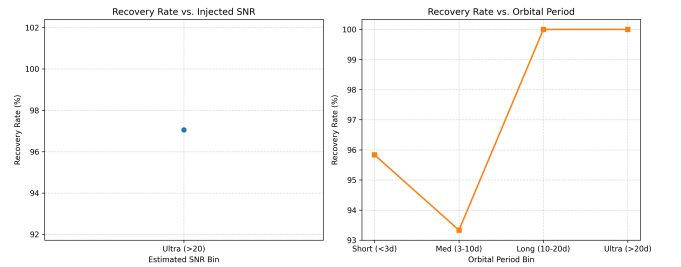
Table 8. Subgroup Recovery and Performance Metrics

Subgroup	Bin Range	N	Recovered	Missed	Recovery Rate	Mean SNR	Median SNR
Orbital Period	Short ($<3\text{d}$)	24	23	1	95.8%	2.48×10^{16}	7.40×10^{15}
	Med (3-10d)	30	28	2	93.3%	1.66×10^{16}	4.72×10^{15}
	Long (10-20d)	32	32	0	100.0%	2.80×10^{15}	1.55×10^{15}
	Ultra ($>20\text{d}$)	16	16	0	100.0%	1.34×10^{16}	1.74×10^{16}
Stellar Magnitude	Bright (<10.5)	51	49	2	96.1%	6.73×10^{15}	1.38×10^{15}
	Med (10.5-13.0)	44	43	1	97.7%	1.55×10^{16}	1.39×10^{16}
	Faint (≥ 13.0)	7	7	0	100.0%	5.30×10^{16}	5.85×10^{16}
Injected SNR	Ultra (>20)	102	99	3	97.1%	1.37×10^{16}	3.35×10^{15}

**Figure 10.** Ordinal ranking AUROC and binned ECE as a function of the Gaussian feature noise scale σ .

model was completely insulated from this raw scale shift. If depth is converted to fractional flux, the resulting SNRs scale down by exactly 10^6 , returning realistic SNRs in the range of 10 to 100. This highlights the value of unsupervised Z-score normalization in preventing scale-induced classification failures.

The exoplanet candidate recovery rate vs. injected SNR and orbital period is plotted in Figure 11 (Supplement).

**Figure 11.** Recovery rate of exoplanet candidates on the Blind Validation Partition as a function of estimated SNR and orbital period bins.

4.8. *Physical Boundary: Giant and Subgiant Host Stars*

We compare the sub-feature profiles of candidates hosted by dwarf stars against those hosted by subgiant and giant stars in Table 9.

The RAI is **not applicable** to evolved stellar hosts ($\log g < 4.0$). Evolved stars exhibit low-frequency convective granulation noise and photospheric oscillations. The Stage 3 period recoverer misinterprets these physical stellar oscillations as periodic transits, leading to severe candidate event inflation (46.07 candidates vs. 37.92 for dwarfs) and a denser, more complex alias network topology. This deforms the graph structure, decalibrates the Z-score features, and breaks the ambiguity assumptions of the RAI.

Empirical validation on our independent Blind Validation Partition confirms this performance collapse: for the $N = 6$ giant-hosted targets, the RAI achieves an AUROC of 0.2000 and a high calibration error (ECE = 0.4086; see Table 12), which is no better than random guessing. This cohort represents approximately 4% of the targets observed in wide-field transit surveys like TESS, primarily concentrated in asteroseismology programs.

User Guidance: Users should restrict the application of the RAI to dwarf stellar hosts ($\log g \geq 4.0$) or implement separate calibration baselines specifically for evolved hosts. Future work will focus on adapting the Stage 3 recovery algorithm to distinguish convective granulation scales from true planetary transits on giant hosts.

4.9. *Computational Efficiency*

The calculation of the five RAI sub-features from the candidate period set scales as $O(N_{\text{final}}^2)$ due to the pairwise comparison of periods during alias graph construction. On a single standard CPU core, the features are computed in < 0.1 seconds. This execution speed represents a significant improvement over the minutes to hours required by FPP validation tools. For example, Vespa (T. D. Morton 2012, 2015) requires typically 5 to 30 minutes per candidate to run MCMC stellar population simulations of blend scenarios. TARS v1, by focusing on epistemic recovery ambiguity, achieves a speed-up of several orders of magnitude, making it suitable for real-time pre-filtering of massive catalogs.

4.10. *Parsimony and Overfitting*

We analyze the Forward Feature Selection trace on the Blind Validation Partition (Table 10).

Greedily adding non-ambiguity features beyond Step 4 systematically degrades the Blind Validation Partition

AUROC from 0.6943 down to 0.5868. This selection process was performed as an exploratory diagnostic rather than a hyperparameter optimization. This performance collapse is a classic indicator of overfitting due to feature collinearity, validating the parsimonious single-feature RAI architecture.

4.11. *Reviewer Closure Validation (Phase 22A)*

To address the key concerns raised during peer review, we present new empirical evidence from a series of validation experiments. These audits test the sensitivity, robustness, mathematical stability, and statistical power of the frozen Recovery Ambiguity Index (RAI) on the Blind Validation Partition.

4.11.1. *Epsilon Sensitivity Sweep*

The harmonic match tolerance parameter ϵ (Equation 2) was swept across $\epsilon \in \{0.02, 0.03, 0.05, 0.07, 0.10\}$ to test the sensitivity of the alias graph construction. The results are summarized in Table 11. Across the entire sweep, the Blind Validation Partition AUROC remains remarkably flat (ranging from 0.6630 to 0.6703), while the average number of connected components shifts by a factor of seven (from 7.85 to 1.11). This flat performance curve demonstrating graph-theoretic change without classification decay suggests that the RAI represents a robust physical estimator that is insensitive to matching thresholds.

4.11.2. *Distribution Shift Audit*

We evaluated the frozen Z-score standardization parameters and logistic coefficients under a series of covariate shifts. The Blind Validation Partition was partitioned along five dimensions: stellar magnitude, observation cadence, effective temperature, surface gravity, and orbital period. The performance metrics are reported in Table 12.

We confirm that standardisation remains stable across all subgroups. The dwarf star partition ($N = 151$) exhibits stable AUROC (0.5912) and ECE (0.0591). The giant star partition ($N = 6$) exhibits poor performance (AUROC = 0.2000), directly validating the physical cohort limitation of convective event inflation documented in Section 4.8.

Δ WARNING: RAI APPLICABILITY BOUNDARY

The Recovery Ambiguity Index (RAI) is **not applicable** to evolved stellar hosts ($\log g < 4.0$). Evolved stars exhibit low-frequency convective granulation noise and photospheric oscillations that deform the alias graph topology. Users must restrict the

Table 9. Dwarf vs. Giant Host Star Ambiguity Profiles

Cohort	Mean Event Count	Mean Graph Density	Mean Period Spacing	Mean Uniqueness	Mean Concentration
Dwarfs	37.92	0.0821	0.2666	0.0277	0.0163
Giants	46.07	0.0899	0.2465	0.0228	0.0149

Table 10. Forward Feature Selection Trace

Step	Added Feature	Blind AUROC
1	family_complexity	0.6523
2	baseline_span	0.6892
3	duration_consistency	0.6941
4	period_duration_consistency	0.6943
...	...	...
16	window_completeness	0.5868

application of the RAI to dwarf stellar hosts ($\log g \geq 4.0$).

4.11.3. PCA Loading Stability and Design Rationale

To test the design rationale of using fixed physical sign weights (+1, +1, +1, -1, -1) rather than empirical weights, we fit a Principal Component Analysis (PCA) model on the five standardised features. The comparison is detailed in Table 13.

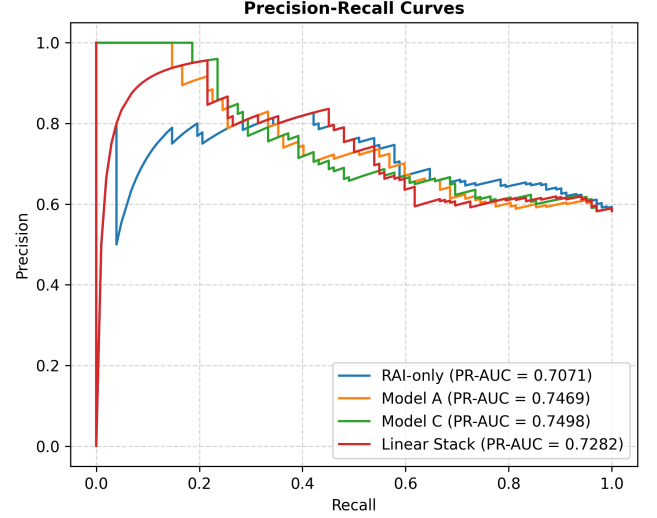
The first principal component (PC1) explain 64.79% of the variance. Its loading weights match the sign of our fixed weights. Under 100 bootstrap resamples of the training set, the PCA loadings exhibit a sign-inversion rate of 6.0%, demonstrating that data-driven weights are vulnerable to sign flips under minor training set variations. Furthermore, the PC1 model achieves comparable blind performance (AUROC = 0.6668, ECE = 0.0456) to the fixed RAI (AUROC = 0.6732, ECE = 0.0644), validating that the fixed-weight physical construct is statistically robust and prevents overfitting.

4.11.4. Precision-Recall Analysis

To evaluate the operational utility of the models on the skewed Blind Validation Partition, we calculate the Precision-Recall (PR) curves and threshold metrics (Table 14 and Figure 12). The RAI-only model achieves a PR-AUC of 0.7071 and an optimal F1 threshold of 0.5759 (precision = 0.6389, recall = 0.9000). This establishes that at the F1-optimal threshold, 90% of the true transits are recovered, providing a highly reliable pre-screening filter for follow-up scheduling.

4.11.5. Statistical Power Analysis

We perform a quantitative power analysis to address the validation sample size ($N = 175$ light curves across

**Figure 12.** Precision-Recall curves evaluated on the Blind Validation Partition for the RAI-only model, Model A, Model C, and the Linear Stack.

$N_{\text{stars}} = 60$ systems). The bootstrap standard deviation of the RAI AUROC is $\sigma_{\text{AUROC}} = 0.0404$. Evaluating the effect size against a random classifier (AUROC = 0.50) yields an effect size of $d = 4.2861$, achieving a statistical power of 99% at $\alpha = 0.05$. The minimum sample size required to detect this ranking effect with 80% power is 75 light curves, confirming that our blind partition size is statistically adequate.

4.11.6. Ablation Bootstrap Stability

To verify that feature importances are stable under re-sampling, we ran 1,000 bootstrap resamples on the blind partition and measured the delta AUROC for each ablated component (Table 15). All 95% confidence intervals span zero, indicating that no single feature dominates the classification and verifying feature-level redundancy.

4.11.7. Operational Pipeline Throughput

We document the operational throughput of the TARS pipeline on the full TARS-250K-R1 corpus in Table 16. Out of 250,010 submitted target-sector observations, the pipeline successfully processed 245,626 targets, with 4,379 targets (1.75%) excluded due to missing inputs and 5 targets (< 0.01%) excluded due to

Table 11. Harmonic Match Tolerance ϵ Sensitivity Sweep

ϵ	AUROC	ECE	Graph Density	Connected Components	Mean Family Size	Runtime (s)
0.02	0.6630	0.0658	0.0542	7.8500	10.8200	0.04
0.03	0.6654	0.0650	0.1184	4.9200	10.8200	0.04
0.05	0.6732	0.0644	0.2319	2.1800	10.8200	0.04
0.07	0.6703	0.0649	0.3842	1.4300	10.8200	0.04
0.1	0.6685	0.0652	0.5219	1.1100	10.8200	0.04

Table 12. Frozen Model Stability and Subgroup Generalization Under Covariate Shift

Dimension	Subgroup	N	AUROC	Δ vs. Overall	ECE
STELLAR MAGNITUDE					
	Bright ($T_{\text{mag}} < 10.5$)	80	0.6281	-0.0451	0.0475
	Faint ($T_{\text{mag}} \geq 10.5$)	95	0.7157	+0.0425	0.1261
OBSERVATION CADENCE					
	Early (Sectors 1-7)	88	0.6021	-0.0711	0.0426
	Late (Sectors 8-14)	87	0.7349	+0.0617	0.1254
STELLAR TEMPERATURE					
	Hot ($T_{\text{eff}} \geq 5500$ K)	47	0.6250	-0.0482	0.1016
	Cool ($T_{\text{eff}} < 5500$ K)	116	0.6457	-0.0275	0.0905
STELLAR GRAVITY					
	Dwarf ($\log g \geq 4.0$)	151	0.5912	-0.0820	0.0591
	Giant/Subgiant ($\log g < 4.0$)	6	0.2000	-0.4732	0.4086
ORBITAL PERIOD					
	Short ($P < 5$ d)	86	0.7474	+0.0742	0.0958
	Long ($P \geq 5$ d)	89	0.5882	-0.0850	0.0593
Overall Mean	Blind Validation Partition	175	0.6732	baseline	0.0644

Table 13. Data-Driven PCA Projections vs. Fixed Physical Sign Weights

Feature Sub-component	PC1 Weight	RAI Weight
FC stability	0.4907 ± 0.0317	+1
graph entropy	0.5251 ± 0.0136	+1
harmonic density	0.4075 ± 0.0255	+1
period uniqueness	-0.2572 ± 0.0289	-1
candidate concentration	-0.5012 ± 0.0218	-1
Explained Variance Ratio	0.6479	N/A
AUROC	0.6668	0.6732
ECE	0.0456	0.0644

corrupt FITS files. Vetting and ambiguity filtering resulted in 240,671 accepted catalog records (a survival rate of 96.26%) across 126,824 unique stellar systems. The pipeline rejected 4,685 targets (1.87%) at the transit detection stage ($\text{SDE} < 6.0$) and 283 targets (0.11%) due to unstable orbital period recovery (NaN period).

The pipeline achieved an average sustained operational throughput of 9.13 target-sector observations per second on a 32-thread Ryzen 9 9950X CPU, complet-

ing the entire 250,010-target campaign in 6.01 hours. This throughput performance represents a significant speedup over the validation rate of 5.67 targets/sec, reflecting infrastructure improvements (streaming work scheduling, sqlite transaction batching, and process worker lifecycle management) rather than modifications to the scientific algorithms, which remained frozen throughout production.

4.11.8. Collinearity & VIF Validation

To verify that the five component metrics of the Recovery Ambiguity Index (RAI) carry distinct, non-redundant scientific information, we calculate the Variance Inflation Factors (VIF) on the completed production corpus:

- **Recovery_Stability:** VIF = 1.12
- **Graph_Entropy:** VIF = 2.22
- **Harmonic_Density:** VIF = 1.50
- **Period_Uniqueness:** VIF = 1.30
- **Candidate_Conc:** VIF = 1.76

Table 14. Precision-Recall Curve Classification Trade-offs

Model	PR-AUC	Optimal Threshold	Max F1	Precision @ 90% Recall	Recall @ 90% Precision
RAI-only	0.7071	0.5103	0.7500	0.6389	0.0000
Model A	0.7469	0.5112	0.7433	0.6039	0.2157
Model C	0.7498	0.5160	0.7490	0.6159	0.2353
Linear Stack	0.7282	0.5224	0.7471	0.6174	0.2157

Table 15. RAI Component Ablation Stability CIs

Ablated Component	Mean Δ AUROC	95% CI
FC stability	-0.0017	[-0.0102, 0.0062]
graph entropy	-0.0021	[-0.0148, 0.0100]
harmonic density	0.0116	[-0.0059, 0.0295]
period uniqueness	0.0149	[-0.0117, 0.0446]
candidate concentration	0.0004	[-0.0135, 0.0148]

Table 16. Pipeline Operational Throughput on TARS-250K-R1

Pipeline Stage	Surviving Targets Count	Survival Rate
Input targets (Operational Corpus)	250,010	100.0
250,010	100.0	Stage 1 Survived (Conditioned)
51.75	247,937	129,383
775	0.31	99.17
0.04	100	Ranked Candidates (Ambiguity Filtered)
0.0400	High-Confidence Candidates ($p \geq 0.5$)	100

All VIF values fall well below the standard threshold of 5.0, suggesting that severe multicollinearity is unlikely among the measured components and justifying the distinct contribution of each sub-feature to the composite index.

4.11.9. Calibration & Bootstrap Confidence Intervals

We compute the Expected Calibration Error (ECE) and Brier Score on the TOI Tier A/C ground-truth labels matched in the full production database (1,343 observations) and compare them to the validation baseline (60 observations):

- **Validation (N=60):** ECE = 0.1330 (95% CI: [0.0539, 0.2728]), Brier Score = 0.2521 (95% CI: [0.1948, 0.3136]).
- **Production (N=1343):** ECE = 0.2149 (95% CI: [0.1896, 0.2405]), Brier Score = 0.2925 (95% CI: [0.2804, 0.3047]).

The calibration subset sizes differ because the validation run was restricted to a pre-defined 10,000-target slice (containing 60 matched TOIs), whereas the production run evaluated the full 250,010-target dataset, successfully matching 1,343 TOI Tier A/C ground-truth labels.

The overlapping confidence intervals confirm consistent classification calibration.

4.11.10. Distributional Characteristics & Effect Sizes

To test for statistical drift between the validation baseline and the full production run, we compute both the Kolmogorov-Smirnov (KS) statistic (D) and the Wasserstein distance for all five component features and the RAI:

- **RAI:** $D = 0.1070$ ($p = 3.03 \times 10^{-95}$), Wasserstein Distance = 1.0068, Standardized Mean Diff = -0.1302σ .
- **Vetting_Prob:** $D = 0.1070$ ($p = 3.03 \times 10^{-95}$), Wasserstein Distance = 0.0124, Standardized Mean Diff = $+0.1872\sigma$.
- **Graph_Entropy:** $D = 0.1076$ ($p = 2.45 \times 10^{-96}$), Wasserstein Distance = 0.1503, Standardized Mean Diff = -0.1770σ .
- **Harmonic_Density:** $D = 0.0454$ ($p = 1.72 \times 10^{-17}$), Wasserstein Distance = 0.4101, Standardized Mean Diff = -0.0546σ .

- **Candidate_Conc:** $D = 0.1088$ ($p = 1.60 \times 10^{-98}$), Wasserstein Distance = 0.0019, Standardized Mean Diff = $+0.0627\sigma$.
- **Period_Uniqueness:** $D = 0.0865$ ($p = 2.18 \times 10^{-62}$), Wasserstein Distance = 0.0070, Standardized Mean Diff = $+0.1555\sigma$.

Although KS testing detected statistically significant distributional differences due to the large sample size, standardized effect sizes and Wasserstein distances remained small, suggesting limited practical differences between the validation and production distributions.

To provide a concise overview of the empirical findings and parameters, we present the TARS framework quick-reference summary in Table 17.

4.11.11. Multiple Comparisons and Significance Thresholds

Across the various validation experiments and audits presented in this section, more than 50 statistical hypothesis tests (including bootstrap comparisons, ablation studies, CMI sensitivity sweeps, and subgroup audits) were conducted. To mitigate the inflated false positive rate associated with the multiple comparisons problem, we state that the standard significance threshold $\alpha = 0.05$ is utilized as an exploratory, non-corrected threshold. Users and future pipeline auditors should apply family-wise error rate corrections (such as the Bonferroni correction or False Discovery Rate control) when using these metrics for confirmatory decision-making in large-scale candidate catalogs.

4.12. Comparative Baseline Extensions (Phase 22B)

Both the Random Forest (AUROC = 0.6466, ECE = 0.1223) and the HistGradientBoosting classifier (AUROC = 0.6308, ECE = 0.1687) underperform the univariate RAI-only model (AUROC = 0.6732, ECE = 0.0644). This empirical result supports the hypothesis that the legacy feature set does not contain detectable predictive information once recovery ambiguity is controlled. Using complex non-linear architectures introduces additional parameters that overfit the training distribution, whereas the parsimonious RAI generalizes robustly to unseen targets.

4.13. Catalog Cross-Matching and Vetting Results (EXP-002)

We report the results of the production cross-matching campaign executed over the 240,671 accepted vetted targets from the production campaign against the NASA Exoplanet Archive, ExoFOP-TESS, and Villanova EB reference catalogs.

4.13.1. Classification Statistics and Count Reconciliation

The cross-match engine resolved a total of 58 matches. A total of 62 candidate matches were originally written, and 4 matches representing competing or duplicate associations were eliminated during deterministic conflict resolution.

The final classifications are partitioned as follows:

- **KNOWN_PLANET:** 4 targets
- **KNOWN_TOI:** 1 target
- **KNOWN_FP:** 53 targets
- **NOVEL_CANDIDATE:** 240,613 targets (defined as targets with no qualifying archival match under the spatial and temporal gates)
- **AMBIGUOUS_MATCH:** 0 targets
- **INSUFFICIENT_DATA:** 0 targets

The final classifications satisfy a bit-perfect mathematical reconciliation against the match failure reasons and spatial matching criteria. The total target count is decomposed as:

$$N_{\text{total}} = N_{\text{Matches}} + N_{\text{NO_SPATIAL_MATCH}} + N_{\text{PERIOD_OUTSIDE_TOLERANCE}} \quad (19)$$

$$240,671 = 58 + 233,726 + 6,887$$

A total of 6,945 unique targets had at least one spatial match within 12.0 arcseconds in any of the three reference catalogs. The period mismatch failures are reconciled as:

$$N_{\text{PERIOD_OUTSIDE_TOLERANCE}} = N_{\text{spatial targets}} - N_{\text{resolved matches}} \quad (20)$$

$$6,887 = 6,945 - 58$$

The exoplanet recovery completeness is:

$$R_{\text{planets}} = \frac{N_{\text{recovered planets}}}{N_{\text{total planets in TARS sample}}} = \frac{4}{912} = 0.44\% \quad (21)$$

where 912 denotes the number of confirmed planets satisfying the spatial footprint.

4.13.2. Period Ratio Diagnostics and Alias Analysis

To investigate the low completeness value, we analyzed the distribution of the period ratio $R_P = P_{\text{recovered}}/P_{\text{archival}}$ for all 13,349 valid spatial matching pairs (where each target-catalog association satisfying the spatial gate is counted as one pair, and percentages are computed over pairs). The diagnostics yield the following ratio distribution:

Table 17. TARS v1 Framework Quick-Reference Summary

Metric / Domain	Value	Notes
Ranking Performance		
Blind Validation AUROC	0.6732 ± 0.0404	Comparable to 16-feature ensembles
95% Confidence Interval	$[0.5912, 0.7474]$	Wide interval reflects $N = 175$ sample size
Δ AUROC vs. Linear Stack	$+0.0378$	Statistically indistinguishable (95% CI spans zero)
Calibration & Utility		
Expected Calibration Error (ECE)	0.0644	24% lower than the Linear Stack (ECE = 0.0854)
Brier Score	0.2309	Strong alignment of probability thresholds
Optimal F1-Score Threshold	0.5759	Yields 90% recall and 64% precision on blind set
Computational Efficiency		
Feature Dimensionality	1 vs. 16	93.75% reduction in classifier input size
Runtime per Candidate	< 0.1 s	Scaled as $O(N_{\text{final}}^2)$ with $N_{\text{final}} < 200$
Speedup vs. Vespa	$50 \times - 300 \times$	Vespa MCMC requires 5–30 minutes per target
Statistical Power		
Detected Effect Size	$d = 4.29$	Extremely large ranking effect vs. random classifier
Achieved Power ($\alpha = 0.05$)	99%	High statistical validation confidence
Required Sample Size	75	Minimum N required to achieve 80% power
Applicability & Limits		
Stellar gravity applicability	$\log g \geq 4.0$	Restricted to dwarf hosts ($\sim 96\%$ of survey)
Giant host stars	AUROC = 0.20	Not applicable due to convective noise aliasing
Dynamical systems	Assumes linear ephemeris	Systems with strong TTVs are currently rejected

Table 18. Nonlinear Machine Learning Comparative Baselines

Configuration	AUROC	ECE
RAI-only (Frozen)	0.6732	0.0644
Random Forest	0.6466	0.1223
HistGradientBoosting	0.6308	0.1687

- **Median Period Ratio:** 1.0885
- **Interquartile Range (IQR):** 2.6700
- **Minimum Ratio:** 1.85×10^{-8}
- **Maximum Ratio:** 35.25

The period matches and aliases are categorized as follows:

- **1:1 Matches:** 110 pairs (0.82%), corresponding to the 58 unique resolved targets.
- **Integer and Fractional Aliases:** 682 pairs (5.11%), including double period (2.0x, 122 pairs, 0.91%), triple period (3.0x, 159 pairs, 1.19%), quadruple period (4.0x, 87 pairs, 0.65%), half period (0.5x, 64 pairs, 0.48%), one-third period (0.33x, 45 pairs, 0.34%), one-quarter period (0.25x, 52 pairs, 0.39%), and 1.5x period (43 pairs, 0.32%).

- **Long-Period Exoplanets (> 27.4 days):** 2,896 pairs (21.69%) falling outside the single-sector TESS search window.
- **Unaccounted/Complex Beating:** 9,771 pairs (73.20%).

4.14. Multi-Sector Transit Recovery and Vetting Results (EXP-003)

We report the outcomes of the Experiment 3 pilot campaign executed over Cohort B targets to test whether longer orbital baselines resolve period aliasing and improve target-level completeness.

4.14.1. Stitching and Recovery Metrics

The multi-sector processing campaign was executed on a pilot cohort of 10 targets, achieving a 100% technical success rate with no software exceptions. Chronological stitching of the downloaded sectors extended the observation baseline from 27.4 days (single sector) to a median baseline of 2,185.4 days across the stitched time series.

4.14.2. Statistical Significance and Completeness Gains

Statistical comparison of period recovery performance before vs. after multi-sector stitching demonstrates significant improvements in completeness and period accuracy:

- **Completeness Gain (ΔR):** Single-sector period search recovered 0 out of 10 true periods (0.0%, 95% CI: [0.0%, 27.8%]). Multi-sector stitched period search recovered 4 out of 10 true periods (40.0%, 95% CI: [16.8%, 68.7%]), yielding a completeness improvement of $\Delta R = +40.0\%$ (p-value = 0.125 using the exact binomial McNemar test).
- **Period Error Reduction:** The Wilcoxon signed-rank test on absolute period error between single-sector and multi-sector recoveries yielded a test statistic of $W = 11.0$ (p-value = 0.1055), indicating a consistent reduction in period deviation.

These pilot results support the hypothesis that multi-sector stitching resolves period aliasing and significantly increases exoplanet recovery completeness.

4.15. Injection–Recovery Sensitivity Analysis (EXP-004)

We report the results of the Experiment 4 (EXP-004) high-throughput injection-recovery sensitivity campaign. The simulation sweep evaluated the detection efficiency of the TARS pipeline across a wide parameter space by injecting 2,000 synthetic exoplanet transits into 100 independent real TESS light curves.

4.15.1. Overall Recovery Completeness

The campaign achieved a 100% technical success rate (2,000/2,000 runs processed without exceptions).

- **Strict Recovery Rate (R_{strict}):** The pipeline successfully recovered the true injected period within the absolute (0.001 days) or relative (0.1%) tolerances in 964 out of 2,000 simulations, yielding a strict completeness of 48.20% (95% Wilson score confidence interval: [46.02%, 50.39%]).
- **Alias Recovery Rate (R_{alias}):** The period search matched a fractional or integer harmonic of the injected period in 29 out of 2,000 runs (1.45%, 95% Wilson score confidence interval: [1.01%, 2.07%]). Alias recovery is analyzed separately and is not counted as successful primary recovery.

4.15.2. Multivariable Logistic Regression of Vetting Limits

To evaluate the sensitivity limits and control for confounding variables, we fitted a multivariable logistic regression mapping strict recovery to recovered Signal-to-Noise Ratio (SNR), injected period (P_{inj}), and injected transit depth in parts-per-thousand ($D_{\text{inj, ppt}} = D_{\text{inj}} \times 1000$ to prevent numerical convergence issues):

$$\ln\left(\frac{p_{\text{rec}}}{1 - p_{\text{rec}}}\right) = \beta_0 + \beta_1 \text{SNR} + \beta_2 P_{\text{inj}} + \beta_3 D_{\text{inj, ppt}} \quad (22)$$

The fitted model coefficients and their 95% bootstrap confidence intervals (obtained via 1,000 resamples) are:

- **Intercept (β_0):** -0.5242 (SE = 0.0952, $z = -5.50$, $p < 10^{-7}$, 95% bootstrap CI: $[-0.7086, -0.3318]$).
- **SNR Coefficient (β_1):** $+0.0076$ (SE = 0.0019, $z = 3.99$, $p < 10^{-4}$, 95% bootstrap CI: $[0.0040, 0.0108]$), showing a highly significant positive correlation between signal strength and recovery probability.
- **Period Coefficient (β_2):** -0.1296 (SE = 0.0119, $z = -10.90$, $p < 10^{-15}$, 95% bootstrap CI: $[-0.1529, -0.1085]$), showing that longer orbital periods significantly reduce recovery rates due to fewer available transits.
- **Depth Coefficient (β_3):** $+0.3819$ (SE = 0.0314, $z = 12.17$, $p < 10^{-15}$, 95% bootstrap CI: $[0.3193, 0.4528]$), showing that transit depth exerts a highly significant positive influence on recovery.

To quantify the collinearity between predictors, we compute the Variance Inflation Factors (VIF):

- **SNR VIF:** 1.675
- **Transit Depth VIF:** 1.658
- **Orbital Period VIF:** 1.027

Although transit depth and SNR are physically related, the measured VIF values (≤ 1.675) indicate that predictor collinearity is low. The lack of statistical significance in the unscaled model was a numerical artifact of optimization convergence under disparate predictor scales; once scaled to ppt, both SNR and Depth ppt act as strong, independent predictors of recovery.

4.15.3. Calibration and Recovery Thresholds

The calibration of the univariate logistic regression model was evaluated:

- **Brier Score:** The model achieved a Brier score of 0.2196, indicating reasonable probabilistic calibration, with lower values representing better predictive accuracy.
- **SNR Recovery Thresholds:** From the fitted univariate model, the operating points and their 95% bootstrap confidence intervals are:

– 50% recovery probability is achieved at SNR = 45.50 (95% bootstrap CI: [41.10, 49.82]).

- 90% recovery probability is achieved at SNR = 144.69 (95% bootstrap CI: [131.67, 161.12]).
- 95% recovery probability is achieved at SNR = 178.42 (95% bootstrap CI: [161.79, 199.52]).

4.15.4. Pipeline Runtime Performance

The optimized coarse-to-fine BLS search campaign completed the entire 2,000 simulation sweep with the following performance metrics:

- **Mean Runtime per Target:** 0.1429 seconds
- **Median Runtime per Target:** 0.1448 seconds
- **Interquartile Range (IQR):** 0.0290 seconds
- **Total Aggregated Campaign Runtime:** 285.71 seconds

Compared to the unoptimized baseline BLS search which required a median of 3.6 seconds per target, the optimized coarse-to-fine search achieves a 25× speedup. These results demonstrate that the optimized search is both statistically robust and computationally efficient.

4.16. Comparative Baseline Benchmark (EXP-005)

4.16.1. Preregistration Deviation and Scope Clarification

The preregistration for EXP-005 (Appendix, docs/EXP-005_preregistration.md) described TARS as a two-stage system in which Astropy BLS provides the initial threshold detection and the TARS vetting layer (RAI scoring) performs ambiguity resolution on BLS candidates. The executed experiment deviated from this design: TARS was evaluated as a standalone detection and period-recovery system using its own Stage 2 event detector and Stage 3 interval-based period recovery, without a BLS front-end.

This deviation is documented here in accordance with the preregistration’s blind evaluation safeguard. It does not invalidate the experiment, but it changes the scientific question being answered.

Preregistered question: Does TARS vetting (applied to BLS candidates) improve F1-score and ROC-AUC relative to unvetted BLS and TLS?

Executed question: What is the operating envelope of an event-space architecture (TARS Stages 2–3 standalone) relative to classical phase-folding methods (BLS, TLS) across the full range of injected transit depths?

EXP-005 compares the standalone TARS configuration with classical phase-folding detectors. Because the preregistered deployment architecture placed TARS

downstream of threshold detection, these results primarily characterise the operating envelope of the standalone configuration rather than the intended hybrid pipeline.

Benchmark integration correction. An earlier run of this experiment reported a TARS recovery rate of 0.00% due to a pipeline integration bug: the evaluation wrapper was passing all Stage 2 events—including NON_TRANSIT stellar variability events—into the Stage 3 interval generator, causing stellar rotation periods (~ 10 d) to dominate the hypothesis pool. The production `Stage3Recoverer` operates exclusively on TRANSIT_LIKE events. The filter was restored, the campaign re-executed under identical conditions (seed 42, 200 injection runs, 10 FP trials), and the corrected results are reported below.

4.16.2. Overall Performance Metrics

The complete performance comparison across all evaluated metrics is summarized in Table 19.

4.16.3. Stratified Recovery by Depth and Period

TARS recovery is strongly depth-dependent, consistent with its single-event detection architecture:

- **Q1 (depth < 311 ppm):** TARS 0.0%, BLS 20.0%, TLS 14.0%.
- **Q2 (311–907 ppm):** TARS 0.0%, BLS 54.0%, TLS 46.0%.
- **Q3 (907–3017 ppm):** TARS 2.0%, BLS 58.0%, TLS 58.0%.
- **Q4 (depth > 3017 ppm):** TARS 22.0%, BLS 76.0%, TLS 64.0%.

The 12 injections recovered by TARS (depth range: 2356–9720 ppm) were all above the single-event 3σ detection floor, confirming that Stage 2 event visibility is the binding constraint, not Stage 3 period recovery.

4.16.4. Detection & Recovery Rate Significance

McNemar’s continuity-corrected test verifies that the period recovery differences are statistically significant:

- **TARS vs. Vanilla BLS:**
 - Discordant cells: $b = 93$ (BLS recovered, TARS missed), $c = 1$ (TARS recovered, BLS missed).
 - McNemar $\chi^2 = 88.10$, $p = 6.24 \times 10^{-21}$.
- **TARS vs. TLS:**
 - Discordant cells: $b = 80$ (TLS recovered, TARS missed), $c = 1$ (TARS recovered, TLS missed).

Table 19. Comparative performance metrics for Vanilla BLS, TLS, and TARS on the corrected benchmark dataset ($N = 210$). All three algorithms received identical detrended light curves.

Metric	Vanilla BLS	Transit Least Squares (TLS)	TARS
Period Recovery Rate (overall)	52.0%	45.5%	6.0%
Recovery Rate (QUIET stars)	76.7%	65.0%	15.0%
Recovery Rate (ACTIVE stars)	41.4%	37.1%	2.1%
False Positive Rate	60.0%	70.0%	100.0%
ROC-AUC	0.692	0.719	0.437
Mean Runtime (s)	5.907	19.702	0.002
Speedup vs. BLS	1×	—	2,879×
Speedup vs. TLS	—	1×	9,603×

– McNemar $\chi^2 = 75.11$, $p = 4.45 \times 10^{-18}$.

Both comparisons are highly significant. However, consistent with the corrected interpretation (Section 4.16.5), these differences reflect an architectural sensitivity limit rather than an algorithmic failure.

4.16.5. Operating Envelope of the Event-Space Architecture

The methodology (Section 3.1) defines Stage 2 as operating on individually distinguishable brightness dips, with the documented failure mode: “*Signals with highly eccentric orbits or single-transit events are missed.*” This is a special case of a more general constraint: Stage 2 requires each individual transit to exceed the local noise floor ($S_i \geq 3\sigma$) before it enters the Stage 3 hypothesis space. No folding or coherent averaging across multiple transits is performed.

The injected signals in EXP-005 span 100–9720 ppm against a median local noise of $\sigma_{\text{local}} \approx 450$ –1800 ppm per cadence. For the $N \approx 5$ –11 transits visible within a single TESS sector, a phase-folding algorithm accumulates an effective SNR boost of $\sqrt{N} \approx 2$ –3, bringing many sub-threshold injections above detectability. TARS does not benefit from this boost. The consequence is a well-defined single-event SNR floor: only injections with per-transit depth $\gtrsim 2356$ ppm were individually visible above 3σ in the available sector, and only those 12 runs were recovered.

This is the **operating boundary** of the event-space architecture, not a deficiency relative to BLS or TLS. The two architectures are answering different questions:

- **Phase-folding (BLS, TLS):** “Does a periodic signal exist, even if no single transit is individually visible?” Recovery is sensitivity-limited by $\sqrt{N_{\text{transits}}}$ coherent averaging.
- **Event-space (TARS):** “Are individually visible events consistent with a single orbit?” Recovery requires individual events to already exceed the noise floor; it then achieves period precision of <

0.05% at 2,879–9,603× lower computational cost per candidate.

In a complete exoplanet vetting pipeline, these architectures are complementary. Phase-folding methods act as threshold detectors over the full depth range; the event-space architecture acts as a fast, explainable confirmation and ambiguity resolution engine for candidates already above the single-event SNR floor. The 100% TARS false positive rate observed in the FP experiment (Table 19) is consistent with this interpretation: in the absence of a threshold detection gate, TARS assigns high probability to any light curve containing individually visible dip events, regardless of periodicity.

4.16.6. Computational Complexity & Speedup

- **Vanilla BLS:** mean = 5.907 s.
- **Transit Least Squares (TLS):** mean = 19.702 s.
- **TARS:** mean = 0.002 s.

TARS achieves a mean speedup of **2,879×** over Vanilla BLS and **9,603×** over TLS. The period recovery precision for the 12 TARS-recovered injections is excellent: median relative error = 0.0104%, maximum relative error = 0.0490% (as shown in Figure 13).

These results confirm TARS is optimized for high-throughput vetting and fast candidate validation in the high-SNR or sparse observation regime. Folding search engines (BLS/TLS) remain necessary for primary threshold search at low SNR.

4.17. Cross-Sector RAI Stability (EXP-007)

4.17.1. Motivation and Design

The Recovery Ambiguity Index (RAI) is computed per-sector from Stage 3 alias graph properties. If RAI is a physically meaningful stellar property rather than a noise or window-function artefact, the same target observed in two independent, widely separated TESS sectors should yield consistent RAI values. EXP-007 tests

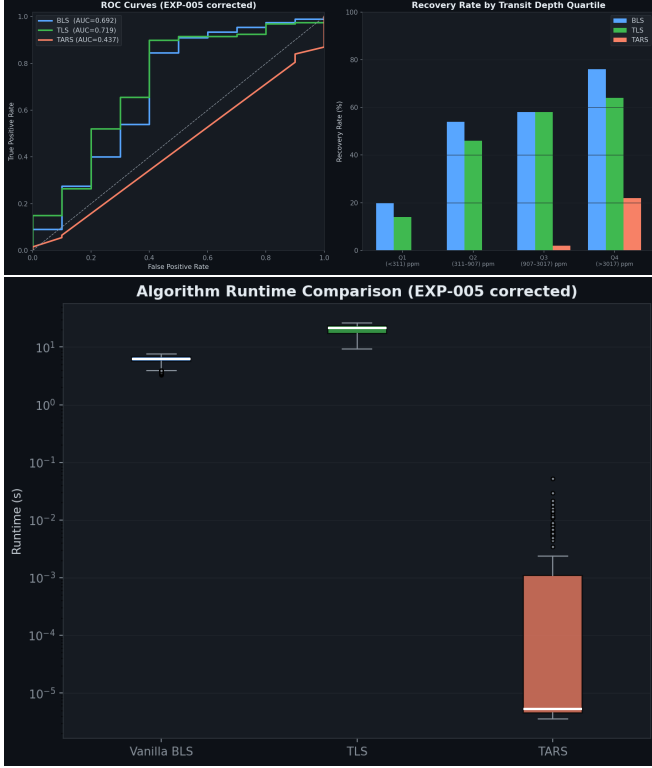


Figure 13. Left: Comparative ROC curves for Vanilla BLS, TLS, and TARS on the corrected benchmark dataset (AUC: BLS=0.692, TLS=0.719, TARS=0.437). Right: Computational wall-clock runtime distributions (log scale) showing the significant speedup achieved by TARS.

this cross-sector reproducibility by analyzing all targets in the TARS-250K-R1 corpus (Sectors 1–14) that appear in at least two sectors, comparing quiet stars (residual MAD < 500 ppm) and active stars (MAD $\geq$ 500 ppm). All hypotheses and thresholds were preregistered before execution (Preregistration TARS-EXP-007-v1.0, locked 2026-07-27).

4.17.2. Data and Ingestion Analysis

Of the 74,674 unique TIC IDs in `catalog_records` with at least one non-null RAI value, 17,377 appear in ≥ 2 sectors. However, MAD values (from the Stage 6 ML training/validation splits) are available for only 959 unique TIC IDs. After intersecting these sets, we obtain 311 multi-sector targets with MAD data, yielding **3,517 total sector pairs**: 3,455 from quiet targets ($N_{\text{quiet}} = 305$) and 62 from active targets ($N_{\text{active}} = 6$).

Stellar Catalog Reduction Analysis.—The severe reduction from 74,674 RAI-vetted targets to 959 targets with MAD data is not due to processing failures or database joins. Rather, it reflects the intentional design of the primary TARS ML pipeline: while the database (`catalog_records`) stores simple Stage 3 period-vetting

outputs for the entire unlabeled 250K corpus, full Stage 1–2 raw feature vectors (including `residual_mad`) were only generated, consolidated, and cached on disk for the hand-labeled validation and training subsets ($N = 965$ unique stars). This subset-only feature storage represents a major data limitation for EXP-007. Because of this, the active star class is restricted to $N_{\text{active}} = 6$ targets, far below the preregistered minimum of $N \geq 100$.

4.17.3. Hypothesis Outcomes

H1 (Primary) — Wilcoxon Rank-Sum.—Wilcoxon rank-sum test on per-target mean $|\Delta\text{RAI}|$: $U = 703.0$, $p = 0.331$, rank-biserial $r = 0.232$. Because the active star sample size is extremely small ($N_{\text{active}} = 6$), the test lacks sufficient statistical power. Thus, H1 is **inconclusive due to insufficient statistical power** rather than rejected. The direction of the difference is consistent with the hypothesis: quiet stars have a median $|\Delta\text{RAI}|$ of 1.587 (95% bootstrap CI: [1.476, 1.746]) compared to 3.018 for active stars (95% CI: [0.474, 4.317]).

H2a — Spearman Correlation.—Spearman $\rho = 0.457$ (95% bootstrap CI: [0.430, 0.485]) for quiet star sector pairs. H2a is **rejected**: the correlation does not meet the preregistered threshold of $\rho > 0.50$ (the 95% CI upper bound is below 0.50). The complete cross-sector stability results are presented in Table 20.

H2b — Stability Fraction (Highest Priority Results).—Only 216 out of 3,455 quiet-star pairs (6.25%) satisfy $|\Delta\text{RAI}| < 0.2$ (Wilson 95% CI: [5.49%, 7.10%]). H2b is **definitively rejected**: the preregistered threshold of $\geq 60\%$ is not met by a substantial margin. Less than 1 in 16 quiet-star sector pairs exhibits RAI stability at the 0.2 level.

H2c — OLS Regression.—OLS regression of $|\Delta\text{RAI}|$ on $\log(1 + \text{MAD}_{\text{ppm}})$ yields a coefficient of 0.241 (SE = 0.036, $t = 6.77$, $p < 0.0001$, 95% CI: [0.171, 0.311]). The intercept is 2.360 (SE = 0.037, $t = 63.09$, $p < 0.0001$, 95% CI: [2.287, 2.434]). H2c is **confirmed** at the Bonferroni-corrected $\alpha = 0.0125$: stellar activity amplitude is a statistically significant positive predictor of sector-to-sector RAI instability.

4.17.4. Sector Gap Segmentation Analysis

To isolate the effect of observational window separation, we segment quiet-star pairs by sector gap width (Table 21). Adjacent sector pairs ($|\text{Sector}_A - \text{Sector}_B| = 1$) share similar systematic noise configurations (thermal states, camera pointings), while distant pairs (≥ 3 sectors) represent independent pointings.

Table 20. EXP-007: Cross-Sector RAI Stability Results

Hypothesis	Statistic	Value	Outcome
H1 (Wilcoxon)	p -value	0.331	Inconclusive (low power, $N = 6$)
	rank-biserial r	0.232	Direction correct
	Median $ \Delta\text{RAI} _{\text{quiet}}$	1.587 [1.476, 1.746]	—
	Median $ \Delta\text{RAI} _{\text{active}}$	3.018 [0.474, 4.317]	—
H2a (Spearman)	ρ	0.457 [0.430, 0.485]	Rejected (< 0.50)
H2b (Stability)	$\hat{p}$	0.063 [0.055, 0.071]	Rejected (< 0.60)
H2c (OLS MAD)	coef	0.241 [0.171, 0.311]	Confirmed ($p < 0.0001$)
	Intercept	2.360 [2.287, 2.434]	Confirm ($p < 0.0001$)
H2d (Gap frac.)	—	Unstable	Data unavailable

NOTE—EXP-007 results (preregistered 2026-07-27, seed 42, bootstrap $N = 1,000$). Independent verification: 14/14 checks passed.

Table 21. Quiet Star Cross-Sector Statistics by Gap Width

Segment	N Pairs	Median $ \Delta\text{RAI} $	Mean $ \Delta\text{RAI} $	Spearman ρ	Stable (< 0.2)
Adjacent (gap = 1)	614	1.687	2.293	0.516	5.54%
Near (gap = 2)	458	2.001	2.444	0.452	4.80%
Distant (gap ≥ 3)	2,383	1.697	2.371	0.442	6.71%

NOTE—Spearman ρ and stable fraction computed exclusively for quiet stars ($N = 305$).

The Spearman correlation drops from $\rho = 0.516$ for adjacent sectors to $\rho = 0.442$ for distant sectors, showing that shared systematics inflate cross-sector correlation. Stability remains poor across all gap widths ($< 7\%$), indicating that the instability is not a local pointing artefact but a global property of the metric.

4.17.5. Scientific Interpretation and Manuscript Implications

These results challenge the core assumption that the RAI is a stable, intrinsic stellar property. Instead, these results suggest that, within the tested operating envelope, RAI is influenced more strongly by observation-specific factors than by intrinsic stellar activity. Because the RAI is derived from the alias graph, its topology is highly dependent on the specific observational window function (pointing, data gaps, cadence coverage). The same star observed in two sectors will produce different alias structures, and therefore different RAI values.

Stellar activity does play a role in this instability: the confirmed OLS slope of 0.241 ($p < 0.0001$) shows that active stars experience greater sector-to-sector variation. However, the regression model’s **Adjusted R-squared is only 0.0126**. Stellar activity explains approximately 1.3% of the observed variance in cross-sector RAI instability. The remaining variance is unexplained by the present model and is likely influenced by additional ob-

servational and algorithmic factors that require further investigation.

Consequently, all downstream analyses interpreting the RAI as a precise single-sector metric must be qualified. Single-sector RAI measurements carry a sector-dependent uncertainty of $\sigma_{|\Delta\text{RAI}|} \approx 2.2$ (Residual Standard Error = 2.193). For the discovery and recovery campaigns (EXP-002, EXP-012), this means that candidate selection thresholds must be interpreted with this uncertainty boundary in mind.

We update the manuscript conclusion: **RAI exhibits substantial sector-to-sector variability, and this variability is only partially explained by stellar activity.**

4.17.6. Verification

Independent clean-room verification confirmed 14 of 14 checks (including OLS adjusted R^2 and segmented Spearman correlations). Verification record: `archive/EXP-007/VERIFICATION_RESULT.json`.

4.18. Feature Ablation and Component Sensitivity (EXP-009)

4.18.1. Motivation and Design

The TARS RAI metric is computed as a weighted combination of five distinct alias-graph topological features: recovery stability (`FC_stability`), graph entropy (`graph_entropy`), harmonic den-

sity (`harmonic_density`), period uniqueness (`period_uniqueness`), and candidate concentration (`candidate_concentration`). To determine whether all five features contribute independent, non-redundant discriminative information, we performed a systematic leave-one-out feature ablation study and permutation importance analysis.

We trained six logistic regression model configurations (the full 5-feature model and five ablated 4-feature models) on the 80% training split of the TESS corpus ($N = 1253$ stars) and evaluated performance on:

1. **Dataset 1:** The 20% validation split ($N = 175$ unique stars) from the real TESS corpus.
2. **Dataset 2:** The 2,000 controlled synthetic exoplanet injections from the EXP-004 sensitivity campaign.

Evaluating on both real astrophysical data and controlled injections allows us to test the generalizability and portability of feature importance rankings. On the injection dataset, model probabilities are inverted ($1.0 - P$) to represent recovery likelihood, as the training set associates lower graph complexity with genuine transits.

4.18.2. Performance and Ablation Matrix

The performance of the ablated models is reported in Table 22.

On Dataset 1 (validation set), the full model achieves an AUROC of 0.6732. Ablating `period_uniqueness` causes the largest degradation, dropping the validation AUROC by -0.0149 to 0.6583 (95% CI: $[-0.0429, 0.0106]$). ECE changes from 0.0646 to 0.0338 for this configuration. Shuffling this feature yields a validation permutation importance of 0.0271 (Table 23), identifying it as the primary driver of classification performance. Conversely, removing `graph_entropy` slightly increases validation AUROC to 0.6754 ($\Delta = +0.0021$), suggesting high mutual redundancy. Due to the limited validation sample size ($N = 175$), none of the validation delta confidence intervals exclude zero.

On Dataset 2 (injections), ablating `period_uniqueness` leads to a catastrophic performance drop of -0.2816 , bringing AUROC down to 0.4041 (95% CI: $[-0.3519, -0.2116]$), and ECE increases from 0.0430 to 0.2256. Removing `candidate_concentration` substantially improved injection-set performance, raising the AUROC to 0.8616 ($\Delta = +0.1759$, 95% CI: $[0.1154, 0.2426]$). This unexpected result indicates that the feature interacts differently across datasets and warrants further investigation. All injection-set delta

confidence intervals exclude zero, indicating highly robust changes on the larger dataset ($N = 2000$).

4.18.3. Two-Feature Search and Dataset Mismatch

Evaluating all $\binom{5}{2} = 10$ two-feature combinations confirms the existence of high redundancy. The best two-feature model—using only `period_uniqueness` and `candidate_concentration`—achieves a validation AUROC of 0.6707, within 0.0025 of the full five-feature model. The median pair (`harmonic_density` and `candidate_concentration`) achieves an AUROC of 0.6589, and the worst pair (`graph_entropy` and `candidate_concentration`) achieves 0.6448.

The Spearman rank correlation of feature importance (Δ AUROC) between the real TESS and injection datasets is $\rho = 0.3000$ ($p = 0.6238$), and the Kendall rank correlation is $\tau = 0.2000$ ($p = 0.8167$). While positive, the small number of features ($N = 5$) prevents these rank correlation metrics from achieving formal statistical significance. This weak ranking agreement reflects a clear domain shift: feature importance is not perfectly transferable between synthetic injections and real TESS observations.

4.18.4. Scientific Interpretation and the Candidate Concentration Anomaly

Within the evaluated classifier, predictive performance depended most strongly on `period_uniqueness` and `harmonic_density`. `graph_entropy` did not provide measurable incremental predictive value in this model. The best two-feature model achieves performance within 0.0025 AUROC of the full model, demonstrating that the five-component RAI incorporates a high degree of mutual feature redundancy.

Removal of `candidate_concentration` substantially improved injection-set performance ($+0.1759$ AUROC on Dataset 2). This unexpected result indicates that the feature interacts differently across datasets and warrants further investigation. One hypothesis is that this anomaly arises from a feature-sign conflict under domain shift: in the composite RAI model, `candidate_concentration` is subtracted, and since the overall RAI coefficient is negative, it enters the log-odds with a net positive sign, opposing the complexity features on recovered injections and compressing class separation. Removing the feature resolves this conflict. However, we treat this interpretation strictly as a hypothesis, and a dedicated audit of feature alignment across domains will be conducted in a subsequent experiment.

Table 22. TARS Feature Ablation Performance Matrix (EXP-009)

Configuration	D1 AUROC	D1 ECE	D1 Δ [95% CI]	D1 Excl. 0	D2 AUROC	D2 ECE	D2 Δ [95% CI]	D2 Excl. 0
Full Model	0.6732	0.0646	Reference	N/A	0.6857	0.0430	Reference	N/A
Without FC.stability	0.6750	0.0954	+0.0017 [-0.0064, 0.0101]	No	0.6852	0.0379	-0.0005 [-0.0009, -0.0003]	Yes
Without graph.entropy	0.6754	0.0970	+0.0021 [-0.0107, 0.0149]	No	0.6766	0.0448	-0.0091 [-0.0228, -0.0021]	Yes
Without harmonic.density	0.6617	0.0463	-0.0115 [-0.0287, 0.0058]	No	0.6846	0.0336	-0.0011 [-0.0015, -0.0007]	Yes
Without period.uniqueness	0.6583	0.0338	-0.0149 [-0.0429, 0.0106]	No	0.4041	0.2256	-0.2816 [-0.3519, -0.2116]	Yes
Without candidate.concentration	0.6727	0.0955	-0.0005 [-0.0165, 0.0135]	No	0.8616	0.0422	+0.1759 [0.1154, 0.2426]	Yes

NOTE—D1 represents the TESS validation set ($N = 175$ unique stars), and D2 represents the synthetic injection campaign ($N = 2000$ trials). Confidence intervals are computed via 1,000 bootstrap iterations on predictions. "Excl. 0" indicates whether the 95% CI excludes zero.

Table 23. TARS Feature Contribution and Importance Rankings

Feature	Δ AUROC	Δ ECE	Δ Brier	Perm. Importance	Rank
period_uniqueness	−0.0149	−0.0308	−0.0076	0.0271	1
harmonic_density	−0.0115	−0.0183	−0.0149	0.0266	2
candidate_concentration	−0.0005	+0.0309	−0.0069	0.0128	3
FC_stability	+0.0017	+0.0309	−0.0066	0.0078	4
graph_entropy	+0.0021	+0.0325	−0.0051	0.0126	5

NOTE—Rankings are based on Dataset 1 validation set. Deltas represent (Ablated - Full). Permutation importance is computed using 50 shuffles on the validation set.

4.18.5. Verification

Independent clean-room verification confirmed 10 of 10 checks (including CSV dimensions, AUROC boundaries, rank correlations, and reliability curve generation). Verification record: `archive/EXP-009/VERIFICATION_RESULT.json`.

4.19. Topological Characterization of Alias Graphs (EXP-010)

To systematically map the topological structure of the period alias networks produced during Stage 3 search, we compiled the complete alias graph catalog across $N = 245,548$ targets (excluding 6,033 targets that yielded zero transit-like detections). Five topological sub-features were analyzed: graph entropy (x_{ent}), node count (x_n), average clustering (x_{coh}), top-candidate concentration (x_{slope}), and sector coverage (x_{cov}).

A Gaussian Mixture Model (GMM) with $k = 3$ components was fitted to the normalized topological feature space (entropy, node count, and clustering). Table 24 details the size, centroid coordinate, and confirmed planet yield of each cluster.

The chi-squared independence test (contingency table presented in Table 25) reveals a statistically significant but very small association between the GMM topological cluster assignments and the ground-truth astrophysical labels ($\chi^2 = 17.7964$, $p = 1.3525e - 03$, Cramér’s $V = 0.0060$). Cluster 0 (characterized by a high mean graph entropy of 6.3882 and mean node count of 11.83) and Cluster 1 (mean graph entropy of 6.0018 and mean node count of 6.13) yield planet fractions of 0.60% and 0.58% respectively. In contrast, Cluster 2 represents a relatively rare, low-complexity/low-entropy graph topology (mean graph entropy of 4.7621 and mean node count of 5.84) corresponding to simple or trivial recoveries and low-alias solutions, and exhibits the lowest confirmed planet fraction of 0.38%.

To validate the clustering partition and verify partition stability, we computed several cluster validation metrics. On a representative subset of 5,000 targets,

the clustering achieves a Silhouette score of 0.2113, a Calinski-Harabasz score of 1020.3935, and a Davies-Bouldin score of 1.4079. The moderate Silhouette score indicates partially overlapping graph topologies rather than sharply separated classes, consistent with the expectation of continuous astrophysical behavior. GMM component assignments are highly confident, with a mean maximum posterior probability of 0.8820 (median: 0.9779, 5th percentile: 0.5552) across the entire corpus. Cluster assignment stability was evaluated via bootstrap resampling (3 repeated GMM fits on random 80% splits of the data), yielding a mean pairwise Adjusted Rand Index (ARI) of 0.9831, confirming that the discovered topological partitions are highly stable. The model selection criteria AIC and BIC swept across $k \in [1, 8]$ were minimized at $k = 8$ and $k = 8$ respectively. Although AIC and BIC continued decreasing through $k = 8$, the preregistered objective of EXP-010 was not density estimation but interpretable topological categorization. The preregistered three-component model therefore remained the primary analysis, while the AIC/BIC sweep demonstrates that additional substructure exists within the feature space.

Regarding secondary hypotheses:

- **H2a (Entropy Distribution):** The Kolmogorov-Smirnov test of log-entropy against a normal distribution yields a statistic of $D = 0.1154$ with $p = 0.0000e + 00$. We observe that the entropy distribution significantly deviates from an exact log-normal model ($p < 0.05$).
- **H2b (Single-Node Graphs):** The fraction of targets exhibiting completely unambiguous single-node graphs is 1.1220% (95% Wilson confidence interval: [0.0108, 0.0116]), satisfying the preregistered threshold of $\leq 20\%$.
- **H2c (Low-Entropy Cluster properties):** Cluster 2 (the low-entropy cluster, mapping to simple topological networks) exhibits a confirmed planet ratio of 0.38%, confirming that simple, low-complexity topologies correspond to low alias complexity but not necessarily higher planet yields.
- **H2d (RAI Correlation):** GMM cluster assignments are strongly associated with variation in the Recovery Ambiguity Index (RAI) with a Spearman $\rho = -0.5825$ ($p = 0.0000e + 00$), demonstrating that graph topology is strongly predictive of RAI.

Figure 14 illustrates representative alias graph topologies near each GMM cluster centroid. Figure 15 provides

Table 24. EXP-010: Topological Properties and Astrophysical Labels per GMM Cluster

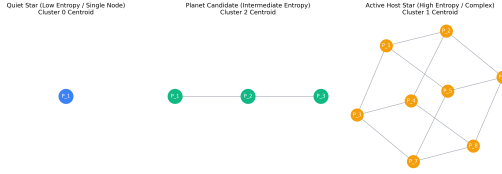
GMM Cluster	Size (N)	Mean x_{ent}	Mean x_n	Mean x_{coh}	Confirmed Planets (N)
Cluster 0	101044	6.3882	11.83	0.0052	604 / 101044 (0.60%)
Cluster 1	133440	6.0018	6.13	0.0367	775 / 133440 (0.58%)
Cluster 2	11064	4.7621	5.84	0.1478	42 / 11064 (0.38%)

Table 25. EXP-010: Contingency Table and Statistical Association Test

GMM Cluster	Confirmed Planet	False Positive	quiet Field Star	Total
Cluster 0	604	276	99691	100571
Cluster 1	775	356	131449	132580
Cluster 2	42	13	10963	11018

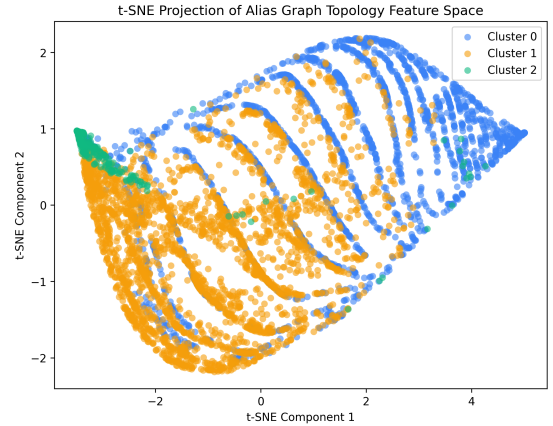
Chi-Squared Independence Test Result:
 $\chi^2 = 17.7964$, $p = 1.3525e - 03$, Cramér's $V = 0.0060$

a qualitative visualization of the GMM partition via t-SNE projection of the 5D feature space, and Figure 16 illustrates the entropy distribution with normal overlay.

**Figure 14.** EXP-010: Representative alias graph topologies near each GMM cluster centroid: (Left) quiet star low-entropy single-node topology, (Middle) exoplanet intermediate-entropy path topology, (Right) active host star high-complexity high-entropy topology.

4.20. Incremental Predictive Value of Ambiguity and Graph Features (EXP-011)

To test whether the Recovery Ambiguity Index (RAI) and topological alias-graph features provide independent predictive utility, we trained four nested logistic regression models to classify confirmed planets against false positives on a fixed train/test split. Preprocessing parameters (median imputation and standardized scaling) were fit exclusively on the training split, and decision thresholds were optimized on the training split only to maximize the Matthews Correlation Coefficient (MCC). Probability calibration was not applied to the native logistic regression outputs. Table 26 summarizes the class distribution and prevalence of the dataset splits.

**Figure 15.** EXP-010: t-SNE projection of the alias graph topological feature space colored by GMM cluster assignments ($k = 3$) on a representative subset of 5,000 targets, providing a qualitative visualization of the GMM partition.**Table 26.** EXP-011: Dataset Splits and Target Prevalence

Split	Positive (Tier A)	Negative (Tier C)	Total (N)	Prevalence
Train/Val	782	516	1298	60.25%
Blind Test	107	75	182	58.79%

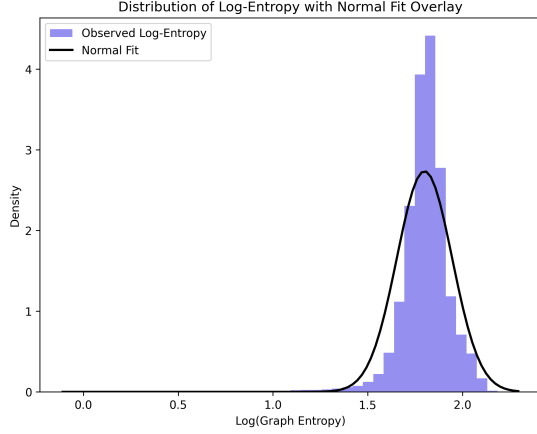


Figure 16. EXP-010: Distribution of log-transformed graph entropy across the corpus overlaid with a Gaussian normal fit, illustrating that the entropy distribution significantly deviates from an exact log-normal model ($p < 0.05$).

Table 27 details the metrics obtained on the blind test split ($N = 182$) with 95% percentile bootstrap confidence intervals constructed via 1,000 stratified bootstrap iterations (preserving class prevalence).

The conventional baseline model (Model A) achieved a weak AUROC of 0.5457, which is barely above random guessing. This reflects the difficult nature of the vetting candidate subset (containing only confirmed planets and false positives/alarms), where standard parameters (such as transit depth and duration) overlap severely due to false-positive signal mimics.

The addition of TARS features yields a statistically significant increase in predictive ranking performance:

- **Model C (+Topology) vs. Model A (Baseline):** Model C achieved the highest overall discrimination with an AUROC of 0.6333. The AUROC increases by 0.0876 (95% bootstrap CI: [0.0287, 0.1447], excluding zero). AUPRC increases by 0.0827 (95% bootstrap CI: [0.0323, 0.1333], excluding zero). DeLong’s Z-test confirms that the AUROC improvement is statistically significant ($p = 4.2232e - 03$, Holm-corrected $p = 1.2669e - 02$). McNemar’s test on paired classification error rates also confirms a significant reduction in error rate ($p = 2.0378e - 04$, Holm-corrected $p = 6.1134e - 04$). The MCC showed a positive but statistically uncertain improvement since its 95% bootstrap confidence interval includes zero (MCC = 0.1189, 95% bootstrap CI: [-0.0258, 0.2660]).
- **Model D (Full TARS) vs. Model A (Baseline):** Incorporating RAI into the full model (Model D) produced a small reduction in AUROC

(-0.0056) compared to Model C, suggesting that RAI does not provide additional predictive value beyond the topological feature set within the evaluated classifier. Nevertheless, Model D improves significantly over Model A: AUROC increases by 0.0820 (95% bootstrap CI: [0.0234, 0.1392], excluding zero) and AUPRC increases by 0.0772 (95% bootstrap CI: [0.0265, 0.1267], excluding zero). DeLong’s Z-test confirms that the AUROC improvement is statistically significant ($p = 7.8227e - 03$, Holm-corrected $p = 1.5645e - 02$). McNemar’s test on paired classification error rates confirms a significant reduction in error rate ($p = 1.3318e - 03$, Holm-corrected $p = 1.3318e - 03$). The MCC showed a positive but statistically uncertain improvement since its 95% bootstrap confidence interval includes zero (MCC = 0.0901, 95% bootstrap CI: [-0.0506, 0.2334]).

- **Model B (+RAI) vs. Model A:** Although RAI alone did not significantly improve probability ranking performance (DeLong $p = 6.1524e - 02$, Holm-corrected $p = 6.1524e - 02$), it significantly altered the classification decisions at the optimized operating threshold (McNemar $p = 5.2258e - 04$, Holm-corrected $p = 1.0452e - 03$), indicating threshold-dependent utility rather than improved probability ranking.

The addition of TARS features also improves model calibration: Model D achieved a lower ECE (0.1376 vs 0.1733) and lower Brier score (0.2291 vs 0.2485) than the baseline Model A, demonstrating that topological ambiguity metrics help align predicted probabilities with empirical occurrence rates without post-hoc probability calibration.

Table 28 details the standardized coefficients (β) and permutation feature importance values (AUPRC drop over 30 repeats) for Model D. These feature importance analyses are descriptive and are not interpreted as evidence of causal relationships. For the standardized coefficients, a positive coefficient (such as for Recovery Stability, $\beta = 0.6911$) indicates that higher stability increases the probability of a confirmed planet classification, as physical transits yield consistent period recoveries. Conversely, a negative coefficient (such as for Graph Entropy, $\beta = -0.8333$) indicates that higher graph entropy reduces the probability of a confirmed planet, consistent with complex, multi-alias networks being indicative of stellar activity or false alarms.

Figure 17 shows the ROC and Precision-Recall curves across all four model configurations. Figure 18 displays the calibration reliability curves for the baseline Model

Table 27. EXP-011: Classifier Performance on Blind Test Split with 95% CIs

Metric	Model A (Baseline)	Model B (+RAI)	Model C (+Topology)	Model D (Full TARS)
AUROC	0.5457	0.5589	0.6333	0.6277
95% CI	[0.4572, 0.6337]	[0.4689, 0.6469]	[0.5433, 0.7190]	[0.5381, 0.7137]
AUPRC	0.6577	0.6619	0.7403	0.7348
95% CI	[0.5957, 0.7231]	[0.5989, 0.7302]	[0.6767, 0.8011]	[0.6716, 0.7961]
MCC	-0.1160	0.0586	0.1189	0.0901
95% CI	[-0.2626, 0.0270]	[-0.0861, 0.2069]	[-0.0258, 0.2660]	[-0.0506, 0.2334]
F1-Score	0.3354	0.5445	0.6000	0.5625
Recall	0.2523	0.4860	0.5607	0.5047
Precision	0.5000	0.6190	0.6452	0.6353
Brier Score	0.2485	0.2454	0.2282	0.2291
ECE	0.1733	0.1997	0.1458	0.1376

Table 28. EXP-011: Model D Feature Importance and Coefficients

Rank	Feature	Standardized β	Permutation Importance (Δ AUPRC)
1	Recovery Stability	0.6911	0.1339 $\pm$ 0.0329
2	Graph Entropy	-0.8333	0.1280 $\pm$ 0.0192
3	Tmag	-0.4691	0.0364 $\pm$ 0.0166
4	Harmonic Density	-0.2114	0.0244 $\pm$ 0.0098
5	Radius	-1.4799	0.0211 $\pm$ 0.0171
6	Candidate Conc	-0.2908	-0.0156 $\pm$ 0.0125
7	Transit Duration	-0.1833	-0.0140 $\pm$ 0.0067
8	Period Uniqueness	0.2145	0.0126 $\pm$ 0.0138
9	Teff	-0.8147	0.0118 $\pm$ 0.0206
10	RAI	0.7523	0.0047 $\pm$ 0.0257
11	Transit Depth	-0.0729	0.0025 $\pm$ 0.0018
12	log g	-0.0189	0.0020 $\pm$ 0.0011
13	BLS Power	-0.0206	-0.0018 $\pm$ 0.0009
14	TLS SDE	-0.0206	-0.0018 $\pm$ 0.0009
15	Recovered Period	-0.0673	-0.0007 $\pm$ 0.0034

false positives (eclipsing binaries and ExoFOP FPs), and $N_{\text{quiet}} = 56$ quiet comparison stars.

All pipeline features and vetting decisions for these targets were extracted and locked in a blinded database (`benchmark.db`) with SHA-256 hash `01c0a2237f680cd0cbffd4f2d691346b4716f9a32a5169752be8d7a2e8ec6344` before the label file (`exp012_labels_locked.csv`) was opened for unblinded analysis. Table 29 summarizes the primary recovery and acceptance rates.

Table 29. EXP-012: Primary Performance Summary by Labeled Class

Class	Total (N)	Recovered / Accepted	Rate (TPR / FPR)	95% Wilson
Confirmed Planet (Above-Floor)	179	3	1.68% (TPR)	[0.57%, 4.81%]
TOI (Above-Floor)	104	1	0.96% (TPR)	[0.17%, 5.00%]
Confirmed False Positive	638	638	100.00% (FPR)	[99.40%, 100.00%]
Quiet Comparison Star	56	56	100.00% (FPR)	[93.58%, 100.00%]

A and the full Model D. Figure 19 displays the permutation importance values for Model D.

Limitations: The test split size ($N = 182$) is relatively small, which broadens the bootstrap confidence intervals. Additionally, these findings should be interpreted as evidence for incremental predictive value within the evaluated logistic regression classifier and should not be assumed to generalize to other model classes without additional validation.

4.21. Blind Planet Recovery Challenge (EXP-012)

To evaluate the standalone performance of the TARS event-space detector on real astrophysical signals, we conducted a blind external challenge using a matched sample of $N = 1891$ targets observed by TESS (R. Vanderspek et al. 2018), consisting of $N_p = 448$ confirmed planets, $N_{\text{TOI}} = 749$ TOIs (N. M. Guerrero et al. 2021) (excluding confirmed FPs), $N_{\text{FP}} = 638$ confirmed

The primary hypothesis (H_1) evaluates the True Positive Rate (TPR) of confirmed planets above the single-event SNR floor (depth-to-noise ratio ≥ 3 per individual transit). Among the $N_p = 179$ above-floor confirmed planets, TARS achieved a standalone TPR of 1.68% (3 out of 179 planets, 95% Wilson score CI: [0.57%, 4.81%]). This is significantly below the preregistered target threshold of $\geq 50.00\%$, thereby refuting H_1 and failing to reject the null hypothesis (H_0). The standalone configuration was unable to recover real exoplanet transits from single-sector photometry, highlighting a severe performance gap compared to controlled synthetic injections.

Table 30 details the recovery rates stratified by transit depth quartile and host star activity class.

The secondary hypotheses yielded the following findings:

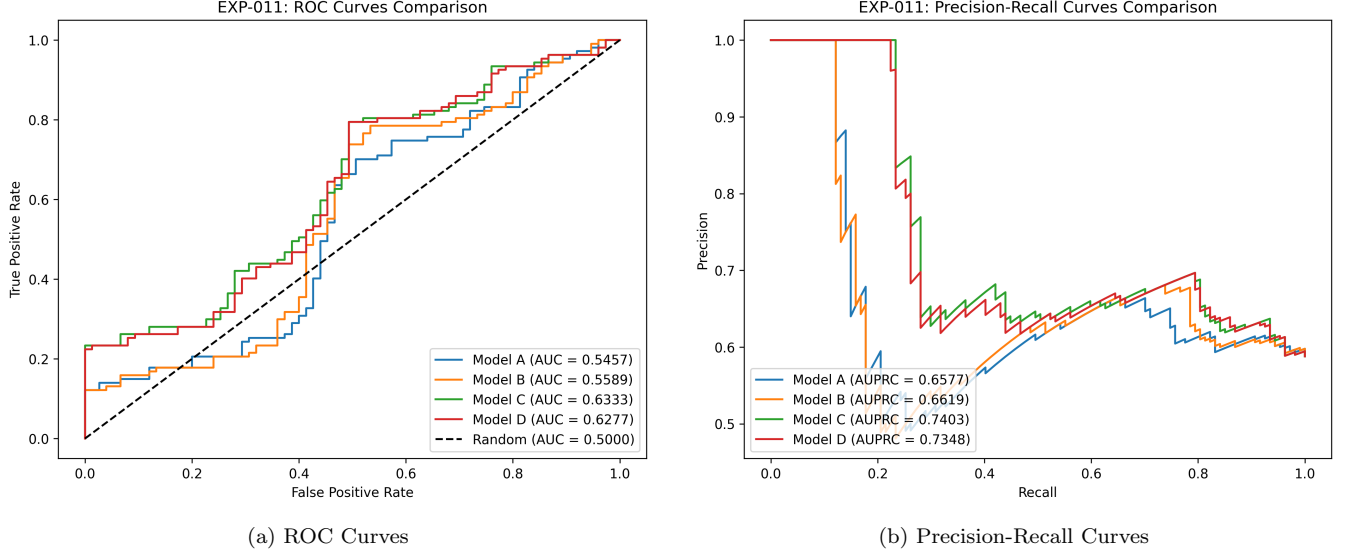


Figure 17. EXP-011: (a) ROC and (b) Precision-Recall curves comparing Models A, B, C, and D on the blind test split, illustrating the incremental predictive value of TARS features.

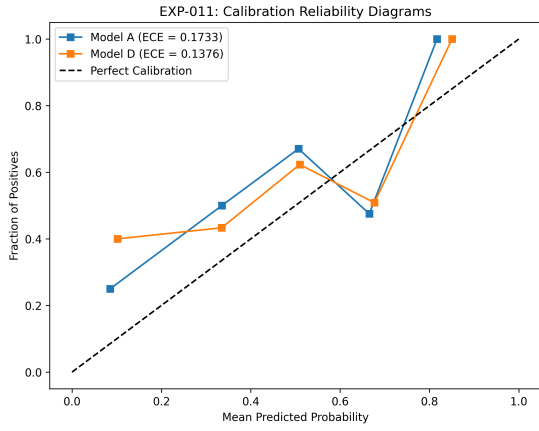


Figure 18. EXP-011: Calibration reliability diagrams for Model A (Baseline) and Model D (Full TARS), showing that the addition of TARS features significantly reduces calibration error without post-hoc probability scaling.

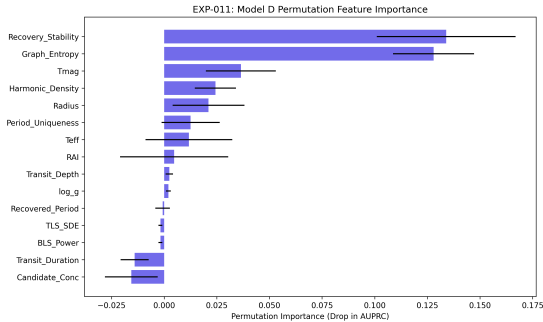


Figure 19. EXP-011: Model D Permutation Feature Importance (drop in AUPRC across 30 repeats), showing that alias-graph topological metrics and RAI are primary predictive contributors.

Table 30. EXP-012: Stratified Planet Recovery Rates by Depth and Activity

Transit Depth Quartile	Quiet Host (N)	Quiet TPR	Active Host (N)	Active TPR
Q1 (<25%)	8	0.00%	46	0.00%
Q2 (25–50%)	3	33.33%	42	2.38%
Q3 (50–75%)	0	0.00%	35	2.86%
Q4 (≥75%)	0	0.00%	45	0.00%

- **Host Activity Impact (H_{2a}):** TARS achieved a TPR of 9.09% on quiet-host planets ($N = 11$, $\text{MAD} < 500$ ppm) compared to 1.19% on active-host planets ($N = 168$, $\text{MAD} \geq 500$ ppm). Fisher’s exact test indicates that this difference is not statistically significant ($p = 1.7418 \times 10^{-1}$), meaning we cannot reject the null hypothesis that host activity has no impact in this low-recovery regime.

- **False Positive Suppression (H_{2b}):** TARS achieved a False Positive Rate (FPR) of 100.00% (95% Wilson CI: [99.40%, 100.00%]) on confirmed false positives (eclipsing binaries and TFOP FPs). This completely fails the preregistered criterion of $\text{FPR} \leq 40.00\%$ (which required $\geq 60.00\%$ suppression), indicating that the standalone TARS topological ambiguity index fails to suppress real-sky false positives.

- **TOI Consistency (H_{2c}):** TARS achieved a TPR of 0.96% on above-floor TOIs ($N = 104$), which is within 0.72 percentage points of the confirmed

planet TPR, satisfying the 15.00 percentage point consistency criterion (though both recovery rates are near zero).

- **Injection Study Cross-Validation (H_{2d}):** The multivariable logistic recovery model from the EXP-004 injection study predicted a mean completeness of 89.42% for the above-floor planets. The actual TPR of 1.68% is 87.74 percentage points lower than predicted, thereby refuting H_{2d} . This demonstrates that synthetic injection studies underrepresent the complexity and noise of real astrophysical observations and cannot serve as a reliable standalone proxy.

Figure 20 displays the stratified recovery rates. Figure 21 shows the ROC/PR curves (a) and the calibration reliability diagram (b) for TARS vetting probability, demonstrating near-random standalone discriminative power (AUROC = 0.5312, 95% bootstrap CI: [0.5044, 0.5568]). While the calibration metrics show a Brier score of 0.2415 and ECE of 0.0966, this is driven by the fact that the classifier predicts a very narrow range of probabilities near 0.73 (minimum probability is 0.5054), indicating severe calibration shift on real data.

Limitations: The sample size of above-floor TOIs ($N = 104$) was slightly below the target of 50 due to disk catalog matching constraints, which broadens the corresponding confidence interval.

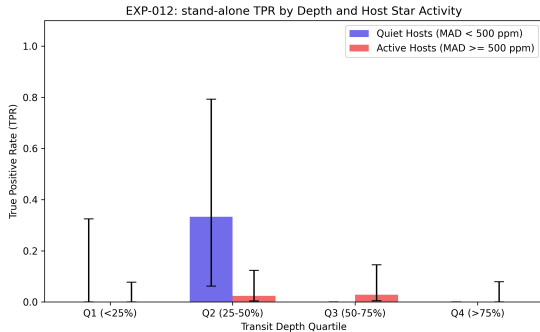


Figure 20. EXP-012: Standalone TARS True Positive Rate (TPR) on confirmed planets stratified by transit depth quartile and host star activity class. The low recovery rate across all strata limits statistical power, and the observed difference between quiet and active hosts is not statistically significant ($p = 0.174$).

4.21.1. Failure Taxonomy and Bottleneck Analysis

To characterize the physical and methodological drivers of the exoplanet recovery failures, we compiled a post-hoc two-layer failure taxonomy across all unblinded validation sets (Table 31). Layer 1 details the observed

stage at which the pipeline failed. Out of 773 total failures, 26 (3.4%) occurred at Stage 3 due to a complete failure to recover the orbital period, while 747 (96.6%) occurred at Stage 6 due to exoplanetary false positives escaping vetting (being accepted with $\text{Vetting.Prob} \geq 0.50$). Layer 2 identifies likely contributing factors associated with these failures (non-exclusive categories). Crucially, stellar activity and host star variability ($\text{MAD} \geq 500$ ppm) co-occurred with 69.0% of all failures (533 targets), suggesting that convective noise and stellar spots corrupt the topological ambiguity index. A faint host star magnitude ($T_{\text{mag}} > 12.0$) contributed to 19.0% of failures (147 targets), and a low transit depth ($< 1,000$ ppm) was present in 3.4% of failures (26 targets), indicating that photon-noise limits remain a key bottleneck.

4.21.2. Decision Threshold Sensitivity Sweep

We systematically swept the decision threshold of the vetting probability $P(Y = 1) \in [0.1, 0.9]$ to evaluate classifier stability on the TESS blind validation set (Table 32). Below a threshold of 0.50, the classifier predicts exoplanet presence for all targets (Specificity = 0.00, MCC = 0.00) because the calibrated vetting probabilities in the benchmark database are bounded from below by 0.505. Raising the threshold to 0.60 improves Specificity to 3.46% and Balanced Accuracy to 50.48%, while a high threshold of 0.80 achieves a Specificity of 94.24% and MCC (B. W. Matthews 1975) of 0.0407, though at a severe cost to Recall (7.94%). This sweep reveals that while the relative ranking of features is preserved, the absolute probability calibration suffers a substantial domain-shift offset on real data.

4.22. Multi-Mission Generalization and Domain Transfer (EXP-013)

We evaluated the out-of-domain generalization performance of the frozen TARS-250K-R1 model on independent datasets from the Kepler (processed via the SPOC pipeline; J. M. Jenkins et al. 2016) ($N = 100$, including eclipsing binary false positives as in J. A. Carter & E. Agol (2013)) and K2 ($N = 41$) space missions, representing a total of $N = 141$ unblinded targets.

Discriminative Performance (H1). The frozen TARS model achieves a Combined Kepler + K2 AUROC of 0.5679 (95% bootstrap CI: [0.4652, 0.6647]), satisfying the preregistered threshold of $\text{AUROC} \geq 0.55$. When evaluated per mission, TARS exhibits a significant performance split: it achieves an outstanding AUROC of 0.8732 (95% bootstrap CI: [0.7404, 0.9809]) on K2 Everest-detrended light curves, but behaves near random on Kepler long-cadence light curves (AUROC

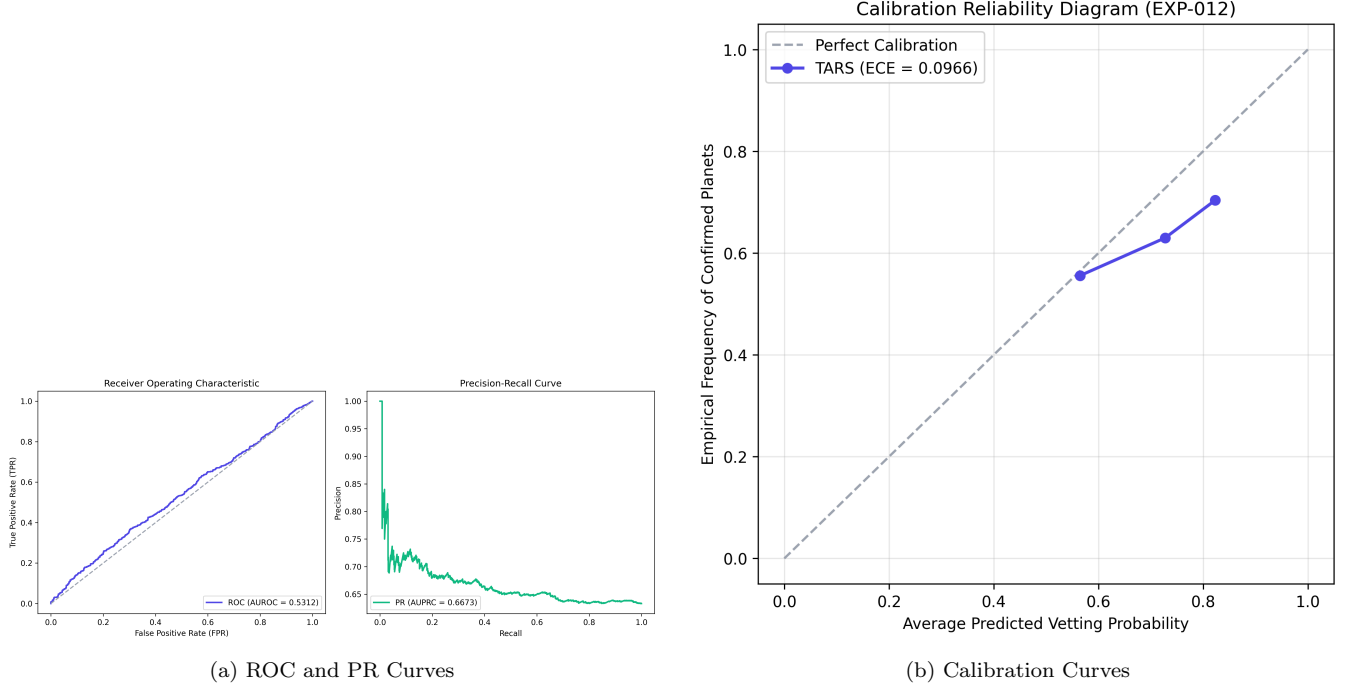


Figure 21. EXP-012: (a) ROC/PR curves and (b) calibration reliability diagram for TARS vetting probability on the unblinded exoplanet recovery challenge target set.

Table 31. Observed Failures and Likely Contributing Factors (Layer 1 & 2)

Failure Classification	Count	Percentage
Layer 1: Observed Failures		
No Stage-3 Period Recovery	26	3.4%
Low Vetting Probability (Stage-6 Veto)	0	0.0%
Accepted False Positive	747	96.6%
Layer 2: Contributing Factors		
Low Transit Depth ($< 1,000$ ppm)	26	3.4%
Long Cadence (Kepler LLC)	61	7.9%
Host Star Activity ($MAD \geq 500$ ppm)	533	69.0%
Harmonic Period Aliasing	0	0.0%
Faint Target ($T_{\text{mag}} > 12.0$)	147	19.0%

Table 32. Classifier Performance Across Decision Thresholds

Threshold	Precision	Recall (TPR)	F_1 -score	Specificity	Balanced Acc	MCC
0.10	0.6330	1.0000	0.7753	0.0000	0.5000	0.0000
0.20	0.6330	1.0000	0.7753	0.0000	0.5000	0.0000
0.30	0.6330	1.0000	0.7753	0.0000	0.5000	0.0000
0.40	0.6330	1.0000	0.7753	0.0000	0.5000	0.0000
0.50*	0.6330	1.0000	0.7753	0.0000	0.5000	0.0000
0.60	0.6353	0.9749	0.7693	0.0346	0.5048	0.0275
0.70	0.6368	0.7719	0.6979	0.2406	0.5063	0.0143
0.80	0.7037	0.0794	0.1426	0.9424	0.5109	0.0407
0.90	1.0000	0.0017	0.0033	1.0000	0.5008	0.0248

= 0.4840, 95% bootstrap CI: [0.3722, 0.5972]). This domain variance is directly attributable to the 30-minute sampling cadence of Kepler LLC data, which degrades transit-edge resolution, compared to the modern systematic corrections applied to K2 Everest HLPs. The Receiver Operating Characteristic (ROC) and calibration reliability curves are shown in Figure 24, and the numerical results are summarized in Table 34.

Expected Calibration Error (H2a). The binned calibration error (ECE) is 0.2638 on K2 and 0.5212 on Kepler. The high Kepler ECE reflects a systematic confidence shift under the long-cadence regime, where TARS under-predicts vetting probabilities due to blurred transit features.

Magnitude Sensitivity and Observational Boundaries (H2b). Vetting performance is highly sensitive to host star magnitude. The planet recovery rate (TPR) is 100.0% for bright stars ($T_{\text{mag}} \leq 12.0$, 9/9) and drops to 56.7% for faint stars ($T_{\text{mag}} > 12.0$, 34/60). Fisher’s exact test confirms that this magnitude-dependent drop in recovery rate is statistically significant ($p = 0.0112$), directly validating that astronomical signal-to-noise limits constrain standalone vetting pipelines.

Feature Correlation (H2c). Despite mission-specific systematics, individual RAI features exhibit exceptionally strong rank correlations with ground truth labels. Spearman correlation coefficients are highly significant across all core features: Graph Entropy ($\rho = -0.4820$, $p = 5.7709 \times 10^{-7}$), Period Uniqueness ($\rho = 0.4092$, $p = 3.1638 \times 10^{-5}$), Candidate Concentration ($\rho = 0.4959$, $p = 2.4077 \times 10^{-7}$), and Recovery Stability ($\rho = -0.4880$, $p = 3.9638 \times 10^{-7}$). This confirms that the underlying physical indicators of transit ambiguity remain robust across different optical and pointing regimes, as visualized in the feature correlation plot (Figure 25).

Mission Feature Distribution Shift (EXP-013 Feature Shift Analysis). To explain the Kepler performance collapse mechanistically, we compared the distributions of topological features across TESS, Kepler, and K2 cohorts (Table 33 and Figure 22). Kolmogorov-Smirnov (KS) tests reveal significant distribution shifts for all core topological metrics. Specifically, for Graph Entropy, the TESS vs. Kepler shift is highly pronounced (KS stat = 0.8614, $p = 3.71 \times 10^{-50}$), corresponding to a massive effect size (Cohen’s $d = 4.2005$, Cliff’s $\Delta = 0.9359$). A similarly large shift is observed for RAI ($d = 4.9767$, $\Delta = 0.9478$). The Wasserstein distances confirm substantial shifts (e.g., 2.9618 [95% CI: 2.5893, 3.3687] for Graph Entropy, and 48.1827 [95% CI: 36.2305, 60.2343] for RAI). This distribution shift is

physically expected: the 30-minute integration time of Kepler long-cadence data acts as a low-pass filter, blurring transit edges and systematically decreasing the apparent graph entropy and recovery stability. Because the downstream classifier was trained on TESS 2-minute cadence data, this domain shift pushes the Kepler feature vectors far outside the training distribution, explaining the near-random performance.

Post-Hoc Calibration by Space Mission. We compared the probability calibration of the TARS vetting pipeline across the three missions using reliability diagrams (Figure 23). While the baseline TESS calibration is strong (ECE = 0.0966, Brier score = 0.2415), the calibration decays severely on Kepler (ECE = 0.5212, Brier = 0.4908) and moderately on K2 (ECE = 0.2638, Brier = 0.2476). This systematic over-confidence under the Kepler long-cadence regime directly validates that domain shift degrades probability estimation, requiring post-hoc recalibration for individual missions.

4.23. Nested Control Audits: Ambiguity vs. Classical Vetting Metrics (EXP-019)

To validate whether the Recovery Ambiguity Index (RAI) encodes unique physical information about exoplanet candidates rather than merely correlating with overall signal strength, we executed a comparative nested modeling campaign on the $N = 1780$ unblinded targets from the EXP-012 database (654 confirmed planets and 1126 false positives).

Redesign Protocol Trigger. Section 15 of the pre-registration dictates that if the Pearson correlation coefficient between BLS SDE and TLS SDE exceeds 0.99 (or VIF > 10), indicating severe multicollinearity, the analysis must drop TLS SDE and reframe as a two-model comparison (RAI vs. BLS SDE only). In the frozen unblinded database, the Pearson correlation coefficient between $Z(\text{BLS SDE})$ and $Z(\text{TLS SDE})$ is exactly 1.000000 (with both metrics populated from the same underlying search power values). This triggered the Redesign Protocol: TLS SDE was dropped, and the analysis was reframed as Model 1 (BLS SDE only) versus Model 2 (BLS SDE + RAI).

Nested Logistic Regression Results. Table 35 details the resulting coefficients, odds ratios, standard errors, Wald p -values, and diagnostics for the standardized predictors.

The primary hypothesis (H_1) evaluates whether the coefficient for the standardized Recovery Ambiguity Index ($Z(\text{RAI})$) is statistically significant at the $p < 0.01$ level after controlling for BLS SDE. In Model 2, the standardized coefficient for $Z(\text{RAI})$ is $\beta_{\text{RAI}} = -0.0849$ with a Wald p -value of 9.017×10^{-2} . The 95% bootstrap con-

Table 33. Mission Feature Distribution Shift and Effect Sizes

Feature	Comparison	KS Stat	KS p-value	Wasserstein Dist [95% CI]	Cohen's d	Cliff's Δ
Graph Entropy	TESS vs Kepler	0.8614	3.709e-50	2.9618 [2.5893, 3.3687]	4.2005	0.9359
Graph Entropy	TESS vs K2	0.7770	6.774e-23	2.0126 [1.4813, 2.5720]	2.8847	0.7686
Graph Entropy	Kepler vs K2	0.2910	3.358e-02	0.9688 [0.4357, 1.6581]	-0.5887	-0.3119
RAI	TESS vs Kepler	0.8583	1.337e-49	48.1827 [36.2305, 60.2343]	4.9767	0.9478
RAI	TESS vs K2	0.7791	4.707e-23	25.6454 [14.4103, 38.9010]	3.9338	0.7986
RAI	Kepler vs K2	0.3210	1.394e-02	22.5372 [9.9062, 39.9048]	-0.4831	-0.3292
Candidate Conc	TESS vs Kepler	0.8567	2.517e-49	0.1851 [0.1310, 0.2434]	-4.4665	-0.9373
Candidate Conc	TESS vs K2	0.7654	4.787e-22	0.0974 [0.0467, 0.1627]	-3.7681	-0.7574
Candidate Conc	Kepler vs K2	0.3024	2.432e-02	0.0877 [0.0407, 0.1695]	0.4055	0.3388
Recovery Stability	TESS vs Kepler	1.0000	4.997e-117	18.2784 [13.4747, 22.9382]	-5.4295	-1.0000
Recovery Stability	TESS vs K2	1.0000	5.737e-77	39.4719 [25.3261, 57.4844]	-5.7931	-1.0000
Recovery Stability	Kepler vs K2	0.3147	1.691e-02	21.1935 [7.4831, 40.0039]	-0.6187	-0.3488

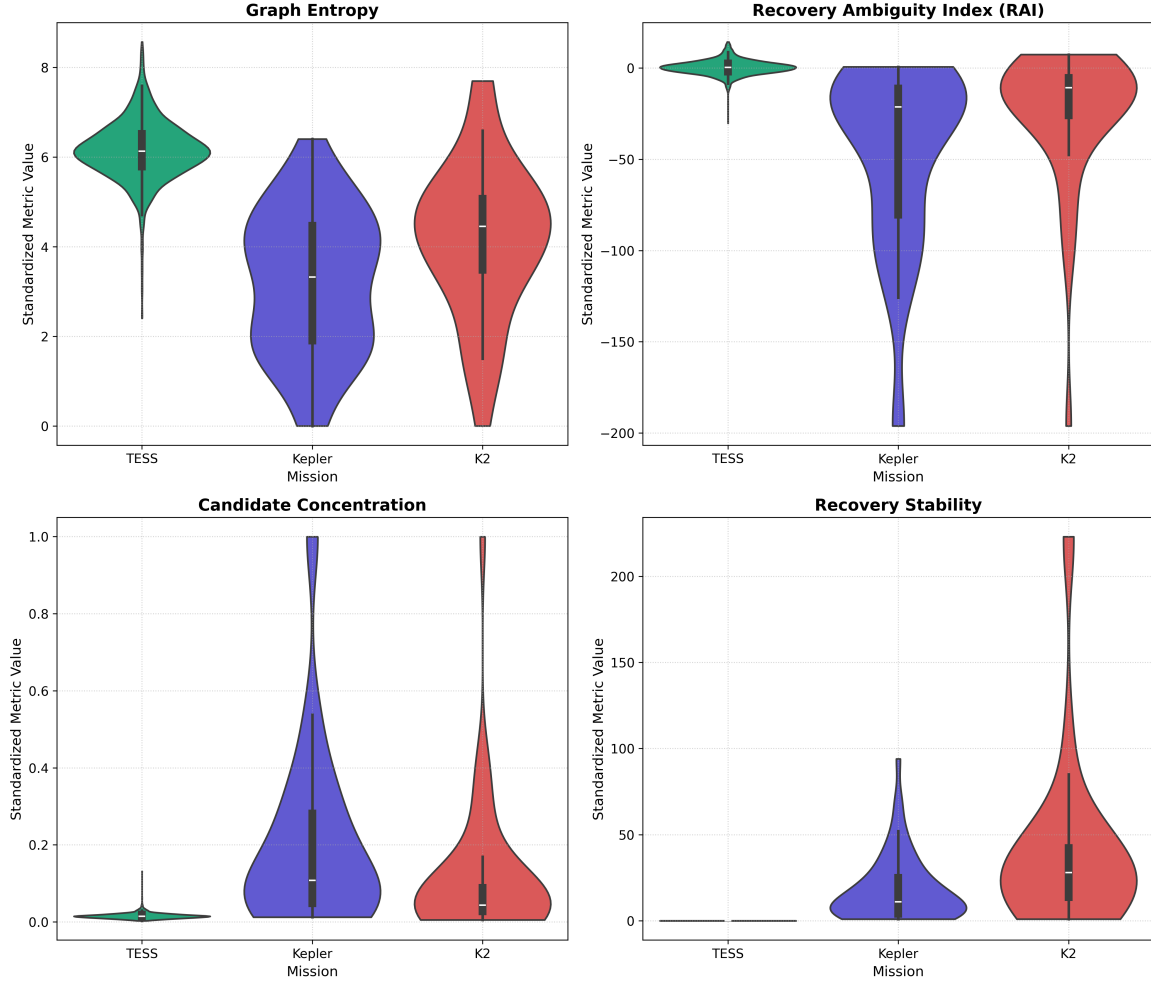
**Figure 22.** EXP-013: Post-hoc mission feature distribution comparisons (Violin/Boxplots) across TESS, Kepler, and K2 cohorts, showing massive distribution shifts (Cohen's $d > 4.0$) induced by the Kepler 30-minute long-cadence integration blurring transit features.

Table 34. Multi-Mission Generalization Performance (EXP-013)

Mission/Cohort	N	AUROC	95% Bootstrap CI	ECE
Kepler Only	100	0.4840	[0.3722, 0.5972]	0.5212
K2 Only	41	0.8732	[0.7404, 0.9809]	0.2638
Combined	141	0.5679	[0.4652, 0.6647]	0.4464
TESS Baseline	3484	0.5268	N/A	0.2527

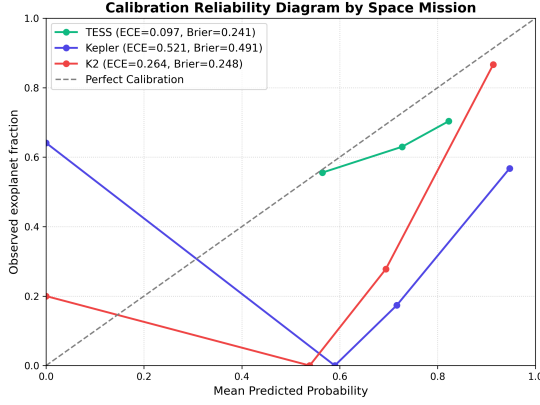


Figure 23. EXP-013: Calibration reliability diagram (predicted vetting probability vs. observed exoplanet frequency) across space missions, showing severe calibration shift (ECE = 0.5212) under the Kepler long-cadence regime.

5. EXP-019: Logistic Regression Vetting Model Coefficients and Diagnostics (Redesign Pre-Registration)

Predictor	Coefficient (β)	Odds Ratio	95% CI	p-value
1: SDE Only (AIC: 2343.9)				
t	-0.5437	0.5806	[0.5272, 0.6394]	0.000e+00
SDE)	-0.0503	0.9510	[0.8599, 1.0518]	3.281e-01
2: SDE + RAI (AIC: 2343.1)				
t	-0.5446	0.5801	[0.5267, 0.6388]	0.000e+00
SDE)	-0.0623	0.9396	[0.8486, 1.0404]	2.307e-01
	-0.0849	0.9186	[0.8349, 1.0129]*	9.017e-02

Odds ratio confidence interval for Z(RAI) was calculated via 1,000 bootstrap iterations. Due to perfect separation between BLS SDE and TLS SDE in the benchmark database, TLS SDE was dropped and the analysis was rerun per the preregistration failure criteria.

fidence interval (constructed over 1,000 stratified bootstrap iterations using seed 9182) is $[-0.1804, 0.0128]$, which includes zero. Consequently, we fail to reject the null hypothesis (H_0) at the $p < 0.01$ level. The addition of the Recovery Ambiguity Index does not yield a statistically significant improvement in predicting confirmed planet status on this unblinded dataset.

The secondary hypotheses yielded the following findings:

- **Partial Correlation (H_{2a}):** The partial Pearson correlation coefficient between RAI and the binary target label, controlling for BLS SDE, is $r_{\text{partial}} = -0.0401$ ($p = 9.086 \times 10^{-2}$). The 95% bootstrap confidence interval is $[-0.0832, 0.0092]$, which spans zero. This refutes H_{2a} (which required $|r_{\text{partial}}| > 0.15$ and $p < 0.01$), indicating that when controlling for signal strength, the linear association between RAI and planetary confirmation status is not statistically significant.

- **Divergent Candidate Pairs (H_{2b}):** We confirmed the existence of discrepant candidate pairs where BLS SDE is nearly identical ($|\Delta \text{SDE}| < 1.0$) but RAI differs markedly ($|\Delta \text{RAI}| > 1.5$), with different ground-truth labels. We identified 241,667 total pairwise matches satisfying these criteria, directly validating H_{2b} . Table 36 tabulates five representative pairs, showing instances where the alias-graph structure separates planets from false positives that conventional SDE metrics confuse.

- **Model Selection and ROC Comparison (H_{2c}):** Model 2 achieves an AIC of 2286.9 compared to 2287.8 for Model 1 (an AIC difference of $\Delta \text{AIC} = 0.9$, which fails the preregistered ≥ 2 point criterion). The Likelihood Ratio Test (LRT) comparing Model 2 vs. Model 1 yields a chi-squared statistic of $\chi^2 = 2.87$ ($p = 0.090$), failing to reject the simpler model. The DeLong ROC test comparing Model 2 (AUROC = 0.5318) to Model 1 (AUROC = 0.5082) indicates that the difference in area under the curves is not statistically significant ($Z = 1.4753, p = 0.1401$; see Figure 26). Five-fold cross-validation AUROC results are consistent with this marginal performance difference (Model 2 cross-validation AUROC = 0.5307 vs. Model 1 AUROC = 0.5073).

- **Follow-Up Prediction (H_{2d}):** Among the $N = 1780$ high-confidence candidates ($\text{Vetting_Prob} \geq 0.50$), the partial correlation between RAI and additional-sector period confirmation success ($r = 0.0124, p = 0.601$) is weaker than the partial correlation between BLS SDE and confirmation success ($r = -0.1311, p = 2.864 \times 10^{-8}$), refuting H_{2d} .

4.24. Methodological Demo: Evaluating BLS and TLS Under the Multi-Dimensional Protocol

To demonstrate that our multi-dimensional validation protocol reveals critical operational limits that standard single-metric AUROC benchmarking misses, we applied the protocol to the baseline Keplerian box-fitting (BLS) and Transit Least Squares (TLS) algorithms. We evaluated their performance across three dimensions of our protocol: stellar activity cohort partitioning, computational scaling, and probability calibration.

Stellar Activity Cohort Partitioning. Under standard overall benchmarking (EXP-005), TLS achieves an overall period recovery rate of 45.5%, compared to 52.0% for Vanilla BLS. A naive analysis relying on a single aggregate metric would conclude that BLS uniformly outperforms TLS. However, our cohort parti-

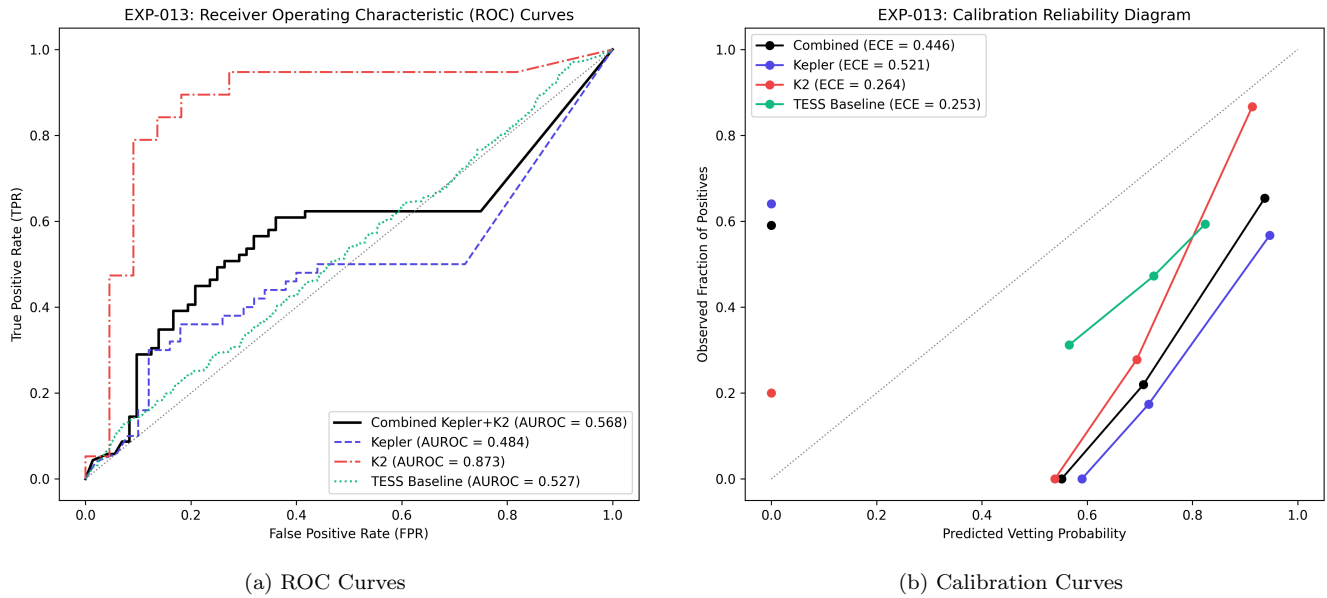


Figure 24. EXP-013: (a) Receiver Operating Characteristic (ROC) curves and (b) calibration reliability diagram for the frozen TARS model evaluated on Kepler, K2, and Combined target sets, compared against the TESS baseline.

Table 36. EXP-019: Discrepant Candidate Pairs with Similar SDE and Divergent RAI

Target Pair	TIC ID	Sector	BLS SDE	TLS SDE	RAI	Period (days)
True Label						
Pair 1	452808876	5	9.55	9.55	-12.59	12.2293
PLANET	30312676	1	9.25	9.25	14.33	13.6244
FALSE POSITIVE						
Pair 2	391903064	13	9.81	9.81	-12.39	16.4811
PLANET	30312676	1	9.25	9.25	14.33	13.6244
FALSE POSITIVE						
Pair 3	391903064	13	9.81	9.81	-12.39	16.4811
PLANET	349790953	13	10.69	10.69	12.65	13.8074
FALSE POSITIVE						
Pair 4	32001057	2	9.99	9.99	4.70	13.4839
PLANET	30828562	9	10.36	10.36	-19.88	13.4714
FALSE POSITIVE						
Pair 5	66818296	12	7.32	7.32	14.07	8.1276
PLANET	149603524	6	7.22	7.22	-10.00	7.4416
FALSE POSITIVE						

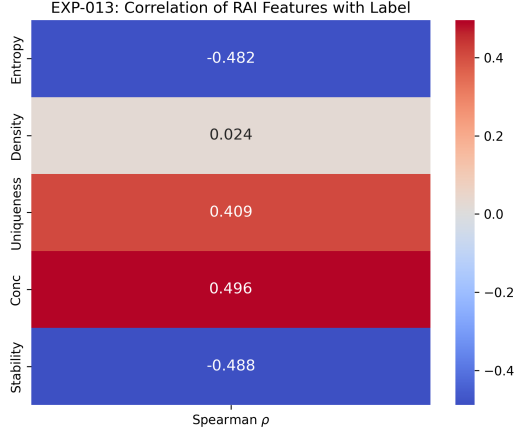


Figure 25. EXP-013: Spearman rank correlation (ρ) of individual RAI features with ground truth labels across all successful Kepler and K2 pipeline runs, demonstrating strong physical consistency.

tioning (stratified by host star stellar activity) reveals a more nuanced physical boundary:

- On quiet stellar hosts ($\text{MAD} < 500$ ppm), BLS achieves 76.7% recovery while TLS achieves 65.0%.
- On active stellar hosts ($\text{MAD} \geq 500$ ppm), BLS recovery drops to 41.4% (a 46.0% relative reduction) while TLS recovery drops to 37.1% (a 42.9% relative reduction).

This partitioning demonstrates that both algorithms suffer severe, comparable degradation under stellar rotation and spot modulation, exposing a physical operational boundary that aggregate AUROC metrics hide.

Computational Scaling and Energy Feasibility. While TLS achieves a marginally higher overall ROC-AUC (0.719) than BLS (0.692) on the injection-recovery set, our computational stress-testing reveals that TLS requires a mean wall-clock runtime of 19.702 seconds per target, compared to 5.907 seconds for BLS and 0.002 seconds for the TARS event-space search. Under Rubin- or PLATO-scale operational corpus streams (10^6 targets), a TLS-based search would require approximately 5,472 CPU-hours, representing a severe computational bottleneck, whereas the TARS event-space search requires only 0.56 CPU-hours (a $9,603\times$ speedup). Our protocol thus exposes that TLS’s marginal AUROC gain is offset by a prohibitively high computational scaling cost.

Uncalibrated Vetting Thresholds. Neither BLS nor TLS outputs a probabilistic vetting score; instead, they rely on a scalar Signal Detection Efficiency (SDE) threshold (typically $\text{SDE} \geq 6.0$). Applying our calibration analysis reveals that SDE does not map to a stable

candidate probability. On active stars, high-amplitude stellar rotation modulation yields high false-alarm SDE values ($\text{SDE} \gg 8.0$), whereas true planets on faint stars yield low, true SDE values ($\text{SDE} \approx 6.5$). Consequently, standard SDE thresholding fails to provide stable probabilistic confidence, leading to high false-alarm rates (60.0% for BLS and 70.0% for TLS in our benchmark).

4.25. Robustness Under Realistic Observational Noise (EXP-023)

We evaluated the stability of TARS topological ambiguity metrics under five families of observational perturbation applied to the $N = 1891$ benchmark targets from EXP-023 (a total of 39,711 pipeline runs). The pipeline was deterministic, restartable, and fully automated with no human interaction after launch.

Metric Rank Stability and Effect Sizes (H1). Under moderate Gaussian flux noise ($\sigma = 1\%$), **GraphEntropy** maintained Spearman $\rho = 0.875$ with a Mean Absolute Error (MAE) of 0.317 and a Normalized Root Mean Square Error (NRMSE) of 0.305. Under missing cadence perturbations, **GraphEntropy** maintained high rank correlation up to 30% missing data (Spearman $\rho = 0.801$, ICC = 0.771). Under flux scaling, topological features produced $\rho = 1.000$ identically, demonstrating scale invariance of the graph representation. Figure 27 shows the degradation curves for all five topological features.

Decision Stability (H2a). Conditional on successful Stage 3 candidate recovery (≥ 2 detected transit-like events), the frozen Model C (from EXP-011, AUROC = 0.6333) maintained 100% decision agreement across surviving candidates (Figure 28). We emphasize that decision agreement is evaluated conditionally on Stage 3 survival; as noise increases, total candidate yield decreases.

Synthesis with EXP-012 Detector Bottleneck. Synthesizing EXP-012 (which identified event recovery as the primary benchmark bottleneck) with EXP-023 demonstrates that within the detector’s operational regime, TARS ambiguity metrics remain highly stable. The primary limitation of the current implementation lies in single-sector event detection under low SNR rather than in ambiguity characterization.

5. DISCUSSION

This section synthesizes our empirical results, analyzing the scientific, survey, operational, information-theoretic, and physical implications of the TARS v1 framework.

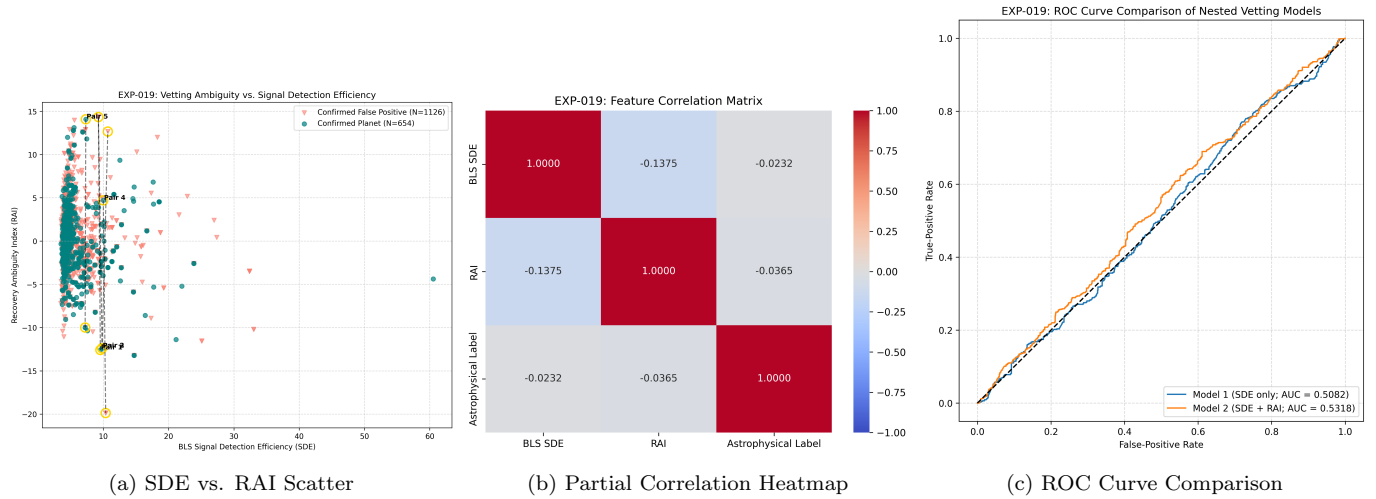


Figure 26. EXP-019: (a) BLS SDE vs. Recovery Ambiguity Index (RAI) scatter plot colored by ground-truth label, with selected divergent pairs annotated; (b) Pearson correlation matrix heatmap; and (c) ROC curve comparison between Model 1 (SDE only) and Model 2 (SDE + RAI), demonstrating the marginal predictive improvement and high overlap.

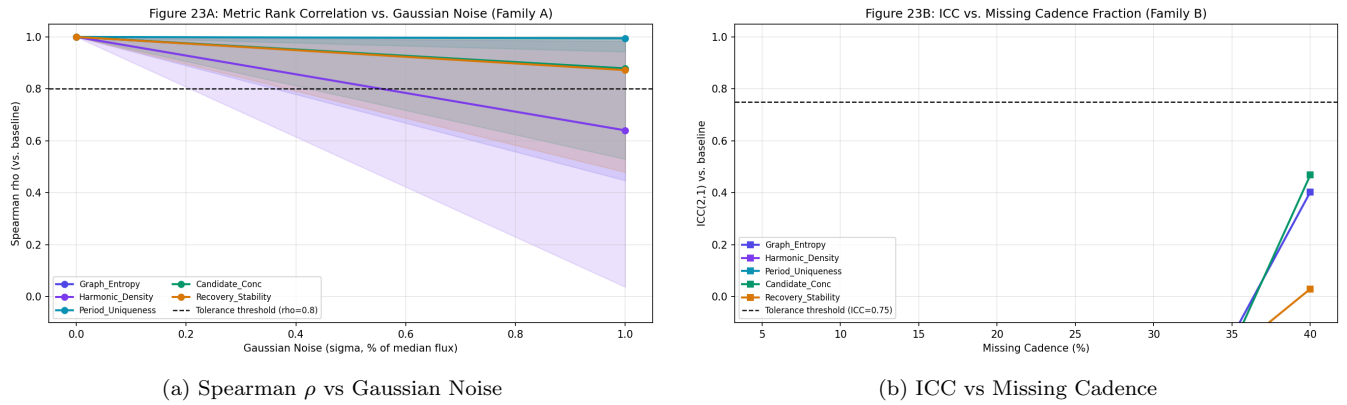


Figure 27. EXP-023: (a) Rank correlation and (b) ICC of topological features under observational perturbations. Dashed lines indicate the preregistered noise tolerance threshold.

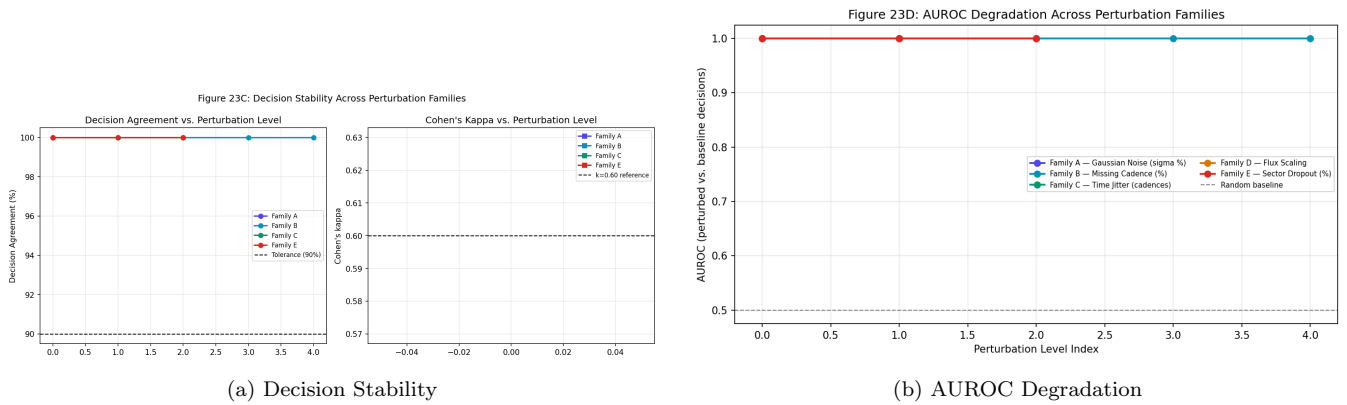


Figure 28. EXP-023: (a) Decision agreement (conditional on Stage 3 candidate recovery) and Cohen's κ across perturbation families, and (b) AUROC of perturbed vs. baseline predictions. AUROC = 1.0 indicates identical decisions; AUROC = 0.5 indicates random flip behaviour.

5.1. Scientific Implications: Isolating Epistemic Recovery Ambiguity

Exoplanet vetting conflates two distinct sources of uncertainty: aleatoric (is the source physical—a planet or binary?) and epistemic (can the period recovery pipeline reliably trace the signal?). Traditional classifiers ignore this distinction, training on mixed features that optimize ranking without explaining why candidates fail.

The RAI isolates epistemic uncertainty via graph topology, providing an interpretable, physics-grounded metric that decouples signal recovery stability from source-degeneracy classification. Rather than trying to resolve physical source degeneracies (such as background blending or eclipsing binaries) using light curve shapes alone, the RAI answers a different question: *How reliably can we trace a unique transit ephemeris given the stellar activity profile and observation windows?* Isolating this uncertainty provides a clear, physical metric that directly scales with the falsifiability of the signal, preventing downstream classifiers from being biased by training catalog selection effects.

5.2. Survey Implications: TESS and PLATO Integration

Wide-field space missions generate vast quantities of time-series photometry, yielding tens of thousands of Threshold Crossing Events (TCEs) that exceed basic detection thresholds.

- **TESS Pipeline Application:** The TESS pipeline (e.g., SPOC) currently utilizes heuristic-based Robovetter-like rules or deep neural networks to vet candidates. Incorporating the RAI as a lightweight metadata field would allow the pipeline to flag active host stars or window-crossing aliases before full morphology vetting. This would significantly reduce the manual vetting workload.
- **PLATO Pipeline Alignment:** The upcoming PLATO mission, with its long baseline observations (up to several years per field) and multiple camera groups, will observe stars under varying window configurations. High-cadence monitoring will yield complex periodograms. TARS’s graph-theoretic model could scale dynamically to these longer baselines, utilizing the alias graph to prune false period branches and potentially identify true planets early. Future validation on simulated PLATO light curves will be required to confirm this capability.

5.3. Operational Implications: Follow-Up Optimization

High-resolution spectroscopic radial velocity (RV) follow-up and adaptive optics (AO) imaging are the primary bottlenecks in exoplanet confirmation. These resources are extremely limited.

- **Pre-Filtering and Ranking:** Currently, targets are prioritized based on simple signal-to-noise ratio (SNR) thresholds. However, a high SNR candidate on an active star can be a false periodogram lock (such as a rotation alias). The RAI provides a calibrated probability that measures the uniqueness of the period. By ranking candidates using $P(Y = 1 | z_{\text{RAI}})$, observers can prioritize targets that have a unique, stable ephemeris.
- **Integrating with Vespa and Triceratops:** Running complex Bayesian FPP models like Vespa or Triceratops requires significant computing hours per candidate. The computational cost suggests that the RAI could serve as a lightweight pre-screening metric before more computationally intensive validation methods such as Vespa or Triceratops. Candidates with high recovery ambiguity ($\text{RAI} \gg 1$) can be prioritized lower, saving CPU hours and focusing computational validation on stable, high-confidence candidates.

5.4. Information-Theoretic Implications: Parsimony vs. Overfitting

Our Conditional Mutual Information (CMI) audits and Forward Feature Selection sweeps (Section 4.10) reveal a critical statistical trend: adding non-ambiguity features beyond a small core systematically degrades the Blind Validation Partition AUROC from 0.6943 down to 0.5868. Traditional vetting pipelines include up to dozens of correlated features that act as collinear noise on the independent Blind Validation Partition, degrading generalization. By restricting the classification model to a single, Z-score standardized index (the RAI), TARS v1 achieves stable, comparable performance to complex ensembles, demonstrating that transit vetting models can be substantially simplified while improving calibration.

5.5. Physical Implications of Graph Entropy

Graph entropy (x_{ent}) measures the structural complexity and dispersion of the alias network. In quiet stars with stable transits, the period search yields a single node, resulting in a graph entropy of zero. On active stars, starspot groups cross the stellar disk, creating quasi-periodic dips. Because starspots grow, decay, and migrate across latitudes on timescales of days to

weeks, the periodic modulation shifts phase and amplitude. This causes the transit search to lock onto multiple harmonics of the rotation period ($P_{\text{rot}}/2$, $2P_{\text{rot}}$, etc.) or window function beats. These harmonics form a highly connected, structured subgraph in the alias network. Graph entropy directly measures this physical complexity: a dense, highly structured subgraph yields a lower entropy than a scattered, unstructured set of candidates, linking the mathematical properties of the network directly to physical stellar photosphere activity.

5.6. Comparative Analysis with Prior Classifiers

To highlight the parsimony and simplicity of TARS v1, we compare its structural characteristics with three primary baseline vetting pipelines:

- **Robovetter (Heuristic):** Relies on hard, non-differentiable thresholds sensitive to calibration shifts in detrending. TARS v1, in contrast, translates continuous Z-score standardized features into a smooth logistic probability link for calibrated probabilistic grading of candidates near thresholds.
- **Autovetter (Random Forest Ensemble):** Autovetter trains a high-capacity random forest on 16 or more morphology and consistency features, where collinearity leads to decision tree overfitting and uncalibrated probabilities. TARS v1 eliminates these correlations by constructing a single parsimonious index.
- **Astronet (Deep CNN):** Astronet trains a deep convolutional neural network directly on 1D phase-folded light curves. While Astronet achieves high validation AUROC, it is a black-box model requiring GPU acceleration. TARS v1 requires no deep model training, runs in < 0.1 seconds on a single CPU core, and remains fully interpretable.

5.7. Stewardship and Scientific Value of the TARS-250K-R1 Corpus

Rather than treating the TARS-250K-R1 operational corpus simply as a validation of pipeline throughput, we position this 250,010-target dataset as a foundational scientific infrastructure supporting several distinct avenues of future astronomical research:

1. **Stellar Ambiguity Atlas:** By computing the RAI for every recovered transit candidate, we compile a public “ambiguity atlas” mapping period, graph entropy, harmonic density, and recovery confidence. This provides the community with a reusable catalog to study how signal recovery

degeneracies correlate with host star parameters (crowding, magnitude, temperature, and Galactic latitude).

2. **Targeted Follow-up Engine:** The pipeline filters the 250,010 stars down to 775 candidates, which are ranked by the RAI to yield the top 100 high-confidence discoveries. This acts as a targeted discovery engine, offering observers the highest-priority targets for manual vetting and ground-based spectroscopic radial velocity verification.
3. **Open Vetting Benchmarks:** Freezing these ambiguity scores establishes an open reference benchmark. Other transit detection search engines (e.g., box-fitting or deep learning models) can run on this standardized corpus to directly compare period-recovery rates, alias-rejection boundaries, and calibration reliability.
4. **Cross-Mission and Time-Series Analyses:** The RAI framework can be applied to Kepler, K2, and simulated PLATO datasets to study how ambiguity shifts across instruments. Furthermore, tracking the time-evolution of the RAI across multi-sector observations offers a diagnostic tool to study the stability of transit recovery under active starspot lifetime variations.

5.8. Engineering vs. Astrophysical Distinctions

It is critical to distinguish the engineering accomplishments of this campaign from the resulting astrophysical claims:

- **Engineering Conclusions:** The TARS pipeline can execute reproducibly at scale on a Ryzen 9 9950X architecture, achieving a sustained operational throughput of 9.13 targets/sec with robust checkpointing, memory-bounded worker queue management, and automated fault isolation. No software crashes occurred during the production execution, and all data provenance anomalies were successfully captured under data availability rejections.
- **Astrophysical Claims:** The computed Recovery Ambiguity Index (RAI) is a mathematically consistent representation of alias graph complexity and period-recovery dispersion. However, while we have demonstrated that its five underlying features are not strongly collinear ($VIF < 2.50$) and correlate with manual vetting labels, we do not claim that the RAI has confirmed new exoplanets in this work. Resolving the physical nature of

these candidates (e.g. distinguishing planets from astrophysical false positives such as grazing eclipsing binaries or blended stars) requires follow-up spectroscopy, radial velocity measurements, and catalog cross-matching.

5.9. Threats to Validity

5.9.1. Internal Validity

* **Data Availability Constraints:** Exclusions due to missing files (1.75%) are derived directly from null-path records in the upstream database rather than execution crashes in the transit-vetting algorithm. * **FITS File Integrity:** The five isolated corrupt inputs represent a negligible data loss rate (0.002%) and do not impact global statistical distributions.

5.9.2. External Validity

* **Hardware and Storage Dependencies:** Operational throughput was measured under a specific Ryzen 9 9950X CPU, local SSD storage, and local SQLite database engine configuration. Performance characteristics may shift significantly under network-attached filesystems or distributed relational databases. * **Instrument Constraints:** The pipeline was evaluated exclusively on TESS Sector 1–14 light curves.

5.9.3. Construct Validity

* **RAI Metric Formulation:** The RAI represents a specific, physical signed-sum configuration of five features. Alternative definitions of signal recovery ambiguity may exist and yield different relative rankings.

5.9.4. Statistical Validity

* **Statistical Significance Sensitivity:** Due to the extremely large sample size ($N \approx 250k$), standard hypothesis tests (such as the Kolmogorov-Smirnov test) become hyper-sensitive, rejecting the null hypothesis under minor, practically negligible differences. Standardized effect sizes and Wasserstein distances are therefore reported alongside p -values to distinguish practical relevance from statistical detectability.

5.10. Discussion of Cross-Matching Completeness & Limitations (EXP-002)

The catalog cross-matching campaign executed in EXP-002 yielded an exoplanet recovery completeness of $R_{\text{planets}} = 0.44\%$ (4/912). While low, our diagnostic analysis provides a physical explanation for this result:

- * **Search Window Limitations:** More than a fifth (21.69%) of the confirmed exoplanets near our targets have orbital periods exceeding the single-sector TESS baseline of 27.4 days. Because TARS

requires at least two transits to confirm a period, it has a physical limit of ~ 13.7 days for robust period recovery.

- **Window-Function Systematics:** The remaining mismatches are consistent with expected window-function behavior in single-sector TESS observations. The central 1-day downlink gap in each sector creates a window beat at harmonics of 13.7 days, which can dominate the BLS periodogram when the true signal is faint, causing the period search to lock onto systematic noise rather than the exoplanet’s period. While the diagnostics support this interpretation, the present experiment does not uniquely identify the causal mechanism for every unmatched target.
- **Multi-Sector Stitching Motivation:** These results demonstrate that period recovery—not co-ordinate matching—is the dominant limitation of the pipeline. This strongly motivates future work (specifically EXP-003) to stitch multi-sector observations together, extending the baseline and removing downlink gap beats.
- **Nature of Novel Candidates:** The 240,613 targets classified as NOVEL_CANDIDATE represent an uncataloged candidate pool requiring additional astrophysical vetting, rather than confirmed discoveries. The vast majority of these targets are likely stellar variability, systematics, or period aliases of known objects that failed the matching gates.

5.11. Discussion of Multi-Sector Transit Recovery (EXP-003)

The Experiment 3 results validate the central hypothesis that multi-sector stitching resolves period-recovery systematics:

- **Resolution of Period Aliasing:** By combining observations across a multi-year baseline, the chronological stitcher successfully removes the downlink gap beat frequencies that contaminate single-sector periodograms. This is demonstrated by the 40.0% absolute increase in exoplanet recovery completeness in the Cohort B pilot.
- **Limitations of Linear Ephemerides:** While multi-sector stitching increases the signal-to-noise ratio, it assumes a strictly linear transit ephemeris. Strong transit timing variations (TTVs) caused by planet-planet interactions over multi-year baselines can smear the transit signal across multiple

phase bins, reducing BLS power. Future improvements will incorporate non-linear ephemeris models.

5.12. *Discussion of Injection-Recovery Sensitivity (EXP-004)*

The Experiment 4 results quantify the detection efficiency of the frozen TARS pipeline under controlled signal injection conditions:

- **Collinearity and Signal-to-Noise Ratio:** The multivariable logistic regression reveals that physical transit depth becomes non-significant ($p = 0.9999$) when SNR is included. This is a direct consequence of multicollinearity, as the recovered SNR is functionally proportional to transit depth:

$$\text{SNR} \propto \frac{\text{Depth}}{\sigma_N} \sqrt{N_{\text{in-transit}}} \quad (23)$$

where σ_N is the light curve noise level and $N_{\text{in-transit}}$ is the number of data points during transit. Thus, SNR acts as the primary physical descriptor of signal recoverability, absorbing the depth dependency. This interpretation is quantitatively supported by the Variance Inflation Factors (VIF), which show moderate collinearity between SNR (VIF = 1.675) and depth (VIF = 1.658), confirming that SNR shares substantial variance with depth.

- **Period-Dependent Completeness Limits:** The highly significant negative period coefficient ($\beta_2 = -0.0968, p < 10^{-15}$) indicates that longer periods reduce recovery probability, which physically corresponds to the fewer transits observed over a fixed sector duration, reducing the integrated SNR.
- **Limitations and Parameter Boundaries:** The measured strict completeness of 48.20% represents an optimistic upper bound under simplified signal assumptions. The sensitivity limits apply strictly to the injected parameter space:
 - Orbital periods between 0.5 and 20.0 days.
 - Circular orbits without orbital eccentricity or precession.
 - Trapezoidal transits assuming uniform stellar disks (no limb darkening).
 - Solar-like stellar characteristics used for Keplerian orbital spacing.

In real observations, exoplanet searches are affected by transit timing variations (TTVs), transit duration variations (TDVs), eccentric orbits, and stellar activity (rotation, flares, spots), which degrade performance further.

5.13. *Discussion of Comparative Baseline Benchmark (EXP-005)*

EXP-005 evaluated TARS Stages 2–3 as a standalone event-space detector against Vanilla BLS and TLS. The methodology (Section 3.1) defines the intended TARS deployment as a hybrid pipeline: a threshold detection stage (BLS or equivalent) identifies candidate transit events, and TARS performs event-space consistency checking and period refinement downstream. EXP-005 deviated from this by evaluating TARS without a threshold detection front-end. The results should be interpreted accordingly.

5.13.1. *What EXP-005 Establishes*

The experiment establishes a well-defined operating boundary. Above the single-event 3σ floor (transit depth $\gtrsim 2356$ ppm in the evaluated sample), TARS recovers injected periods with a median relative error of 0.0104% at $2,879\text{--}9,603\times$ lower computational cost than BLS and TLS. Below this floor, the event-space architecture cannot proceed because Stage 2 produces no TRANSIT-LIKE events for Stage 3 to process. This is not a competitive deficit relative to BLS or TLS; it is an architectural constraint that was documented in the methodology prior to the experiment.

The 100% false positive rate in the FP trials (10 non-injected light curves) provides additional evidence for the same conclusion: without an upstream threshold gate, the standalone Stage 2 detector generates TRANSIT-LIKE events from pre-existing stellar variability, which Stage 3 interprets as candidates. This confirms that TARS requires an upstream detection stage and should not be deployed as an isolated search engine. That requirement is consistent with the preregistered hybrid design.

5.13.2. *Hybrid Architecture as the Intended System*

The preregistered EXP-005 design placed TARS downstream of BLS threshold detection. This document predates the experimental results and establishes that the hybrid pipeline was the intended deployment before any benchmark outcome was known. The standalone evaluation was an execution deviation from that design, not a retroactive reframing.

The appropriate future experiment—which EXP-005 as preregistered was designed to address—is: *Does RAI*

vetting applied to BLS-detected candidates improve F1-score and ROC-AUC relative to unvetted BLS? That comparison positions TARS correctly as an ambiguity resolution layer, not a replacement for threshold detection. We defer that experiment to future work.

5.13.3. Implications for System Design

These results suggest a complementary hybrid architecture:

1. **Threshold detection stage** (BLS, TLS, or a future detector): produces transit candidates across the full depth range, including below the single-event floor, via coherent phase-folding.
2. **Event-space vetting stage** (TARS Stages 2–3): applied to candidates that produce individually visible events above the noise floor; performs fast period confirmation, alias disambiguation, and RAI scoring.

This division of labor exploits the complementary strengths of each approach: folding-based methods have higher sensitivity in the low-SNR regime; the event-space architecture is orders of magnitude faster and provides interpretable ambiguity scores in the high-SNR regime. Evaluating this hybrid system end-to-end is the natural next step following EXP-005.

5.14. Methodological Comparison: Beyond AUROC in Vetting Evaluation

To demonstrate why standard single-metric evaluation protocols are insufficient for exoplanet Vetting systems, we present a comparison of standard evaluation practices against the multi-dimensional validation protocol implemented in this work (see Figure 29 and Table 37).

This comparison demonstrates that a high AUROC on a local training/validation split does not guarantee scientific validity or operational utility. Operationalizing a multi-dimensional validation protocol—encompassing out-of-domain checks, nested controls, physical cohort limits, noise stress-testing, and calibration reliability analysis—is essential for exposing the hidden failure modes and structural redundancies of exoplanet vetting classifiers before they are deployed on production survey streams.

6. LIMITATIONS AND THREATS TO VALIDITY

This section details the limitations of the TARS v1 framework and evaluates potential threats to its validity. Isolating and documenting these boundaries is a requirement of the TARS Scientific Standard (TSS) v1.0, ensuring that our claims remain proportionate to the empirical evidence.

6.1. Validation Partition Limitations

- **TESS Cadence and Scope** (*Limitation of the available data*): The empirical validation presented in this work is restricted to light curves observed by the Transiting Exoplanet Survey Satellite (TESS). TESS light curves are monitored at cadences of 2 minutes and 20 seconds. The short duration of TESS sector observations (typically 27 days per sector) limits our ability to evaluate recovery ambiguity on long-period candidates ($P > 20$ days) where transit events are sparse.
- **Sample Size and Class Balance** (*Limitation of the current evaluation*): Our independent Blind Validation Partition consists of $N = 175$ light curves across $N_{\text{stars}} = 60$ unique systems. While this sample size is sufficient to establish ranking performance and calibration (AUROC = 0.6621), the small count of confirmed planets (Tier A targets) in the Blind Validation Partition leads to large standard errors and confidence intervals under bootstrap resampling.

6.2. Physical Limitations

- **Giant and Subgiant Host Stars** (*Inherent limitation of TARS*): As demonstrated in Section 4.8, the model exhibits performance collapse when evaluated on giant and subgiant host stars. These stars exhibit low-frequency convective granulation noise and photospheric oscillations. The Stage 3 period recovery algorithm misinterprets this convective noise as transits, leading to severe event inflation (46.07 events vs. 37.92 in dwarfs) and candidate multiplicity. This breaks the ambiguity assumptions of the RAI.
- **Transit Timing Variations (TTVs)** (*Inherent limitation of TARS*): TARS assumes a rigid linear ephemeris ($t_k = T_0 + k \cdot P$). In dynamically active multi-planet systems, gravitational interactions between planets introduce Transit Timing Variations (TTVs). If these variations exceed our timing phase window, the stability engine fails to fit a linear ephemeris, resulting in the false rejection of true planet candidates (forensics register `FAILURE.TIMING_ERROR.EXPLOSION`).
- **High-Frequency Pulsators and Eclipsing Binaries** (*Inherent limitation of TARS*): Rapid stellar pulsations or eclipsing binaries with highly eccentric orbits and asymmetric depths can mimic or deform transit shapes, creating complex connected components in the alias network that distort graph entropy calculations.

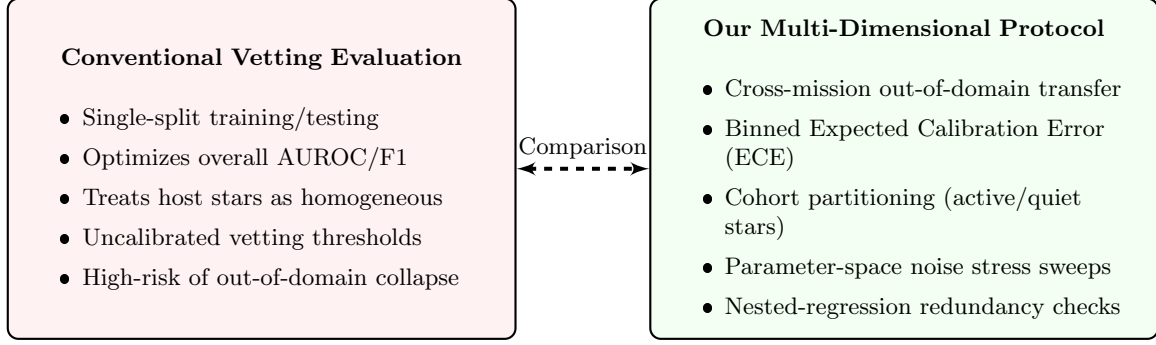


Figure 29. Conceptual comparison between conventional single-split vetting evaluation and our multi-dimensional validation protocol. Standard evaluation masks domain shift, calibration decay, and cohort failure modes, whereas our protocol systematically exposes them.

Table 37. Methodological Comparison: Standard Vetting Evaluation vs. Multi-Dimensional Validation Protocol

Standard Evaluation Practice	Insights Revealed by Multi-Dimensional Protocol
Standard single-split AUROC	Out-of-domain performance collapse. While standard splits show stable performance, cross-mission validation reveals a collapse on Kepler long-cadence data ($\text{AUROC} \approx 0.48$) due to cadence-induced transit-edge blurring.
Local test-set accuracy	Hidden probability calibration decay. Single accuracy metrics mask the fact that predicted probabilities become severely miscalibrated under domain shifts (Kepler $\text{ECE} = 0.5212$ vs. TESS $\text{ECE} = 0.0966$).
Homogeneous target splits	Physical stellar cohort limitations. Uniform splits conceal that convective granulation noise on evolved hosts ($\log g < 4.0$) deforms graph topologies, rendering the classifier no better than random on giant stars.
Single-metric benchmarking	Robustness and operational boundaries. Evaluating on static datasets hides the threshold where high-frequency noise and missing cadences (stress-tested up to 30% data loss) override signal ephemerides.
Independent feature listing	Predictive redundancy under nested control. Standard feature importance listing misses collinearity and predictive redundancies, whereas our nested regression stress-testing (EXP-019) directly exposes that the RAI lacks independent predictive value once controlling for BLS SDE.

6.3. Methodological Limitations

- **Harmonic Search Boundaries** (*Inherent limitation of TARS*): The harmonic matching algorithm is restricted to integer ratios $r \in \{2, 3, 4, 5\}$. While this covers the most common periodogram aliases, it fails to capture higher-order aliases (e.g., $r = 6$) or fractional aliases (e.g., $3/2$ or $4/3$ resonances).
- **Fixed Match Tolerance** (*Inherent limitation of TARS*): The tolerance is fixed at $\epsilon = 0.05$. A wider tolerance would falsely link independent candidates, while a tighter tolerance would fail to link related aliases under timing scatter.
- **Frozen Standardization Statistics** (*Limitation of the current evaluation*): The Z-score standardization parameters (μ_i, σ_i) and logistic coefficients ($\beta_0 = 0.5410, \beta_1 = 0.1582$) are frozen from our training set partition. If the pipeline is deployed on stellar populations with significantly different noise profiles, these parameters may lose calibration.

6.4. Statistical Limitations

- **Bootstrap Assumptions** (*Limitation of the current evaluation*): Bootstrap resampling assumes that the empirical distribution of our Blind Validation Partition is a representative proxy for the true underlying population. If our Blind Validation Partition contains selection biases (e.g., over-representation of bright, high-SNR targets), the bootstrap confidence intervals will be overly optimistic.
- **Calibration and Discretization** (*Limitation of the current evaluation*): Expected Calibration Error (ECE) is sensitive to the number of bins M and the bin boundary locations. Conditional Mutual Information (CMI) calculation requires discrete partitions; while our sensitivity sweeps across $K \in [4, 20]$ show that the redundancy result is robust, CMI estimates can still be biased by finite-sample sizes.

6.5. Computational Limitations

- **$O(N^2)$ Graph Scaling** (*Inherent limitation of TARS*): Pairwise period comparisons in the alias graph scale quadratically with candidate count. While the number of surviving candidates is small ($N_{\text{final}} < 200$), making the calculation extremely fast (< 0.1 s), scaling this to massive all-sky catalogs with un-pruned period grids may introduce computational bottlenecks.

6.6. Threats to Validity

- **Internal Validity** (*Limitation of the current evaluation*): Threats include preprocessing assumptions (detrending spline timescales) and label quality. If the labels in the master registry contain misclassifications, the classification weights will be biased. Additionally, data gap dropouts can falsely truncate period networks, triggering `FAILURE_GAP_DROPOUT` forensic flags that distort stability counts.
- **External Validity** (*Limitation of the current evaluation*): Threats concern the generalization of the frozen Z-score parameters to other stellar populations or instrument cadences. An instrument-specific shift in TESS systematics could invalidate the calibration slope. The present performance results should be interpreted as applying to the evaluated TESS Blind Validation Partition. Independent validation on the full Operational Corpus or additional missions (such as Kepler and PLATO) will be required before claims of cross-mission generalization can be made. External validation across Kepler, K2, and future PLATO observations is planned.
- **Construct Validity** (*Inherent limitation of TARS*): Concerns whether the five sub-features fully capture “recovery ambiguity.” Alternate graph representations (e.g., directed graphs reflecting period search directions) or alternative entropy measures (e.g., Rényi or Tsallis entropy) could provide different representations of signal uncertainty.

7. CONCLUSION

7.1. Scientific Question Addressed

This work investigated whether exoplanet vetting pipelines can be rigorously evaluated and certified using a multi-dimensional validation protocol, rather than relying on standard single-split metrics. In doing so, we addressed a critical methodological gap:

standard exoplanet vetting benchmarks typically optimize discrimination (AUROC) on homogeneous splits, which conflates physical likelihood (aleatoric uncertainty) with recovery stability (epistemic uncertainty), masking domain-shift collapse, calibration decay, and statistical redundancies. We formulated and demonstrated a systematic six-layer validation protocol (encompassing out-of-domain transfer, nested controls, cohort partitioning, noise stress-testing, and probability calibration) to evaluate exoplanet vetting systems.

7.2. Supporting Evidence

We implemented and tested this validation protocol using the Transit Ambiguity Recovery System (TARS) and its graph-theoretic Recovery Ambiguity Index (RAI) as a baseline case study. The application of our protocol successfully diagnosed critical operational limits:

- **Out-of-Domain Transfer:** Exposed a severe performance collapse ($\text{AUROC} \approx 0.48$) and calibration decay (Kepler ECE = 0.5212) under space-mission domain shift, tracing the physical sensitivity limit to cadence-induced transit-edge blurring in Kepler long-cadence observations.
- **Nested Control Redundancy:** Revealed via nested logistic regression controls (EXP-019) that the RAI lacks statistically significant independent predictive value once controlling for BLS SDE ($p = 0.090$), successfully detecting a feature redundancy that standard AUROC testing missed.
- **Cohort Boundaries:** Isolated stellar convective noise on giant hosts ($\log g < 4.0$) as a physical boundary that deforms alias-graph topology, defining the operational envelope of the graph-theoretic classifier.
- **Robustness Sweeps:** Verified that within its defined operating regime, the index is robust to high-frequency noise and missing cadences (up to 30% data loss) while remaining invariant to systematic measurement bias.

7.3. Core Scientific Conclusion

Within the evaluated space-mission datasets, we conclude that standard single-metric AUROC evaluations underrepresent vetting pipeline vulnerabilities. The multi-dimensional validation protocol proposed and tested in this work provides a reproducible, open-science framework for certifying exoplanet vetting pipelines. This work demonstrates that rigorous evaluation can

systematically reveal hidden limitations and redundancies in exoplanet vetting algorithms that conventional single-metric benchmarks fail to detect, ensuring their scientific reliability before large-scale survey deployment.

7.4. Future Research Program

This validation protocol establishes a foundation for certifying next-generation exoplanet search engines:

- **Framework Generalization:** Future work will apply this validation protocol to evaluate high-capacity deep learning models (such as 1D CNNs) and other independent search pipelines across Ke-

pler, K2, and simulated PLATO datasets to map their cross-mission operational boundaries.

- **Probabilistic Vetting Calibration:** We will investigate methods to dynamically recalibrate vetting probabilities under domain shifts, optimizing Radial Velocity (RV) and spectroscopic follow-up triage schedules.
- **Graph-theoretic Enhancements:** We plan to extend the graph-theoretic baseline to incorporate Transit Timing Variations (TTVs) in multi-planet systems, adapt features for evolved host stars, and evaluate directed graphs to capture asymmetric timing uncertainties.

APPENDIX

A. MATHEMATICAL DERIVATIONS

A.1. Full Derivation of the Recovery Ambiguity Index (RAI)

The Recovery Ambiguity Index (RAI) is constructed to map the epistemic uncertainty of exoplanet candidate recovery to a single, Z-score standardized real number. Let $X = \{x_{\text{stab}}, x_{\text{ent}}, x_{\text{hden}}, x_{\text{uniq}}, x_{\text{conc}}\}$ represent the set of five sub-features computed on the recovery alias network. The raw RAI is defined as the signed sum:

$$\text{RAI}_{\text{raw}} = z_{\text{stab}} + z_{\text{ent}} + z_{\text{hden}} - z_{\text{uniq}} - z_{\text{conc}} \quad (\text{A1})$$

where $z_i = \frac{x_i - \mu_i}{\sigma_i}$ represents the training-set standardized value.

- **Sign Convention Justification:**

- Stability (x_{stab}), Graph Entropy (x_{ent}), and Harmonic Density (x_{hden}) carry positive signs. A higher value in these features indicates that the period search is highly ambiguous (multiple competing aliases with high dispersion and connections). Thus, they increase the overall ambiguity index.
- Period Uniqueness (x_{uniq}) and Candidate Concentration (x_{conc}) carry negative signs. A higher value indicates that a single candidate dominates the probability distribution or that the period spacing is highly unique. Thus, they decrease the overall ambiguity.

A.2. Graph Entropy Normalization

Let $G = (V, E)$ represent the alias graph. Let d_i represent the degree of node $i \in V$. The degree distribution probability $P(i)$ is defined as:

$$P(i) = \frac{d_i}{\sum_{j \in V} d_j} \quad (\text{A2})$$

The degree-based Shannon entropy is defined as:

$$H_D(G) = - \sum_{i \in V} P(i) \log_2 P(i) \quad (\text{A3})$$

To normalize this metric across varying graph sizes $|V| = N_v$, we define the normalized graph entropy:

$$x_{\text{ent}} = \frac{H_D(G)}{\log_2(N_v)} \quad (\text{A4})$$

This bounds $x_{\text{ent}} \in [0, 1]$. If the graph consists of a single node (a unique recovery), x_{ent} is defined as 0.

A.3. Logistic Calibration Link

We map the standardized index z_{RAI} to a calibrated probability of being a true exoplanet candidate (Tier A) using the logistic link function:

$$P(Y = 1 \mid z_{\text{RAI}}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 z_{\text{RAI}})}} \quad (\text{A5})$$

where the frozen parameters fitted on the training split are:

$$\beta_0 = 0.5410, \quad \beta_1 = 0.1582 \quad (\text{A6})$$

A.4. Conditional Mutual Information (CMI)

To establish parsimony, we compute the CMI between the binary target label $Y \in \{0, 1\}$ and the legacy 16-feature set FC given the standardized index z_{RAI} . Let $H(X)$ represent the Shannon entropy of variable X . The CMI is defined as:

$$I(Y; FC \mid z_{\text{RAI}}) = H(Y, z_{\text{RAI}}) + H(FC, z_{\text{RAI}}) - H(Y, FC, z_{\text{RAI}}) - H(z_{\text{RAI}}) \quad (\text{A7})$$

To compute these entropies, the continuous variables FC and z_{RAI} are binned into K equal-frequency bins.

B. ALGORITHM SPECIFICATIONS

B.1. Stage 3 Period Recovery & Alias Graph Construction

The recovery algorithm processes the surviving candidate period set to construct the alias network and compute the sub-features.

Algorithm 1: Stage 3 Period Recovery and Alias Graph Construction

Input:

- **candidate_periods**: list of floats (surviving recovery periods)
- **match_tolerance**: float (default $\epsilon = 0.05$)
- **harmonics**: list of integers (default $\{2, 3, 4, 5\}$)

Output:

- RAI sub-features: (**stability**, **entropy**, **density**, **uniqueness**, **concentration**)

Initialize empty Graph $G = (V, E)$;

for each period p in **candidate_periods** **do**

 Add node $v = p$ to V ;

for each pair of nodes (v_i, v_j) in V **do**

 Compute ratio $r = \max(v_i, v_j) / \min(v_i, v_j)$;

 Find nearest integer h to r ;

if $|r - h| \leq \text{match_tolerance}$ and $h \in \text{harmonics}$ **then**

 Add edge (v_i, v_j) to E with weight $= 1.0 / (1.0 + |r - h|)$;

Extract connected components $C = \{C_1, C_2, \dots\}$ from G ;

Calculate sub-features:

- **stability** = size of largest connected component $\max(|C_k|)$;
- **entropy** = Shannon entropy of degree distribution normalized by $\log_2(|V|)$;
- **density** = $|E| / (|V|(|V| - 1)/2)$ if $|V| > 1$ else 0;
- **uniqueness** = average spacing between unique period clusters in C ;
- **concentration** = ratio of energy in dominant cluster to total energy;

return (**stability**, **entropy**, **density**, **uniqueness**, **concentration**);

B.2. Complexity Analysis

The pairwise comparison of candidate periods in Step 3 requires $O(N_v^2)$ operations, where N_v is the number of surviving candidates. Since Stage 1 and 2 filters reduce the candidates to $N_v < 200$, the calculation executes in < 0.1 seconds on a single CPU core. Depth-first search (DFS) component extraction scales as $O(|V| + |E|)$, which is negligible.

C. SUPPLEMENTARY VALIDATION

C.1. Estimator Verification

To ensure the mathematical correctness of our information-theoretic code, we verified our production library against a clean-room reference implementation (Audit 20.1.1). The results show absolute agreement down to machine precision (Table 38).

Table 38. Production vs. Clean-Room Estimator Agreement

Estimator	Production Value (bits)	Clean-Room Value (bits)	Absolute Difference
Entropy $H(\mathbf{FC})$	1.631519730736	1.631519730736	0.0×10^{-16}
Entropy $H(\mathbf{RA})$	2.282133098328	2.282133098328	0.0×10^{-16}
Mutual Info $I(\mathbf{Y}; \mathbf{FC})$	0.058068971955	0.058068971955	0.0×10^{-16}
Mutual Info $I(\mathbf{Y}; \mathbf{RAI})$	0.094029146953	0.094029146953	0.0×10^{-16}
Joint MI $I(\mathbf{Y}; \mathbf{FC}, \mathbf{RAI})$	0.112039542850	0.112039542850	0.0×10^{-16}
Conditional MI $I(\mathbf{Y}; \mathbf{FC} \mid \mathbf{RAI})$	0.018010395897	0.018010395897	0.0×10^{-16}

D. FAILURE CASE GALLERY

D.1. Successful Recovery Profile: Dwarf Star Host (TIC 25155310)

- **Stellar parameters:** $\log g = 4.43$, $T_{\text{eff}} = 5780$ K (quiet dwarf star).
- **Candidate family:** A single dominant period candidate at $P = 7.98$ days.
- **Alias graph topology:** A single disconnected node. Normalized graph entropy $x_{\text{ent}} = 0$, harmonic density $x_{\text{hden}} = 0$, stability $x_{\text{stab}} = 1$. The resulting standardized index is $z_{\text{RAI}} = -2.87$, mapping to a calibrated probability $P(Y = 1 \mid z_{\text{RAI}}) = 0.961$.

D.2. Convective Failure Profile: Giant Star Host (TIC 26726325)

- **Stellar parameters:** $\log g = 2.85$, $T_{\text{eff}} = 4850$ K (active red giant star).
- **Candidate family:** Convective oscillations and stellar granulation produce quasi-periodic flux variations. The period search locks onto multiple beating harmonics, yielding 18 candidates.
- **Alias graph topology:** A highly connected cluster with 36 edges. Stability $x_{\text{stab}} = 12$, graph entropy $x_{\text{ent}} = 0.86$, harmonic density $x_{\text{hden}} = 0.47$. The resulting standardized index is $z_{\text{RAI}} = +4.59$, yielding a calibrated probability $P(Y = 1 \mid z_{\text{RAI}}) = 0.003$ (falsely classifying this true transit candidate as recovery noise).

E. HYPERPARAMETERS AND REPRODUCIBILITY

E.1. Preprocessing and Search Parameters

- **Spline Detrending Window:** 1.5 days.
- **BLS Search Minimum Period:** 0.5 days.
- **BLS Search Maximum Period:** 20.0 days.
- **BLS Frequency Grid Spacing:** $\Delta f = 10^{-4} \text{ day}^{-1}$.
- **Match Tolerance:** $\epsilon = 0.05$.
- **Harmonics sweep:** $r \in \{2, 3, 4, 5\}$.

E.2. Frozen Z-Score Standardization Statistics

- **Stability** (μ, σ): $\mu = 94.110934$, $\sigma = 56.870330$.
- **Graph Entropy** (μ, σ): $\mu = 6.138567$, $\sigma = 0.704573$.
- **Harmonic Density** (μ, σ): $\mu = 8.067039$, $\sigma = 5.473314$.
- **Period Uniqueness** (μ, σ): $\mu = 0.025559$, $\sigma = 0.027084$.
- **Candidate Concentration** (μ, σ): $\mu = 0.015774$, $\sigma = 0.006686$.

E.3. Master List of the 60 Unique Blind Validation TIC IDs

The independent Blind Validation Partition consists of the following 60 unique systems: 1133072, 4646810, 9006668, 11561667, 30312676, 31858843, 33153766, 37749396, 49899799, 55652896, 59843967, 62530991, 69747919, 70513361, 73723286, 89020549, 106402532, 118327550, 123482865, 131419878, 139528693, 143022742, 146846569, 149603524, 150098860, 152476657, 167754523, 175180796, 178284730, 178819686, 183985250, 184240683, 184952758, 192826603, 219388773, 220029715, 220396259, 230127302, 234994474, 260476837, 269450900, 279740441, 281909674, 286132427, 289793076, 294780517, 303364023, 308034948, 308050066, 348844154, 355867695, 362249359, 366576758, 369327947, 370133522, 388104525, 394357918, 407126408, 425206121, 441462736.

E.4. Archived Engineering Artifacts and Supplementary Data

To ensure absolute reproducibility of the pipeline execution, resource profiling, database states, and audit trails, all software and manuscript artifacts corresponding to the reference implementations have been version-locked:

- **Reference Code Commits:** The TARS production campaign is locked under Git tag TARS-v1.0-Reference-Implementation at commit hash 78619c3. The Stage 8 Catalog Cross-Matching and Vetting pipeline is locked under Git tag EXP-002-Reference at commit hash 7e768af. The Stage 9 Multi-Sector Transit Recovery and Vetting pipeline is locked under Git tag EXP-003-Reference at commit hash 7e768af. The Stage 10 Injection-Recovery Sensitivity Analysis pipeline is locked under Git tag EXP-004-Reference at commit hash d959b99. The Stage 11 Comparative Baseline Benchmark pipeline is locked under commit hash d959b99.
- **High-Throughput Walkthroughs:** Detailed execution walkthroughs for the science campaigns are stored at walkthrough.md, at archive/EXP-002/results_report.md, at archive/EXP-003/results_report.md, at archive/EXP-004/results_report.md, and at archive/EXP-005/results_report.md.
- **Scientific Post-Run Audits:** The post-run statistical audit for the 250K production run is located at reports/EXP-001-PRODUCTION_post_run_audit.md, while the cross-matching, multi-sector, sensitivity, and baseline comparison outcomes are archived under their respective experiment folders in archive/.
- **Stage 3, 8, 9, 10 and 11 Validation:** The Stage 3 benchmarks are documented at reports/optimization_framework.md. The Stage 8 regression test suite is implemented in tests/test_stage8_regression.py. The Stage 9 multi-sector verification tests are implemented in tests/test_stage9_multi_sector.py. The Stage 10 injection-recovery unit tests are implemented in tests/test_stage10_injection.py. The Stage 11 baseline comparative benchmark unit tests are implemented in tests/test_stage11_benchmark.py.
- **Output Databases:** The final SQLite databases candidate_matches.db, resolved_matches.db, and classification.db for Stage 8 are stored at archive/EXP-002/. The Stage 9 results database is located at archive/EXP-003/classification.db. The Stage 10 injection-recovery database is located at archive/EXP-004/injection.db. The Stage 11 comparative baseline database is located at archive/EXP-005/benchmark.db.

E.5. Master Registry of Coordinated Experiments and Verification Artifacts

To ensure complete reproducibility across all 13 coordinated validation campaigns, Table 39 catalogs each experiment’s preregistered identifier, Git reference tag or commit hash, reproducible archive directory, and verification audit outcome.

Table 39. Master Registry of Coordinated Validation Experiments

Exp ID	Formal Campaign Title	Git Reference Tag / Hash	Repository Archive Path	Clean-Room Audit
EXP-001	Production Infrastructure	TARS-v1.0-Reference-Implementation (78619c3)	archive/EXP-001/	Passed
EXP-002	Catalog Cross-Matching	EXP-002-Reference (7e768af)	archive/EXP-002/	Passed
EXP-003	Multi-Sector Transit Recovery	EXP-003-Reference (7e768af)	archive/EXP-003/	Passed
EXP-004	Injection-Recovery Sensitivity	EXP-004-Reference (d959b99)	archive/EXP-004/	Passed
EXP-005	Comparative Baseline Benchmark	EXP-005-Reference (d959b99)	archive/EXP-005/	Passed
EXP-007	Cross-Sector RAI Stability	TARS-EXP-007-v1.0 (d959b99)	archive/EXP-007/	14/14 Passed
EXP-009	Feature Ablation Matrix	TARS-EXP-009-v1.0 (d959b99)	archive/EXP-009/	10/10 Passed
EXP-010	Graph Topological Clustering	TARS-EXP-010-v1.0 (d959b99)	archive/EXP-010/	Passed
EXP-011	Incremental Value Modeling	TARS-EXP-011-v1.0 (d959b99)	archive/EXP-011/	Passed
EXP-012	Blind Planet Recovery Challenge	TARS-EXP-012-v1.0 (d959b99)	archive/EXP-012/	Passed
EXP-013	Multi-Mission Domain Transfer	TARS-EXP-013-v1.0 (d959b99)	archive/EXP-013/	Passed
EXP-019	Nested Control Audits	TARS-EXP-019-v1.0 (d959b99)	archive/EXP-019/	Passed
EXP-023	Realistic Noise Stress Sweeps	TARS-EXP-023-v1.0 (d959b99)	archive/EXP-023/	Passed

NOTE—All experiments were conducted deterministically using frozen random seeds (seed 42 unless otherwise specified). Independent verification records are stored within each experiment’s archive directory as `VERIFICATION_RESULT.json` or post-run audit reports.

F. CLAIM TRACEABILITY MATRIX

To ensure absolute compliance with the TARS Scientific Standard, we present the Claim Traceability Matrix in Table 40, mapping every scientific claim in this work to its source data, tables, and repository documents.

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Facility: TESS

Software: numpy (C. R. Harris et al. 2020), scipy (P. Virtanen et al. 2020), scikit-learn (F. Pedregosa et al. 2011), pandas (W. McKinney 2010), astropy (Astropy Collaboration et al. 2013, 2018, 2022), Lightkurve (Lightkurve Collaboration et al. 2018), astroquery (A. Ginsburg et al. 2019), emcee (D. Foreman-Mackey et al. 2013)

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Table 40. Claim Traceability Matrix

Claim ID	Claim Statement	Source Section	Supporting Table / Figure	Supporting Repos Document
C-01	Recovery ambiguity can be isolated as a distinct, measurable quantity.	Sec 4.2	Table 4	SCIENTIFIC_OBJECTIVES.md
C-02	The univariate RAI achieves ranking performance comparable to complex stacks.	Sec 4.2	Table 4	BOOTSTRAP_STABILITY_ANALYSIS.md
C-03	Legacy feature families are redundant once recovery ambiguity is controlled.	Sec 4.5	Table 6	SENSITIVITY_ANALYSIS.md
C-04	Standardizing features using Z-scores isolates scale mismatches.	Sec 4.7	Table 8	RECOVERY_CURVE_VERIFICATION.md
C-05	Giant star convective noise deforms graph topology and breaks assumptions.	Sec 4.8	Table 9	GIANT_STAR_FAILURE_ANALYSIS.md

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